Detailed Subject Syllabi for the Ten Courses of the Faculty of Geoscience and Technology

Subject catalogue with chapters, sections, topics, laboratory and field practice, recommended textbooks (open-access PDFs with contents) and the most similar subjects at Tsinghua University, China University of Geosciences, MIT and Colorado School of Mines

Draft for discussion | 12 November 2026 | companion to 'Proposed Subjects for the Ten Courses'

How this document is organised

- Part A lists every course's subjects semester by semester and gives each subject's catalogue number (S001, S002...).
- Part B is the Subject Catalogue. Each subject is described once, even when several courses use it. Subjects with the same or almost the same name and content are merged into one entry (for example GIS for Geoscientists, GIS for Planning and Geographic Information Systems). The 'Used in' and 'Versions' lines show which course uses which version and how many credits.
- Syllabus: chapters, numbered sections and topics (in italics), followed by laboratory, field, computer or project practice and the assessment scheme. Contact hours follow the credit rule of the curriculum (1 credit = 16 lecture hours, 32 laboratory hours or 1 field week).
- Textbooks: open-access books come first, with the link to the free PDF or online edition; their tables of contents are in Appendix A. Only legally free editions are linked. Commercial textbooks have no legal free PDF; for these a WorldCat library-lookup link is given instead.
- Benchmark subjects: MIT (OpenCourseWare) and Colorado School of Mines (official catalogue) publish subject syllabi, and these are summarised with links. Tsinghua University and China University of Geosciences publish course names, credits, hours and semesters in their training plans but not subject syllabi, so their matching courses are named with a link to the plan.

Note on accuracy. Chapter lists of open textbooks marked 'not verified' in Appendix A could not be checked for this report; open the link for the current contents. Edition numbers of commercial books should be checked against your library.

Contents

Contents 2

Part A. Subjects of each course by semester 7

Part B. Subject Catalogue 14

S001 General education (languages, humanities, physical education) . . . 14
S002 Calculus I . . . 14
S003 Linear Algebra . . . 15
S004 General Physics I (Mechanics, Heat, Waves) . . . 15
S005 General Chemistry with Laboratory . . . 16
S006 Programming for Engineers (Python) . . . 16
S007 Introduction to Geoscience and Technology (seminar) . . . 17
S008 Calculus II (Multivariable and Vector Calculus) . . . 17
S009 General Physics II (Electromagnetism, Optics, Modern Physics) . . . 18
S010 Physics Laboratory . . . 19
S011 Physical Geology (with laboratory) . . . 19
S012 Engineering Mechanics (Statics and Dynamics) . . . 20
S013 Engineering Graphics and CAD . . . 20
S014 Geological Recognition Field Practice (2 weeks) . . . 21
S015 Differential Equations and Mathematical Physics . . . 21
S016 Probability and Statistics . . . 22
S017 Mechanics of Materials . . . 22
S018 Fluid Mechanics . . . 23
S019 Electrical Engineering Fundamentals . . . 23
S020 Earth Materials (Mineralogy and Petrology I) . . . 24
S021 Electronic Engineering and Instrumentation . . . 24
S022 Numerical Methods and Geodata Analysis . . . 25
S023 Petrology and Optical Petrography . . . 25
S024 Structural Geology . . . 26
S025 Paleontology, Stratigraphy and Historical Geology . . . 27
S026 Geological Mapping Field School (4 weeks) . . . 27
S027 Ore Deposit Geology . . . 28
S028 Geochemistry . . . 29
S029 Ore Microscopy and Ore Petrography . . . 29
S030 Applied Geophysics for Exploration . . . 30
S031 Remote Sensing and Digital Image Processing . . . 30
S032 Sedimentology, Stratigraphy and Sedimentary Basins . . . 31
S033 Geographic Information Systems (GIS) . . . 31
S034 Mineral Exploration Methods . . . 32
S035 Exploration Geochemistry . . . 33
S036 Exploration Drilling and Sampling . . . 33
S037 Geostatistics and Mineral Resource Estimation . . . 34
S038 Introduction to Mining and Mineral Processing . . . 34
S039 Hydrogeology . . . 35
S040 Metallogeny and Mineral Prospectivity . . . 35

S041 Mineral Economics and Project Evaluation . . . 36
S042 Mineral Exploration Design (capstone) . . . 36
S043 Reporting Codes, Exploration Standards and Mineral Law . . . 36
S044 3D Geological Modelling . . . 37
S045 Coal and Industrial Minerals Geology . . . 37
S046 Environmental Geology, Mine Wastes and Closure . . . 37
S047 Professional English and Technical Communication . . . 38
S048 Elective (from any course of the faculty) . . . 38
S049 Structural Analysis I . . . 38
S050 Construction Materials (with laboratory) . . . 39
S051 Soil Mechanics . . . 39
S052 Engineering Surveying . . . 40
S053 Surveying and Engineering Geology Field Practice (4 weeks) . . . 40
S054 Structural Analysis II (Matrix and Finite Element Methods) . . . 40
S055 Reinforced Concrete Design I . . . 41
S056 Steel Structures . . . 41
S057 Foundation Engineering . . . 41
S058 Engineering Geology and Site Investigation . . . 42
S059 Engineering Hydraulics and Hydrology . . . 42
S060 Building Architecture and Design . . . 43
S061 Reinforced Concrete Design II (Buildings) . . . 43
S062 Structural Dynamics and Earthquake Engineering . . . 43
S063 Rock Mechanics . . . 44
S064 Construction Technology and Management . . . 44
S065 Road and Bridge Engineering . . . 45
S066 Engineering Economics and Project Feasibility . . . 45
S067 BIM and Computer-Aided Structural Design . . . 45
S068 Integrated Civil Engineering Design (capstone) . . . 46
S069 Slope Stability and Geohazard Mitigation . . . 46
S070 Building Services and Green Building . . . 47
S071 Construction Contracts, Codes and Law . . . 47
S072 Tunnelling and Underground Space Engineering . . . 47
S073 Fundamentals of Geophysics . . . 48
S074 Signals, Systems and Digital Signal Processing . . . 48
S075 Electromagnetic Field and Wave Theory . . . 48
S076 Applied Geology for Geophysicists and Drilling Engineers . . . 49
S077 Geological and Geophysical Field Practice (4 weeks) . . . 49
S078 Gravity and Magnetic Prospecting . . . 50
S079 Electrical Prospecting (Resistivity, IP, SP) . . . 50
S080 Seismic Exploration . . . 51
S081 Petrophysics and Rock Physics . . . 51
S082 Geophysical Inverse Theory . . . 52
S083 Geophysical Instrumentation . . . 52
S084 Electromagnetic Prospecting (TEM, MT, CSAMT, GPR) . . . 52
S085 Seismic Data Processing . . . 53

S086 Well Logging and Formation Evaluation . . . 53
S087 Geophysical Equipment Design . . . 54
S088 Airborne and UAV Geophysics . . . 54
S089 Numerical Modelling in Geophysics . . . 54
S090 Seismic Interpretation and Prospect Evaluation . . . 55
S091 Integrated Geophysical Interpretation and Survey Design (capstone) . . . 55
S092 Engineering and Environmental Geophysics . . . 55
S093 Radiometric and Nuclear Prospecting . . . 56
S094 Geophysical Equipment Development Project . . . 56
S095 Analytical Chemistry (Quantitative and Instrumental) . . . 56
S096 Instrumental Geoanalysis . . . 57
S097 Isotope Geochemistry and Geochronology . . . 57
S098 Aqueous Geochemistry and Geochemical Thermodynamics . . . 58
S099 Geochemical Data Processing and Multivariate Statistics . . . 58
S100 Biogeochemical and Gas Methods . . . 58
S101 Hydrogeochemistry . . . 59
S102 Environmental Geochemistry . . . 59
S103 Organic Geochemistry . . . 59
S104 Exploration Geochemistry II (Anomaly Interpretation and Regional Mapping) . . . 60
S105 Geochemical Survey Design (capstone) . . . 60
S106 QA/QC and Laboratory Management . . . 60
S107 Rock Mechanics and Rock Breaking . . . 61
S108 Machine Elements and Mechanical Design . . . 61
S109 Hydraulic and Pneumatic Drives . . . 62
S110 Drilling Equipment and Geology Field Practice (4 weeks) . . . 62
S111 Drilling Engineering I (Methods) . . . 62
S112 Drilling Fluids and Hydraulics . . . 63
S113 Drilling Machinery and Equipment . . . 63
S114 Drill Bits and Diamond Tools . . . 64
S115 Engineering Thermodynamics and Heat Transfer . . . 64
S116 Electrical Drives and Automation . . . 64
S117 Drilling Engineering II (Core Drilling for Exploration) . . . 65
S118 Directional and Horizontal Drilling . . . 65
S119 Well Design: Casing, Cementing and Completion . . . 66
S120 Water Well Drilling and Groundwater . . . 66
S121 Geotechnical Drilling, In-situ Testing and Piling . . . 66
S122 Drilling Instrumentation and Monitoring (MWD/LWD) . . . 67
S123 Drilling Project Management and Cost Engineering . . . 67
S124 Drilling Accidents, Well Control and Safety . . . 67
S125 Drilling Project Design (capstone) . . . 68
S126 Trenchless Technology and Mechanised Tunnelling . . . 68
S127 Deep, Scientific and Geothermal Drilling . . . 68
S128 Drilling Automation and Intelligent Drilling . . . 69
S129 Systematic Mineralogy . . . 69
S130 Physical Chemistry . . . 69

S131 Mineral Physics and Crystal Chemistry . . . 70
S132 Analytical Mineralogy . . . 70
S133 Mineral Processing I (Comminution and Physical Separation) . . . 70
S134 Materials Science and Engineering . . . 71
S135 Industrial Minerals and Rocks . . . 71
S136 Process Mineralogy . . . 72
S137 Mineral Processing II (Flotation and Separation) . . . 72
S138 Non-metallic Mineral Materials (Ceramics, Glass, Cement) . . . 72
S139 Rock Mechanics and Dimension Stone . . . 73
S140 Construction Aggregates and Building Stone Technology . . . 73
S141 Gemmology . . . 73
S142 Introduction to Extractive Metallurgy . . . 74
S143 Mineral Materials Development Project (capstone) . . . 74
S144 Functional Mineral Materials . . . 74
S145 Testing and Standards for Mineral Products . . . 75
S146 Environmental Mineralogy and Tailings Utilisation . . . 75
S147 Plant Design and Process Economics . . . 75
S148 Database Systems . . . 76
S149 Data Structures and Algorithms . . . 76
S150 Surveying and Geodesy . . . 76
S151 Cartography and Geological Map Compilation . . . 77
S152 Software Engineering . . . 77
S153 Spatial Databases and Geological Data Models . . . 78
S154 Photogrammetry and UAV Mapping . . . 78
S155 GNSS Positioning and Satellite Geodesy . . . 78
S156 Spatial Analysis and Mineral Prospectivity Mapping . . . 79
S157 Machine Learning for Geoscience . . . 79
S158 WebGIS and Geospatial Application Development . . . 80
S159 Geological Information System Design (capstone) . . . 80
S160 Big Data and Cloud Computing for Geoscience . . . 80
S161 Hyperspectral, Microwave and SAR Remote Sensing . . . 81
S162 Digital Mine and Smart Mining . . . 81
S163 Geohazard Monitoring and Early-Warning Systems . . . 81
S164 Introduction to Space Science and Astronomy . . . 82
S165 Atmospheric Physics . . . 82
S166 Observatory and Meteorological Station Practice (4 weeks) . . . 82
S167 Orbital Mechanics . . . 83
S168 Atmospheric Dynamics . . . 83
S169 Physics of Remote Sensing . . . 84
S170 Space Physics (Sun, Magnetosphere, Ionosphere) . . . 84
S171 Atmospheric Chemistry . . . 84
S172 Planetary Science . . . 85
S173 Spacecraft Systems Engineering . . . 85
S174 Satellite Communications and Telemetry . . . 85
S175 Weather Analysis and Numerical Weather Prediction . . . 86

S176 Climate System Science . . . 86
S177 Satellite Meteorology . . . 86
S178 Space Mission Design (CubeSat capstone) . . . 87
S179 Planetary Geology and Space Resources . . . 87
S180 Space Weather . . . 87
S181 Data Assimilation . . . 88
S182 Geomorphology and Landscape Geology . . . 88
S183 Principles of Tourism and Geotourism . . . 88
S184 Urban and Regional Planning Principles . . . 89
S185 Landscape and Geoheritage Field Survey (4 weeks) . . . 89
S186 Tourism Planning . . . 89
S187 Geoheritage Assessment and Geopark Management . . . 90
S188 Landscape Architecture and Site Design . . . 90
S189 Environmental Impact Assessment . . . 91
S190 Tourism Infrastructure Engineering . . . 91
S191 Tourism Economics and Market Analysis . . . 91
S192 Tourist Facility Design . . . 92
S193 Geohazards and Visitor Safety . . . 92
S194 Natural and Cultural Heritage Conservation . . . 92
S195 Destination Management and Marketing . . . 93
S196 Structures for Leisure Facilities . . . 93
S197 Tourist District Master Plan Studio (capstone) . . . 93
S198 Sustainable Tourism and Carrying Capacity . . . 93
S199 Tourism Law, Policy and Investment . . . 94
S200 Heritage Interpretation and Visitor-Centre Design . . . 94
S201 Sedimentary Petrology . . . 94
S202 Petroleum Geology . . . 95
S203 Reservoir Rock and Fluid Properties . . . 95
S204 Sedimentary Basin Analysis . . . 96
S205 Petroleum Drilling Engineering . . . 96
S206 Reservoir Engineering . . . 96
S207 Production Engineering . . . 97
S208 Petroleum Machinery and Equipment . . . 97
S209 Well Testing . . . 97
S210 Petroleum Exploration and Development Design (capstone) . . . 98
S211 Enhanced Oil Recovery . . . 98
S212 Petroleum Refining and Processing . . . 98
S213 Natural Gas and Unconventional Resources . . . 99
S214 Reservoir Simulation . . . 99
S215 Petroleum Economics and HSE . . . 99
S216 Production Internship (5 weeks) . . . 100
S217 Graduation Design / Bachelor Thesis . . . 100
S218 Frontiers Seminar of the Discipline . . . 100
Appendix A. Open-access textbooks: links and tables of contents **101**
Appendix B. Benchmark sources **104**

Part A. Subjects of each course by semester

Common faculty foundation (all courses)

Semester	No.	Subject (as named in the course)	Cr	Catalogue entry
Sem 1	S001	General education (languages, humanities, physical education)	3	-
Sem 1	S002	Calculus I	5	-
Sem 1	S003	Linear Algebra	3	-
Sem 1	S004	General Physics I (Mechanics, Heat, Waves)	4	-
Sem 1	S005	General Chemistry with Laboratory	3	-
Sem 1	S006	Programming for Engineers (Python)	3	-
Sem 1	S007	Introduction to Geoscience and Technology (seminar)	1	-
Sem 2	S001	General education (languages, humanities, physical education)	2	-
Sem 2	S008	Calculus II (Multivariable and Vector Calculus)	5	-
Sem 2	S009	General Physics II (Electromagnetism, Optics, Modern Physics)	4	-
Sem 2	S010	Physics Laboratory	1	-
Sem 2	S011	Physical Geology (with laboratory)	4	-
Sem 2	S012	Engineering Mechanics (Statics and Dynamics)	4	-
Sem 2	S013	Engineering Graphics and CAD	2	-
Summer 1	S014	Geological Recognition Field Practice (2 weeks)	2	-
Sem 3	S001	General education (languages, humanities, physical education)	2	-
Sem 3	S015	Differential Equations and Mathematical Physics	3	-
Sem 3	S016	Probability and Statistics	3	-
Sem 3	S017	Mechanics of Materials	3	-
Sem 3	S018	Fluid Mechanics	3	-
Sem 3	S019	Electrical Engineering Fundamentals	3	-
Sem 3	S020	Earth Materials (Mineralogy and Petrology I)	4	-
Sem 4	S001	General education (languages, humanities, physical education)	2	-
Sem 4	S021	Electronic Engineering and Instrumentation	3	-
Sem 4	S022	Numerical Methods and Geodata Analysis	3	-

Mineral Exploration course

Semester	No.	Subject (as named in the course)	Cr	Catalogue entry
Sem 4	S023	Petrology and Optical Petrography	4	-
Sem 4	S024	Structural Geology	4	-
Sem 4	S025	Paleontology, Stratigraphy and Historical Geology	4	-
Summer 2	S026	Geological Mapping Field School (4 weeks)	4	-
Sem 5	S027	Ore Deposit Geology	4	-
Sem 5	S028	Geochemistry	3	-
Sem 5	S029	Ore Microscopy and Ore Petrography	3	-
Sem 5	S030	Applied Geophysics for Exploration	3	-
Sem 5	S031	Remote Sensing Geology	2	Remote Sensing and Digital Image Processing
Sem 5	S032	Sedimentology and Sedimentary Basins	3	Sedimentology, Stratigraphy and Sedimentary Basins
Sem 5	S033	GIS for Geoscientists	2	Geographic Information Systems (GIS)
Sem 6	S034	Mineral Exploration Methods	4	-
Sem 6	S035	Exploration Geochemistry	3	-
Sem 6	S036	Exploration Drilling and Sampling	2	-
Sem 6	S037	Mineral Resource Estimation and Geostatistics	3	Geostatistics and Mineral Resource Estimation
Sem 6	S038	Introduction to Mining and Mineral Processing	2	-

Semester	No.	Subject (as named in the course)	Cr	Catalogue entry
Sem 6	S039	Hydrogeology	2	-
Sem 6	S040	Metallogeny and Mineral Prospectivity	2	-
Sem 6	S041	Mineral Economics and Project Evaluation	2	-
Summer 3	S216	Production Internship (5 weeks)	5	-
Sem 7	S042	Mineral Exploration Design (capstone)	3	-
Sem 7	S043	Reporting Codes, Exploration Standards and Mineral Law	2	-
Sem 7	S044	3D Geological Modelling	2	-
Sem 7	S045	Coal and Industrial Minerals Geology	3	-
Sem 7	S046	Environmental Geology and Mine Closure	2	Environmental Geology, Mine Wastes and Closure
Sem 7	S047	Professional English and Technical Communication	2	-
Sem 7	S048	Elective (from any course of the faculty)	4	-
Sem 8	S217	Graduation Design / Bachelor Thesis	8	-
Sem 8	S218	Frontiers Seminar of the Discipline	2	-

Civil Engineering course

Semester	No.	Subject (as named in the course)	Cr	Catalogue entry
Sem 4	S049	Structural Analysis I	4	-
Sem 4	S050	Construction Materials (with laboratory)	3	-
Sem 4	S051	Soil Mechanics (with laboratory)	3	Soil Mechanics
Sem 4	S052	Engineering Surveying	2	-
Summer 2	S053	Surveying and Engineering Geology Field Practice (4 weeks)	4	-
Sem 5	S054	Structural Analysis II (Matrix and Finite Element Methods)	3	-
Sem 5	S055	Reinforced Concrete Design I	4	-
Sem 5	S056	Steel Structures	3	-
Sem 5	S057	Foundation Engineering	3	-
Sem 5	S058	Engineering Geology and Site Investigation	3	-
Sem 5	S059	Engineering Hydraulics and Hydrology	2	-
Sem 5	S060	Building Architecture and Design	2	-
Sem 6	S061	Reinforced Concrete Design II (Buildings)	3	-
Sem 6	S062	Structural Dynamics and Earthquake Engineering	3	-
Sem 6	S063	Rock Mechanics	3	-
Sem 6	S064	Construction Technology and Management	3	-
Sem 6	S065	Road and Bridge Engineering	3	-
Sem 6	S066	Engineering Economics	2	Engineering Economics and Project Feasibility
Sem 6	S067	BIM and Computer-Aided Structural Design	3	-
Summer 3	S216	Production Internship (5 weeks)	5	-
Sem 7	S068	Integrated Civil Engineering Design (capstone)	4	-
Sem 7	S069	Slope Stability and Geohazard Mitigation	3	-
Sem 7	S070	Building Services and Green Building	2	-
Sem 7	S071	Construction Contracts, Codes and Law	2	-
Sem 7	S072	Tunnelling and Underground Space Engineering	3	-
Sem 7	S047	Professional English and Technical Communication	2	-
Sem 7	S048	Elective (from any course of the faculty)	2	-
Sem 8	S217	Graduation Design / Bachelor Thesis	8	-
Sem 8	S218	Frontiers Seminar of the Discipline	2	-

Geophysical Exploration course

Semester	No.	Subject (as named in the course)	Cr	Catalogue entry
Sem 4	S073	Fundamentals of Geophysics	3	-
Sem 4	S074	Signals, Systems and Digital Signal Processing	3	-
Sem 4	S075	Electromagnetic Field Theory	3	Electromagnetic Field and Wave Theory
Sem 4	S076	Petrology and Structural Geology for Geophysicists	3	Applied Geology for Geophysicists and Drilling Engineers
Summer 2	S077	Geological and Geophysical Field Practice (4 weeks)	4	-
Sem 5	S078	Gravity and Magnetic Prospecting	4	-
Sem 5	S079	Electrical Prospecting (Resistivity, IP, SP)	4	-
Sem 5	S080	Seismic Exploration	4	-
Sem 5	S081	Petrophysics and Rock Physics	2	-
Sem 5	S082	Geophysical Inverse Theory	3	-
Sem 5	S083	Geophysical Instrumentation	3	-
Sem 6	S084	Electromagnetic Prospecting (TEM, MT, CSAMT, GPR)	4	-
Sem 6	S085	Seismic Data Processing	4	-
Sem 6	S086	Well Logging and Formation Evaluation	3	-
Sem 6	S027	Ore Deposit Geology	2	-
Sem 6	S087	Geophysical Equipment Design	3	-
Sem 6	S088	Airborne and UAV Geophysics	2	-
Sem 6	S089	Numerical Modelling in Geophysics	2	-
Summer 3	S216	Production Internship (5 weeks)	5	-
Sem 7	S090	Seismic Interpretation	3	Seismic Interpretation and Prospect Evaluation
Sem 7	S091	Integrated Geophysical Interpretation and Survey Design (capstone)	3	-
Sem 7	S092	Engineering and Environmental Geophysics	3	-
Sem 7	S093	Radiometric and Nuclear Prospecting	2	-
Sem 7	S094	Geophysical Equipment Development Project	3	-
Sem 7	S047	Professional English and Technical Communication	2	-
Sem 7	S048	Elective (from any course of the faculty)	2	-
Sem 8	S217	Graduation Design / Bachelor Thesis	8	-
Sem 8	S218	Frontiers Seminar of the Discipline	2	-

Geochemical Exploration course

Semester	No.	Subject (as named in the course)	Cr	Catalogue entry
Sem 4	S023	Petrology and Optical Petrography	4	-
Sem 4	S024	Structural Geology	4	-
Sem 4	S095	Analytical Chemistry (Quantitative and Instrumental)	4	-
Summer 2	S026	Geological Mapping Field School (4 weeks)	4	-
Sem 5	S028	Geochemistry	4	-
Sem 5	S027	Ore Deposit Geology	4	-
Sem 5	S096	Instrumental Geoanalysis	3	-
Sem 5	S097	Isotope Geochemistry and Geochronology	3	-
Sem 5	S098	Aqueous Geochemistry and Geochemical Thermodynamics	3	-
Sem 5	S030	Applied Geophysics for Exploration	3	-
Sem 6	S035	Exploration Geochemistry I (Survey Methods)	4	Exploration Geochemistry
Sem 6	S099	Geochemical Data Processing and Multivariate Statistics	3	-
Sem 6	S100	Biogeochemical and Gas Methods	2	-
Sem 6	S101	Hydrogeochemistry	2	-
Sem 6	S102	Environmental Geochemistry	3	-
Sem 6	S103	Organic Geochemistry	2	-
Sem 6	S034	Mineral Exploration Methods	2	-
Sem 6	S033	GIS for Geoscientists	2	Geographic Information Systems (GIS)
Summer 3	S216	Production Internship (5 weeks)	5	-

Semester	No.	Subject (as named in the course)	Cr	Catalogue entry
Sem 7	S104	Exploration Geochemistry II (Anomaly Interpretation and Regional Mapping)	3	-
Sem 7	S105	Geochemical Survey Design (capstone)	3	-
Sem 7	S106	QA/QC and Laboratory Management	2	-
Sem 7	S037	Mineral Resource Estimation and Geostatistics	2	Geostatistics and Mineral Resource Estimation
Sem 7	S046	Geochemistry of the Mining Environment and Remediation	3	Environmental Geology, Mine Wastes and Closure
Sem 7	S047	Professional English and Technical Communication	2	-
Sem 7	S048	Elective (from any course of the faculty)	3	-
Sem 8	S217	Graduation Design / Bachelor Thesis	8	-
Sem 8	S218	Frontiers Seminar of the Discipline	2	-

Drill Engineering course

Semester	No.	Subject (as named in the course)	Cr	Catalogue entry
Sem 4	S107	Rock Mechanics and Rock Breaking	3	-
Sem 4	S108	Machine Elements and Mechanical Design	4	-
Sem 4	S109	Hydraulic and Pneumatic Drives	3	-
Sem 4	S076	Geology for Drilling Engineers	2	Applied Geology for Geophysicists and Drilling Engineers
Summer 2	S110	Drilling Equipment and Geology Field Practice (4 weeks)	4	-
Sem 5	S111	Drilling Engineering I (Methods)	4	-
Sem 5	S112	Drilling Fluids and Hydraulics	3	-
Sem 5	S113	Drilling Machinery and Equipment	4	-
Sem 5	S114	Drill Bits and Diamond Tools	3	-
Sem 5	S115	Engineering Thermodynamics and Heat Transfer	2	-
Sem 5	S116	Electrical Drives and Automation	2	-
Sem 5	S051	Geotechnical Engineering Basics	2	Soil Mechanics
Sem 6	S117	Drilling Engineering II (Core Drilling for Exploration)	3	-
Sem 6	S118	Directional and Horizontal Drilling	3	-
Sem 6	S119	Well Design: Casing, Cementing and Completion	3	-
Sem 6	S120	Water Well Drilling and Groundwater	3	-
Sem 6	S121	Geotechnical Drilling, In-situ Testing and Piling	3	-
Sem 6	S122	Drilling Instrumentation and Monitoring (MWD/LWD)	3	-
Sem 6	S123	Drilling Project Management and Cost Engineering	2	-
Summer 3	S216	Production Internship (5 weeks)	5	-
Sem 7	S124	Drilling Accidents, Well Control and Safety	3	-
Sem 7	S125	Drilling Project Design (capstone)	3	-
Sem 7	S126	Trenchless Technology and Mechanised Tunnelling	3	-
Sem 7	S127	Deep, Scientific and Geothermal Drilling	2	-
Sem 7	S128	Drilling Automation and Intelligent Drilling	2	-
Sem 7	S047	Professional English and Technical Communication	2	-
Sem 7	S048	Elective (from any course of the faculty)	3	-
Sem 8	S217	Graduation Design / Bachelor Thesis	8	-
Sem 8	S218	Frontiers Seminar of the Discipline	2	-

Mineral Engineering course

Semester	No.	Subject (as named in the course)	Cr	Catalogue entry
Sem 4	S023	Optical Mineralogy and Petrography	4	Petrology and Optical Petrography
Sem 4	S129	Systematic Mineralogy	3	-
Sem 4	S130	Physical Chemistry	3	-
Sem 4	S024	Structural Geology	2	-
Summer 2	S026	Geological Mapping Field School (4 weeks)	4	-
Sem 5	S131	Mineral Physics and Crystal Chemistry	3	-
Sem 5	S132	Analytical Mineralogy	4	-
Sem 5	S133	Mineral Processing I (Comminution and Physical Separation)	4	-
Sem 5	S134	Materials Science and Engineering	3	-
Sem 5	S135	Industrial Minerals and Rocks	3	-
Sem 5	S136	Process Mineralogy	3	-
Sem 6	S137	Mineral Processing II (Flotation and Separation)	4	-
Sem 6	S138	Non-metallic Mineral Materials (Ceramics, Glass, Cement)	4	-
Sem 6	S139	Rock Mechanics and Dimension Stone	3	-
Sem 6	S140	Construction Aggregates and Building Stone Technology	3	-
Sem 6	S141	Gemmology	3	-
Sem 6	S142	Introduction to Extractive Metallurgy	3	-
Summer 3	S216	Production Internship (5 weeks)	5	-
Sem 7	S143	Mineral Materials Development Project (capstone)	3	-
Sem 7	S144	Functional Mineral Materials	3	-
Sem 7	S145	Testing and Standards for Mineral Products	2	-
Sem 7	S146	Environmental Mineralogy and Tailings Utilisation	3	-
Sem 7	S147	Plant Design and Process Economics	3	-
Sem 7	S047	Professional English and Technical Communication	2	-
Sem 7	S048	Elective (from any course of the faculty)	2	-
Sem 8	S217	Graduation Design / Bachelor Thesis	8	-
Sem 8	S218	Frontiers Seminar of the Discipline	2	-

Geological Information course

Semester	No.	Subject (as named in the course)	Cr	Catalogue entry
Sem 4	S148	Database Systems	3	-
Sem 4	S149	Data Structures and Algorithms	3	-
Sem 4	S024	Structural Geology	3	-
Sem 4	S150	Surveying and Geodesy	3	-
Summer 2	S026	Geological Mapping Field School (4 weeks)	4	-
Sem 5	S033	Geographic Information Systems	4	Geographic Information Systems (GIS)
Sem 5	S031	Remote Sensing and Digital Image Processing	4	-
Sem 5	S151	Cartography and Geological Map Compilation	3	-
Sem 5	S037	Geostatistics	3	Geostatistics and Mineral Resource Estimation
Sem 5	S152	Software Engineering	3	-
Sem 5	S027	Mineral Deposits and Exploration for Information Engineers	3	Ore Deposit Geology
Sem 6	S153	Spatial Databases and Geological Data Models	3	-
Sem 6	S044	3D Geological Modelling	4	-
Sem 6	S154	Photogrammetry and UAV Mapping	3	-
Sem 6	S155	GNSS Positioning	2	GNSS Positioning and Satellite Geodesy
Sem 6	S156	Spatial Analysis and Mineral Prospectivity Mapping	3	-
Sem 6	S157	Machine Learning for Geoscience	3	-
Sem 6	S158	WebGIS and Geospatial Application Development	2	-
Summer 3	S216	Production Internship (5 weeks)	5	-
Sem 7	S159	Geological Information System Design (capstone)	3	-
Sem 7	S160	Big Data and Cloud Computing for Geoscience	3	-

Semester	No.	Subject (as named in the course)	Cr	Catalogue entry
Sem 7	S161	Hyperspectral and Radar (SAR) Remote Sensing	3	Hyperspectral, Microwave and SAR Remote Sensing
Sem 7	S162	Digital Mine and Smart Mining	2	-
Sem 7	S163	Geohazard Monitoring and Early-Warning Systems	3	-
Sem 7	S047	Professional English and Technical Communication	2	-
Sem 7	S048	Elective (from any course of the faculty)	2	-
Sem 8	S217	Graduation Design / Bachelor Thesis	8	-
Sem 8	S218	Frontiers Seminar of the Discipline	2	-

Space course

Semester	No.	Subject (as named in the course)	Cr	Catalogue entry
Sem 4	S164	Introduction to Space Science and Astronomy	3	-
Sem 4	S075	Electromagnetism and Electromagnetic Waves	3	Electromagnetic Field and Wave Theory
Sem 4	S074	Signals, Systems and Digital Signal Processing	3	-
Sem 4	S165	Atmospheric Physics	3	-
Summer 2	S166	Observatory and Meteorological Station Practice (4 weeks)	4	-
Sem 5	S167	Orbital Mechanics	4	-
Sem 5	S168	Atmospheric Dynamics	4	-
Sem 5	S169	Physics of Remote Sensing	4	-
Sem 5	S170	Space Physics (Sun, Magnetosphere, Ionosphere)	4	-
Sem 5	S171	Atmospheric Chemistry	2	-
Sem 5	S172	Planetary Science	2	-
Sem 6	S173	Spacecraft Systems Engineering	4	-
Sem 6	S174	Satellite Communications and Telemetry	3	-
Sem 6	S175	Weather Analysis and Numerical Weather Prediction	3	-
Sem 6	S176	Climate System Science	3	-
Sem 6	S177	Satellite Meteorology	3	-
Sem 6	S155	Satellite Geodesy and GNSS	2	GNSS Positioning and Satellite Geodesy
Sem 6	S031	Remote Sensing Image Processing	2	Remote Sensing and Digital Image Processing
Summer 3	S216	Production Internship (5 weeks)	5	-
Sem 7	S178	Space Mission Design (CubeSat capstone)	4	-
Sem 7	S179	Planetary Geology and Space Resources	3	-
Sem 7	S180	Space Weather	2	-
Sem 7	S161	Microwave and SAR Remote Sensing	3	Hyperspectral, Microwave and SAR Remote Sensing
Sem 7	S181	Data Assimilation	2	-
Sem 7	S047	Professional English and Technical Communication	2	-
Sem 7	S048	Elective (from any course of the faculty)	2	-
Sem 8	S217	Graduation Design / Bachelor Thesis	8	-
Sem 8	S218	Frontiers Seminar of the Discipline	2	-

Tourist District Development course

Semester	No.	Subject (as named in the course)	Cr	Catalogue entry
Sem 4	S182	Geomorphology and Landscape Geology	3	-
Sem 4	S183	Principles of Tourism and Geotourism	3	-
Sem 4	S184	Urban and Regional Planning Principles	3	-
Sem 4	S052	Engineering Surveying	3	-
Summer 2	S185	Landscape and Geoheritage Field Survey (4 weeks)	4	-
Sem 5	S186	Tourism Planning	4	-
Sem 5	S187	Geoheritage Assessment and Geopark Management	3	-
Sem 5	S188	Landscape Architecture and Site Design	4	-
Sem 5	S060	Architectural Design Basics	3	Building Architecture and Design
Sem 5	S189	Environmental Impact Assessment	3	-
Sem 5	S033	GIS for Planning	3	Geographic Information Systems (GIS)

Semester	No.	Subject (as named in the course)	Cr	Catalogue entry
Sem 6	S190	Tourism Infrastructure Engineering	4	-
Sem 6	S191	Tourism Economics and Market Analysis	3	-
Sem 6	S192	Tourist Facility Design	3	-
Sem 6	S193	Geohazards and Visitor Safety	3	-
Sem 6	S194	Natural and Cultural Heritage Conservation	3	-
Sem 6	S195	Destination Management and Marketing	2	-
Sem 6	S196	Structures for Leisure Facilities	2	-
Summer 3	S216	Production Internship (5 weeks)	5	-
Sem 7	S197	Tourist District Master Plan Studio (capstone)	5	-
Sem 7	S198	Sustainable Tourism and Carrying Capacity	3	-
Sem 7	S199	Tourism Law, Policy and Investment	2	-
Sem 7	S200	Heritage Interpretation and Visitor-Centre Design	2	-
Sem 7	S066	Project Finance and Feasibility Studies	2	Engineering Economics and Project Feasibility
Sem 7	S047	Professional English and Technical Communication	2	-
Sem 7	S048	Elective (from any course of the faculty)	2	-
Sem 8	S217	Graduation Design / Bachelor Thesis	8	-
Sem 8	S218	Frontiers Seminar of the Discipline	2	-

Petroleum Exploration and Engineering course

Semester	No.	Subject (as named in the course)	Cr	Catalogue entry
Sem 4	S032	Sedimentology and Stratigraphy	4	Sedimentology, Stratigraphy and Sedimentary Basins
Sem 4	S024	Structural Geology (Petroleum Applications)	3	Structural Geology
Sem 4	S115	Engineering Thermodynamics and Heat Transfer	3	-
Sem 4	S201	Sedimentary Petrology	2	-
Summer 2	S026	Geological Mapping Field School (4 weeks)	4	-
Sem 5	S202	Petroleum Geology	4	-
Sem 5	S103	Organic Geochemistry	2	-
Sem 5	S080	Seismic Exploration	3	-
Sem 5	S086	Well Logging and Formation Evaluation	4	-
Sem 5	S203	Reservoir Rock and Fluid Properties	4	-
Sem 5	S204	Sedimentary Basin Analysis	3	-
Sem 6	S205	Petroleum Drilling Engineering	4	-
Sem 6	S206	Reservoir Engineering	4	-
Sem 6	S207	Production Engineering	4	-
Sem 6	S090	Seismic Interpretation and Petroleum Exploration	3	Seismic Interpretation and Prospect Evaluation
Sem 6	S208	Petroleum Machinery and Equipment	3	-
Sem 6	S209	Well Testing	2	-
Summer 3	S216	Production Internship (5 weeks)	5	-
Sem 7	S210	Petroleum Exploration and Development Design (capstone)	3	-
Sem 7	S211	Enhanced Oil Recovery	3	-
Sem 7	S212	Petroleum Refining and Processing	3	-
Sem 7	S213	Natural Gas and Unconventional Resources	3	-
Sem 7	S214	Reservoir Simulation	2	-
Sem 7	S215	Petroleum Economics and HSE	2	-
Sem 7	S047	Professional English and Technical Communication	2	-
Sem 8	S217	Graduation Design / Bachelor Thesis	8	-
Sem 8	S218	Frontiers Seminar of the Discipline	2	-

Part B. Subject Catalogue

S001 General education (languages, humanities, physical education)

Used in: All ten courses - Sem 1 (3 cr), Sem 2 (2 cr), Sem 3 (2 cr), Sem 4 (2 cr)

Syllabus

Chapter 1. Foreign language - **1.1** Academic reading: *scientific texts, note taking* **1.2** Writing: *summaries, short reports* **1.3** Listening and speaking: *lectures, presentations*

Chapter 2. History and social sciences - **2.1** National history and institutions: *per national curriculum* **2.2** Ethics and society: *professional responsibility*

Chapter 3. Physical education - **3.1** Fitness and sport: *per semester programme* **3.2** Health and safety: *first aid basics*

Chapter 4. Writing and communication - **4.1** Structure of technical writing: *paragraphs, argument* **4.2** Oral presentation: *slides, delivery*

Assessment: Per university regulations

Recommended textbooks

No textbook: practice-based subject (faculty guidelines, logbook and report templates).

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH university general education (47 cr); CUGB general education (39 cr required + 16 elective). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page.

S002 Calculus I

Used in: All ten courses - Sem 1 (5 cr)

Syllabus

Chapter 1. Functions and models - **1.1** Functions and their graphs: *domain, range, composition, inverse* **1.2** Elementary functions: *polynomial, rational, trigonometric, exponential, logarithmic* **1.3** Mathematical modelling: *fitting functions to data*

Chapter 2. Limits and continuity - **2.1** Limit of a function: *numerical, graphical, epsilon-delta idea* **2.2** Limit laws: *squeeze theorem, one-sided and infinite limits* **2.3** Continuity: *intermediate value theorem*

Chapter 3. Derivatives - **3.1** Definition and interpretation: *tangent, rate of change* **3.2** Differentiation rules: *product, quotient, chain rule* **3.3** Implicit and logarithmic differentiation **3.4** Derivatives of inverse and transcendental functions **3.5** Related rates

Chapter 4. Applications of differentiation - **4.1** Linear approximation and differentials **4.2** Extreme values and optimisation: *closed interval method, first/second derivative tests* **4.3** Curve sketching **4.4** L'Hopital's rule **4.5** Newton's method

Chapter 5. Integrals - **5.1** Antiderivatives **5.2** Riemann sums and the definite integral **5.3** Fundamental theorem of calculus **5.4** Substitution rule

Chapter 6. Techniques of integration - **6.1** Integration by parts **6.2** Trigonometric integrals and substitution **6.3** Partial fractions **6.4** Improper integrals **6.5** Numerical integration: *trapezoidal, Simpson*

Chapter 7. Applications of integration - **7.1** Areas between curves **7.2** Volumes: *disc, shell* **7.3** Arc length **7.4** Work, hydrostatic force, centre of mass **7.5** Average value

Computer practice: Plotting functions and numerical derivatives/integrals in Python (weekly exercise sheet)

Assessment: Weekly problem sets 20%; two midterm tests 30%; final examination 50%

Recommended textbooks

- **Calculus, Volumes 1-3** - Strang, Herman et al. (OpenStax, Rice University). *CC BY-NC-SA; free PDF and web.* PDF/online: https://openstax.org/details/books/calculus-volume-1. Table of contents: Appendix A, item A01.
- Stewart, Calculus: Early Transcendentals (Cengage). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 18.01SC Single Variable Calculus (OCW, Fall 2010).** Syllabus: Units: 1 Differentiation; 2 Applications of differentiation; 3 The definite integral and its applications; 4 Techniques of integration; 5 Exploring the infinite (series). Link: https://www.ocw.mit.edu/courses/18-01sc-single-variable-calculus-fall-2010/pages/syllabus
- **Colorado School of Mines - MATH111 Calculus for Scientists and Engineers I (4).** Syllabus: Functions, limits and continuity; derivatives and applications; definite and indefinite integrals; plane geometry. Link: https://catalog.mines.edu/coursesaz/math/math.pdf
- **Tsinghua University / China University of Geosciences:** TH 10421055 Calculus A(1); CUGB Advanced Mathematics A1 (S1). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page.

S003 Linear Algebra

Used in: All ten courses - Sem 1 (3 cr)

Syllabus

Chapter 1. Linear systems - **1.1** Systems of linear equations: *row reduction, echelon forms* **1.2** Gaussian and Gauss-Jordan elimination **1.3** Existence and uniqueness of solutions

Chapter 2. Matrices - **2.1** Matrix operations and algebra **2.2** Inverse matrices **2.3** LU factorisation **2.4** Special matrices: *symmetric, triangular, orthogonal*

Chapter 3. Determinants - **3.1** Definition and properties **3.2** Cofactor expansion **3.3** Cramer's rule **3.4** Geometric meaning: *area, volume*

Chapter 4. Vector spaces - **4.1** Subspaces, span and linear independence **4.2** Basis and dimension **4.3** Column space, null space and rank: *rank-nullity theorem*

Chapter 5. Linear transformations - **5.1** Matrix of a transformation **5.2** Rotation, projection, reflection **5.3** Change of basis

Chapter 6. Orthogonality and least squares - **6.1** Inner product and norms **6.2** Orthogonal projections **6.3** Gram-Schmidt and QR **6.4** Least-squares fitting of data

Chapter 7. Eigenvalues and eigenvectors - **7.1** Characteristic equation **7.2** Diagonalisation **7.3** Symmetric matrices and spectral theorem **7.4** Introduction to singular value decomposition

Computer practice: NumPy linear algebra: solving systems, least-squares regression of field data, eigen-analysis of a stress tensor

Assessment: Problem sets 20%; midterm 30%; final 50%

Recommended textbooks

- **Linear Algebra** - Jim Hefferon. *Free (CC BY-SA), PDF via OpenIntro.* PDF/online: https://hefferon.net/linearalgebra/. Table of contents: Appendix A, item A02.
- Strang, Introduction to Linear Algebra (Wellesley-Cambridge Press). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 18.06SC Linear Algebra (OCW, Fall 2011).** Syllabus: Unit I Ax = b and the four subspaces; Unit II least squares, determinants and eigenvalues; Unit III positive definite matrices and applications. Link: https://www.ocw.mit.edu/courses/18-06sc-linear-algebra-fall-2011/pages/syllabus
- **Tsinghua University / China University of Geosciences:** TH 10421324 Linear Algebra; CUGB Linear Algebra (S3). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page.

S004 General Physics I (Mechanics, Heat, Waves)

Used in: All ten courses - Sem 1 (4 cr)

Syllabus

Chapter 1. Measurement and vectors - **1.1** Units, dimensions and significant figures **1.2** Vector algebra and components

Chapter 2. Kinematics - **2.1** Motion in one dimension: *displacement, velocity, acceleration* **2.2** Motion in two and three dimensions: *projectiles, circular motion* **2.3** Relative motion

Chapter 3. Newton's laws - **3.1** Force and mass **3.2** Applications: *friction, tension, inclined planes* **3.3** Circular dynamics and drag

Chapter 4. Work and energy - **4.1** Work and kinetic energy **4.2** Potential energy and conservation of energy **4.3** Power

Chapter 5. Momentum and collisions - **5.1** Impulse and linear momentum **5.2** Elastic and inelastic collisions **5.3** Centre of mass

Chapter 6. Rotation - **6.1** Rotational kinematics **6.2** Moment of inertia and torque **6.3** Angular momentum **6.4** Static equilibrium and elasticity: *stress, strain, moduli*

Chapter 7. Gravitation and fluids - **7.1** Newton's law of gravitation, orbits **7.2** Pressure, buoyancy, Bernoulli's equation

Chapter 8. Oscillations and waves - **8.1** Simple harmonic motion and pendulums **8.2** Damped and forced oscillations **8.3** Mechanical waves, superposition, standing waves **8.4** Sound

Chapter 9. Thermodynamics - **9.1** Temperature, heat and thermal expansion **9.2** Kinetic theory of gases **9.3** First and second laws, heat engines, entropy

Assessment: Homework 20%; lab reports (in Physics Laboratory); midterm 30%; final 50%

Recommended textbooks

- **University Physics, Volume 1** - Ling, Sanny, Moebs et al. (OpenStax). *CC BY; free PDF and web.* PDF/online: https://openstax.org/books/university-physics-volume-1/pages/preface. Table of contents: Appendix A, item A03.
- Halliday, Resnick and Walker, Fundamentals of Physics (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 8.01SC Classical Mechanics (OCW, Fall 2016).** Syllabus: Vectors; kinematics; Newton's laws; circular motion; drag and constraints; momentum and impulse; continuous mass transfer; work and

kinetic energy; potential energy and energy conservation; collisions; rotational motion; angular momentum; rolling. Link: https://www.ocw.mit.edu/courses/8-01sc-classical-mechanics-fall-2016

- **Colorado School of Mines - PHGN100 Physics I - Mechanics (4).** Syllabus: Mechanics with vectors and calculus: kinematics and dynamics of particles and systems, Newton's laws, energy and momentum, rotation, oscillations and waves. Link: https://catalog.mines.edu/coursesaz/physics
- **Tsinghua University / China University of Geosciences:** TH 10431174 University Physics CE; CUGB College Physics (1). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page.

S005 General Chemistry with Laboratory

Used in: All ten courses - Sem 1 (3 cr)

Syllabus

Chapter 1. Matter and measurement - **1.1** Classification of matter **1.2** Units, precision and accuracy

Chapter 2. Atoms, molecules and ions - **2.1** Atomic structure and isotopes **2.2** Periodic table **2.3** Chemical formulas and nomenclature

Chapter 3. Stoichiometry - **3.1** The mole **3.2** Chemical equations **3.3** Limiting reagent and yield **3.4** Solution concentration

Chapter 4. Electronic structure and bonding - **4.1** Quantum numbers and orbitals **4.2** Periodic trends **4.3** Ionic, covalent and metallic bonding **4.4** Molecular geometry: *VSEPR*

Chapter 5. Thermochemistry - **5.1** Energy, enthalpy and Hess's law **5.2** Calorimetry

Chapter 6. States of matter and solutions - **6.1** Gases: *ideal gas law* **6.2** Liquids and solids: *crystal lattices, unit cells* **6.3** Solubility and colligative properties

Chapter 7. Chemical equilibrium - **7.1** Equilibrium constant **7.2** Le Chatelier's principle **7.3** Solubility product: *mineral dissolution*

Chapter 8. Acids, bases and redox - **8.1** pH and buffers **8.2** Oxidation states and balancing redox **8.3** Electrochemical cells: *corrosion, Eh*

Chapter 9. Kinetics - **9.1** Rate laws **9.2** Activation energy and catalysis

Laboratory practice:

- Laboratory safety and glassware
- Density and measurement
- Stoichiometry of a reaction
- Acid-base titration
- Calorimetry
- Solubility and precipitation of minerals
- Redox titration
- Spectrophotometric determination of iron

Assessment: Lab reports 25%; quizzes 15%; midterm 20%; final 40%

Recommended textbooks

- **Chemistry 2e** - Flowers, Theopold, Langley, Robinson (OpenStax). *CC BY; free PDF and web (also on LibreTexts).* PDF/online: https://chem.libretexts.org/Bookshelves/General_Chemistry/Chemistry_2e_%28OpenStax%29. Table of contents: Appendix A, item A04.
- Brown, LeMay, Bursten et al., Chemistry: The Central Science (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 10440012 University Chemistry B; CUGB College Chemistry + Chemistry Experiments. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page.

S006 Programming for Engineers (Python)

Used in: All ten courses - Sem 1 (3 cr)

Syllabus

Chapter 1. Getting started - **1.1** Programs, interpreters and the Python environment: *Jupyter, IDE* **1.2** Values, variables and expressions

Chapter 2. Control flow - **2.1** Conditionals **2.2** Loops: *for, while* **2.3** Functions, parameters and return values **2.4** Recursion

Chapter 3. Data structures - **3.1** Strings **3.2** Lists and tuples **3.3** Dictionaries and sets **3.4** Files and exceptions

Chapter 4. Program design - **4.1** Modules and packages **4.2** Debugging and testing **4.3** Introduction to classes and objects

Chapter 5. Scientific Python - **5.1** NumPy arrays and vectorised computation **5.2** pandas data frames: *reading CSV, cleaning, grouping* **5.3** Matplotlib plotting: *line, scatter, histograms, maps*

Chapter 6. Applications in geoscience and engineering - **6.1** Processing GPS and field-notebook data **6.2** Unit conversion and simple calculators **6.3** Version control with Git: *basics*

Computer practice:

- Weekly programming labs (2 h)
- mini-project: read, clean and plot a borehole or weather-station dataset

Assessment: Labs 40%; mini-project 20%; final practical exam 40%

Recommended textbooks

- **Think Python: How to Think Like a Computer Scientist (2nd ed.)** - Allen B. Downey (Green Tea Press). *CC BY-NC; free online (3rd ed. free online as Jupyter notebooks).* PDF/online: https://greenteapress.com/thinkpython2/html/. Table of contents: Appendix A, item A05.
- Downey, Think Python (O'Reilly). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- McKinney, Python for Data Analysis (O'Reilly). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 6.100L Introduction to CS and Programming using Python (OCW, Fall 2022).** Syllabus: Python 3 programming; the role of computation in solving problems; writing small programs; application to computational modelling and understanding data. Link: https://ocw.mit.edu/courses/6-100l-introduction-to-cs-and-programming-using-python-fall-2022/pages/syllabus/
- **Massachusetts Institute of Technology (MIT) - 12.010 Computational Methods of Scientific Programming (OCW, Fall 2024).** Syllabus: Python, Julia and C/C++; code design; algorithm development and verification; Jupyter; machine learning; cloud and HPC workflows; data analysis and modelling; visualisation; numerical analysis; reproducibility and version control. Link: https://www.ocw.mit.edu/courses/12-010-computational-methods-of-scientific-programming-fall-2024/pages/syllabus
- **Tsinghua University / China University of Geosciences:** TH 30250023 Computer Language and Programming; CUGB Computer Language Programming. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page.

S007 Introduction to Geoscience and Technology (seminar)

Used in: All ten courses - Sem 1 (1 cr)

Syllabus

Chapter 1. The Faculty of Geoscience and Technology - **1.1** Institutes and divisions **1.2** The ten courses and graduate profiles

Chapter 2. Careers and industry - **2.1** Exploration, mining, petroleum, civil, space and tourism sectors **2.2** Professional registration and ethics

Chapter 3. Safety - **3.1** Field safety: *planning, communication, hazards* **3.2** Laboratory safety: *chemical, electrical, radiation*

Chapter 4. Study skills - **4.1** Using the library and databases **4.2** Academic integrity and citation

Field practice: Visits to two faculty laboratories and one industry site

Assessment: Attendance and reflective report (pass/fail)

Recommended textbooks

No textbook: practice-based subject (faculty guidelines, logbook and report templates).

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEOL102 Introduction to Geological Engineering (1).** Syllabus: Case studies by faculty and professionals giving an overview of geology and geological engineering careers; short paper on an academic/professional path. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** TH 20030181 Intro to Civil, Hydraulic and Ocean Eng.; CUGB Intro to the profession. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page.

S008 Calculus II (Multivariable and Vector Calculus)

Used in: All ten courses - Sem 2 (5 cr)

Syllabus

Chapter 1. Sequences and series - **1.1** Convergence tests **1.2** Power series and radius of convergence **1.3** Taylor and Maclaurin series and approximations

Chapter 2. Vectors and geometry of space - **2.1** Vectors in 3D, dot and cross products **2.2** Lines and planes **2.3** Cylindrical and spherical coordinates

Chapter 3. Vector functions - **3.1** Curves in space, arc length, curvature **3.2** Velocity and acceleration: *motion in space*

Chapter 4. Partial derivatives - **4.1** Functions of several variables, limits **4.2** Partial derivatives and tangent planes **4.3** Chain rule, directional derivative and gradient **4.4** Maxima and minima, Lagrange multipliers

Chapter 5. Multiple integrals - **5.1** Double integrals in rectangular and polar coordinates **5.2** Triple integrals in cylindrical and spherical coordinates **5.3** Applications: *mass, centroid, moment of inertia*

Chapter 6. Vector calculus - **6.1** Vector fields **6.2** Line integrals and conservative fields **6.3** Green's theorem **6.4** Divergence and curl **6.5** Surface integrals **6.6** Stokes' and divergence (Gauss) theorems

Computer practice: Visualising surfaces, gradients and vector fields (Python/Matplotlib) - topographic surface example

Assessment: Problem sets 20%; midterm 30%; final 50%

Recommended textbooks

- **Calculus, Volumes 1-3** - Strang, Herman et al. (OpenStax, Rice University). *CC BY-NC-SA; free PDF and web.* PDF/online: https://openstax.org/details/books/calculus-volume-1. Table of contents: Appendix A, item A01.
- Stewart, Calculus: Early Transcendentals (Cengage). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 18.02SC Multivariable Calculus (OCW, Fall 2010).** Syllabus: Units: 1 Vectors and matrices; 2 Partial derivatives; 3 Double integrals and line integrals in the plane; 4 Triple integrals and surface integrals in 3-space (physics applications). Link: https://www.ocw.mit.edu/courses/18-02sc-multivariable-calculus-fall-2010/pages/syllabus
- **Colorado School of Mines - MATH112 Calculus for Scientists and Engineers II (4).** Syllabus: Vectors; techniques and applications of integration; infinite series; multivariate functions and surfaces. Link: https://catalog.mines.edu/coursesaz/math/math.pdf
- **Colorado School of Mines - MATH213 Calculus for Scientists and Engineers III (4).** Syllabus: Partial derivatives; multiple integrals; vector calculus. Link: https://catalog.mines.edu/coursesaz/math/math.pdf
- **Tsinghua University / China University of Geosciences:** TH 10421065 Calculus A(2); CUGB Advanced Mathematics A2 (S2). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page.

S009 General Physics II (Electromagnetism, Optics, Modern Physics)

Used in: All ten courses - Sem 2 (4 cr)

Syllabus

Chapter 1. Electric charge and field - **1.1** Coulomb's law **1.2** Electric field of charge distributions **1.3** Gauss's law

Chapter 2. Electric potential and capacitance - **2.1** Potential and potential energy **2.2** Capacitors and dielectrics

Chapter 3. Current and circuits - **3.1** Current, resistance, resistivity of materials **3.2** DC circuits and Kirchhoff's rules **3.3** RC circuits

Chapter 4. Magnetism - **4.1** Magnetic force and field **4.2** Sources of magnetic field: *Biot-Savart, Ampere* **4.3** Magnetic properties of materials: *para-, dia-, ferromagnetism*

Chapter 5. Electromagnetic induction - **5.1** Faraday's and Lenz's laws **5.2** Inductance and RL circuits **5.3** AC circuits basics

Chapter 6. Maxwell's equations and waves - **6.1** Displacement current **6.2** Electromagnetic waves and the spectrum **6.3** Energy and momentum of EM waves

Chapter 7. Optics - **7.1** Reflection, refraction, polarisation **7.2** Interference and diffraction

Chapter 8. Modern physics - **8.1** Photons and the photoelectric effect **8.2** Atomic structure and spectra **8.3** Nuclear physics and radioactivity: *decay laws, half-life*

Assessment: Homework 20%; midterm 30%; final 50%

Recommended textbooks

- Halliday, Resnick and Walker, Fundamentals of Physics (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 8.02 Physics II: Electricity and Magnetism (OCW, Spring 2019).** Syllabus: Three parts: electrostatics; magnetic fields and forces; Maxwell's equations and electromagnetic radiation. Link: https://ocw.mit.edu/courses/8-02-physics-ii-electricity-and-magnetism-spring-2019/
- **Colorado School of Mines - PHGN200 Physics II - Electromagnetism and Optics (4).** Syllabus: Laws of electricity and magnetism; electromagnetic devices and materials; simple circuits; electromagnetic radiation; introductory optics. Link: https://catalog.mines.edu/coursesaz/physics
- **Tsinghua University / China University of Geosciences:** CUGB College Physics (2) (S3). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S010 Physics Laboratory

Used in: All ten courses - Sem 2 (1 cr)

Syllabus

Chapter 1. Measurement and errors - **1.1** Uncertainty, propagation of errors **1.2** Graphing and linear fits

Laboratory practice:

- Free fall and g
- Simple pendulum
- Hooke's law and Young's modulus
- Specific heat
- Ohm's law and resistivity
- RC time constant
- Magnetic field of a coil
- Electromagnetic induction
- Refraction and lenses
- Diffraction grating
- Radioactive decay counting

Assessment: 10 lab reports 80%; practical test 20%

Recommended textbooks

- **Lessons in Electric Circuits (6 volumes)** - Tony R. Kuphaldt. *Design Science / Creative Commons licence; free PDF per volume.* PDF/online: https://ibiblio.org/kuphaldt/electricCircuits/index.htm. Table of contents: Appendix A, item A06.

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 10430782 Physics Experiments A(1); CUGB Physics Experiments (S3). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page.

S011 Physical Geology (with laboratory)

Used in: All ten courses - Sem 2 (4 cr)

Syllabus

Chapter 1. Earth as a system - **1.1** Origin and internal structure of the Earth **1.2** Geological time and the rock cycle

Chapter 2. Plate tectonics - **2.1** Evidence: *palaeomagnetism, seafloor spreading* **2.2** Plate boundaries and associated processes

Chapter 3. Minerals - **3.1** Mineral properties and identification **3.2** Silicate and non-silicate groups

Chapter 4. Igneous processes - **4.1** Magma generation **4.2** Intrusive and volcanic rocks **4.3** Volcanoes and volcanic hazards

Chapter 5. Sedimentary processes - **5.1** Weathering and soils **5.2** Sediment transport and deposition **5.3** Sedimentary rocks and environments

Chapter 6. Metamorphism - **6.1** Agents of metamorphism **6.2** Metamorphic rocks and facies

Chapter 7. Structures and earthquakes - **7.1** Folds, faults and joints **7.2** Earthquakes, seismic waves and hazard

Chapter 8. Surface processes - **8.1** Mass wasting **8.2** Streams and floods **8.3** Groundwater **8.4** Glaciers, deserts and coasts

Chapter 9. Geological time and resources - **9.1** Relative and absolute dating **9.2** Mineral and energy resources **9.3** Geology and the environment

Laboratory practice:

- Mineral identification (2 sessions)
- Igneous, sedimentary and metamorphic rock identification (3 sessions)
- Topographic maps
- Geological maps and simple cross-sections
- Structure contours
- Relative dating exercises

Field practice: One-day local geology excursion

Assessment: Labs 30%; midterm 25%; final 45%

Recommended textbooks

- **Physical Geology (2nd ed.)** - Steven Earle (BCcampus Open Education). *CC BY; free PDF and web.* PDF/online: https://yc-prod-openlibrary-repo.ecampusontario.ca/jspui/bitstream/123456789/713/3/Physical-Geology-2nd-Edition-1591817334.pdf. Table of contents: Appendix A, item A07.
- Marshak, Earth: Portrait of a Planet (W.W. Norton). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Grotzinger and Jordan, Understanding Earth (W.H. Freeman). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.001 Introduction to Geology (OCW, Fall 2013).** Syllabus: Mineral and rock identification; origin of continents; geologic mapping; plate tectonics; erosion by rivers and glaciers; history of life. Three 2-hour lecture/lab sessions per week; mandatory weekend field trip; field report and labs 60%, quizzes 35%. Link: https://ocw.mit.edu/courses/12-001-introduction-to-geology-fall-2013/pages/syllabus/
- **Colorado School of Mines - GEGN101 Earth and Environmental Systems (4 sem. hrs).** Syllabus: Nature, composition and evolution of lithosphere, hydrosphere, atmosphere and biosphere; human interactions with natural systems; energy and mass cycling, natural hazards, land use, waste disposal, resources and water. 3 h lecture + 3 h lab. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Geosciences (S2). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S012 Engineering Mechanics (Statics and Dynamics)

Used in: All ten courses - Sem 2 (4 cr)

Syllabus

Chapter 1. Introduction and vectors - **1.1** Units and idealisations **1.2** Force vectors and components

Chapter 2. Equilibrium of a particle - **2.1** Free-body diagrams **2.2** Coplanar and 3D force systems

Chapter 3. Moments and equivalent systems - **3.1** Moment of a force and couple **3.2** Equivalent force systems and resultants **3.3** Distributed loads

Chapter 4. Rigid body equilibrium - **4.1** Supports and reactions **4.2** 2D and 3D equilibrium **4.3** Statical determinacy

Chapter 5. Structural analysis - **5.1** Plane trusses: *joints, sections* **5.2** Frames and machines

Chapter 6. Internal forces - **6.1** Shear and bending moment diagrams in beams

Chapter 7. Friction - **7.1** Dry friction, wedges, belts

Chapter 8. Centroids and moments of inertia - **8.1** Centroids of areas and volumes **8.2** Parallel-axis theorem

Chapter 9. Kinematics - **9.1** Particle kinematics: *rectilinear, curvilinear* **9.2** Rigid-body planar kinematics

Chapter 10. Kinetics - **10.1** Newton's second law for particles and rigid bodies **10.2** Work-energy method **10.3** Impulse-momentum method **10.4** Vibrations: *introduction*

Assessment: Problem sets 20%; two tests 30%; final 50%

Recommended textbooks

- **Engineering Statics: Open and Interactive** - Daniel Baker, William Haynes. *CC BY-NC-SA; free web and PDF.* PDF/online: https://engineeringstatics.org/pdf/statics.pdf. Table of contents: Appendix A, item A08.
- Hibbeler, Engineering Mechanics: Statics and Dynamics (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - CEEN241 Statics (3).** Syllabus: Force and moment systems, equilibrium, centroids and second moments, hydrostatic pressure, friction, vector methods applied to structures. Link: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf
- **Tsinghua University / China University of Geosciences:** TH 20310334 Theoretical Mechanics. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S013 Engineering Graphics and CAD

Used in: All ten courses - Sem 2 (2 cr)

Syllabus

Chapter 1. Drawing standards - **1.1** Lines, lettering, scales **1.2** Drawing sheets and title blocks

Chapter 2. Projection - **2.1** Orthographic projection **2.2** Isometric and oblique views **2.3** Auxiliary views

Chapter 3. Sections and dimensions - **3.1** Section views **3.2** Dimensioning and tolerances

Chapter 4. CAD - **4.1** 2D CAD commands and layers **4.2** 3D solid modelling basics **4.3** Plotting and printing

Chapter 5. Geological and engineering drawings - **5.1** Topographic profiles and geological cross-sections **5.2** Site plans and borehole logs

Computer practice:

- Weekly CAD exercises
- final drawing set (machine part, site plan, geological section)

Assessment: Drawing assignments 60%; CAD practical test 40%

Recommended textbooks

- Giesecke et al., Technical Drawing with Engineering Graphics (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 20120152 Fundamentals of Engineering Graphics + 20900012 Engineering Computer Drafting. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S014 Geological Recognition Field Practice (2 weeks)

Used in: All ten courses - Summer 1 (2 cr)

Syllabus

Chapter 1. Preparation - **1.1** Field safety, equipment and clothing **1.2** Topographic maps and GNSS

Chapter 2. Field observation - **2.1** Describing outcrops and rocks **2.2** Recognising bedding, folds, faults and joints **2.3** Landforms and surface processes

Chapter 3. Field recording - **3.1** Field notebook layout **3.2** Sketching **3.3** Compass-clinometer measurement of planes and lines **3.4** Photograph logging

Field practice:

- 10 field days at representative localities (igneous, sedimentary, metamorphic, structural and geomorphological stops)
- daily notebook checks

Assessment: Field notebook 50%; final field report 40%; conduct and safety 10%

Recommended textbooks

- **Physical Geology (2nd ed.)** - Steven Earle (BCcampus Open Education). *CC BY; free PDF and web.* PDF/online: https://yc-prod-openlibrary-repo.ecampusontario.ca/jspui/bitstream/123456789/713/3/Physical-Geology-2nd-Edition-1591817334.pdf. Table of contents: Appendix A, item A07.
- Coe (ed.), Geological Field Techniques (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN217 Geologic Field Methods (2).** Syllabus: Field description of rocks; topographic maps and profiles; structural measurements; mapping contacts; 1:1 cross-sections; interpreting geological histories; weekend field exercises. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Beidaihe Geological Field Trip (2 cr); TH 40030711 Cognition Practice. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page.

S015 Differential Equations and Mathematical Physics

Used in: All ten courses - Sem 3 (3 cr)

Syllabus

Chapter 1. First-order equations - **1.1** Separable and linear equations **1.2** Exact equations and integrating factors **1.3** Applications: *decay, mixing, cooling, population*

Chapter 2. Second-order linear equations - **2.1** Homogeneous equations with constant coefficients **2.2** Undetermined coefficients and variation of parameters **2.3** Mechanical vibrations and resonance

Chapter 3. Systems of equations - **3.1** Matrix methods and eigenvalues **3.2** Phase plane and stability

Chapter 4. Laplace transform - **4.1** Definition and properties **4.2** Solving IVPs **4.3** Step and impulse functions **4.4** Convolution and transfer functions

Chapter 5. Fourier series - **5.1** Periodic functions **5.2** Sine and cosine series **5.3** Convergence

Chapter 6. Partial differential equations - **6.1** Separation of variables **6.2** Heat (diffusion) equation **6.3** Wave equation **6.4** Laplace equation: *potential fields*

Chapter 7. Numerical methods - **7.1** Euler and Runge-Kutta methods **7.2** Stability and error

Computer practice:

- Solving ODEs with SciPy
- 1D groundwater diffusion and heat flow examples

Assessment: Problem sets 20%; midterm 30%; final 50%

Recommended textbooks

- **Notes on Diffy Qs: Differential Equations for Engineers** - Jiri Lebl. *CC BY-SA / CC BY-NC-SA; free PDF.* PDF/online: https://jirka.org/diffyqs/diffyqs.pdf. Table of contents: Appendix A, item A09.
- Kreyszig, Advanced Engineering Mathematics (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 18.03SC Differential Equations (OCW, Fall 2011).** Syllabus: Unit I first-order ODEs (geometric/numerical methods, integrating factors, autonomous equations); Unit II second-order constant-coefficient equations (damped oscillators, resonance, frequency response); Unit III Fourier series and Laplace transform (step/delta functions, convolution, transfer functions); Unit IV first-order systems (matrix methods, phase portraits, linearisation, limit cycles). Link: https://ocw.mit.edu/courses/18-03sc-differential-equations-fall-2011/pages/syllabus/syllabus/

- **Colorado School of Mines - MATH225 Differential Equations (3).** Syllabus: First- and higher-order equations and systems; Laplace transforms; phase-plane and stability analysis; applications in physics, mechanics, electrical engineering and environmental science. Link: https://catalog.mines.edu/coursesaz/math/math.pdf
- **Tsinghua University / China University of Geosciences:** TH 10420262 Intro to Mathematical-Physical Equations. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S016 Probability and Statistics

Used in: All ten courses - Sem 3 (3 cr)

Syllabus

Chapter 1. Descriptive statistics - **1.1** Data types, histograms, box plots **1.2** Measures of centre and spread

Chapter 2. Probability - **2.1** Sample spaces and events **2.2** Conditional probability and Bayes' theorem

Chapter 3. Random variables - **3.1** Discrete distributions: *binomial, Poisson* **3.2** Continuous distributions: *normal, lognormal, exponential* **3.3** Expectation and variance

Chapter 4. Sampling - **4.1** Central limit theorem **4.2** Sampling distributions

Chapter 5. Estimation - **5.1** Point and interval estimation **5.2** Confidence intervals for means and proportions

Chapter 6. Hypothesis testing - **6.1** t, chi-square and F tests **6.2** Type I and II errors **6.3** Nonparametric tests

Chapter 7. Regression and correlation - **7.1** Simple and multiple linear regression **7.2** Residual analysis

Chapter 8. Spatial sampling introduction - **8.1** Random, systematic and stratified sampling of the ground **8.2** Lognormal grade distributions

Computer practice:

- Statistical analysis in Python (pandas, SciPy)
- assay data exercise

Assessment: Assignments 25%; midterm 25%; final 50%

Recommended textbooks

- **OpenIntro Statistics (4th ed.)** - Diez, Cetinkaya-Rundel, Barr. *CC BY-SA; free PDF (pay-what-you-want, $0 minimum).* PDF/online: https://leanpub.com/os. Table of contents: Appendix A, item A10.
- Montgomery and Runger, Applied Statistics and Probability for Engineers (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - MATH201 Introduction to Statistics (3).** Syllabus: Experimental design and data collection; summary and display of data; propagation of error; interval estimation; hypothesis testing; linear regression. Link: https://catalog.mines.edu/coursesaz/math/math.pdf
- **Tsinghua University / China University of Geosciences:** TH 10420803 Probability and Statistics; CUGB Probability and Mathematical Statistics A (S3). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page.

S017 Mechanics of Materials

Used in: All ten courses - Sem 3 (3 cr)

Syllabus

Chapter 1. Stress and strain - **1.1** Normal and shear stress **1.2** Strain and Hooke's law **1.3** Material properties and testing

Chapter 2. Axial loading - **2.1** Deformation of axially loaded members **2.2** Statically indeterminate members **2.3** Thermal stress

Chapter 3. Torsion - **3.1** Shafts, angle of twist **3.2** Power transmission

Chapter 4. Bending - **4.1** Flexure formula **4.2** Composite beams **4.3** Shear stress in beams

Chapter 5. Stress transformation - **5.1** Plane stress, principal stresses **5.2** Mohr's circle **5.3** Failure theories

Chapter 6. Deflection of beams - **6.1** Elastic curve **6.2** Integration and superposition methods

Chapter 7. Columns - **7.1** Euler buckling **7.2** Effective length

Laboratory practice:

- Tension test of steel
- Compression test of concrete/rock
- Torsion test
- Beam bending and deflection
- Strain-gauge measurement

Assessment: Problem sets 15%; labs 15%; midterm 25%; final 45%

Recommended textbooks

- Hibbeler, Mechanics of Materials (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - CEEN311 Mechanics of Materials (3).** Syllabus: Stress, strain and material properties; axial, torsional, bending and combined loads; beam deflection; indeterminate systems; Euler buckling; stress transformation; pressure vessels; stress concentrations. Link: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf
- **Tsinghua University / China University of Geosciences:** TH 20310343 Mechanics of Materials. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S018 Fluid Mechanics

Used in: All ten courses - Sem 3 (3 cr)

Syllabus

Chapter 1. Fluid properties - **1.1** Density, viscosity, surface tension **1.2** Newtonian and non-Newtonian fluids: *drilling muds*

Chapter 2. Fluid statics - **2.1** Pressure and manometers **2.2** Forces on submerged surfaces **2.3** Buoyancy

Chapter 3. Fluid kinematics - **3.1** Streamlines, continuity equation

Chapter 4. Energy and momentum - **4.1** Bernoulli equation and applications **4.2** Momentum equation and forces on bends/jets

Chapter 5. Dimensional analysis - **5.1** Buckingham Pi **5.2** Similitude and models

Chapter 6. Pipe flow - **6.1** Laminar and turbulent flow **6.2** Friction factor and Moody diagram **6.3** Minor losses **6.4** Pumps and pipe networks

Chapter 7. Open-channel flow - **7.1** Uniform flow: *Manning* **7.2** Specific energy and hydraulic jump

Chapter 8. Flow in porous media - **8.1** Darcy's law **8.2** Permeability and hydraulic conductivity

Laboratory practice:

- Hydrostatic force
- Bernoulli apparatus
- Pipe friction losses
- Flow over weirs
- Darcy permeameter

Assessment: Labs 20%; problem sets 15%; midterm 25%; final 40%

Recommended textbooks

- **University Physics, Volume 1** - Ling, Sanny, Moebs et al. (OpenStax). *CC BY; free PDF and web.* PDF/online: https://openstax.org/books/university-physics-volume-1/pages/preface. Table of contents: Appendix A, item A03.
- White, Fluid Mechanics (McGraw-Hill). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN351 Geological Fluid Mechanics (3).** Syllabus: Fluid properties; Bernoulli, momentum and mass equations; laminar and turbulent flow in pipes, channels and earth materials; Darcy's law; geostrophic flow; sediment transport. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** TH 30040493 Fluid Mechanics. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S019 Electrical Engineering Fundamentals

Used in: All ten courses - Sem 3 (3 cr)

Syllabus

Chapter 1. Circuit fundamentals - **1.1** Charge, current, voltage, power **1.2** Ohm's and Kirchhoff's laws **1.3** Series/parallel circuits

Chapter 2. Circuit analysis - **2.1** Node and mesh analysis **2.2** Thevenin and Norton equivalents

Chapter 3. Transients - **3.1** Capacitors and inductors **3.2** RC and RL transients

Chapter 4. AC circuits - **4.1** Sinusoids and phasors **4.2** Impedance **4.3** AC power and power factor **4.4** Three-phase systems

Chapter 5. Magnetic circuits and transformers - **5.1** Magnetic circuits **5.2** Ideal and real transformers

Chapter 6. Electrical machines - **6.1** DC motors **6.2** Induction motors **6.3** Generators and site power supply

Chapter 7. Electrical safety - **7.1** Earthing, protection devices **7.2** Safety on drill rigs and field camps

Laboratory practice:

- Measurement with multimeter and oscilloscope
- Kirchhoff's laws
- RC transient
- AC power measurement
- Transformer test
- Induction motor start-up

Assessment: Labs 25%; midterm 25%; final 50%

Recommended textbooks

- **Lessons in Electric Circuits (6 volumes)** - Tony R. Kuphaldt. *Design Science / Creative Commons licence; free PDF per volume.* PDF/online: https://ibiblio.org/kuphaldt/electricCircuits/index.htm. Table of contents: Appendix A, item A06.
- Hambley, Electrical Engineering: Principles and Applications (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed (covered through physics at Mines/MIT).

S020 Earth Materials (Mineralogy and Petrology I)

Used in: All ten courses - Sem 3 (4 cr)

Syllabus

Chapter 1. Crystallography - **1.1** Symmetry, crystal systems and forms **1.2** Miller indices **1.3** Unit cells

Chapter 2. Crystal chemistry - **2.1** Bonding, ionic radii, coordination **2.2** Solid solution and polymorphism

Chapter 3. Mineral properties - **3.1** Physical properties for identification **3.2** Introduction to the polarising microscope

Chapter 4. Silicate minerals - **4.1** Structural classes: *nesosilicates to tectosilicates* **4.2** Rock-forming silicates

Chapter 5. Non-silicate minerals - **5.1** Oxides, sulfides, carbonates, sulfates, halides, native elements **5.2** Ore minerals

Chapter 6. Igneous rocks - **6.1** Classification: *QAPF, TAS* **6.2** Textures

Chapter 7. Sedimentary rocks - **7.1** Clastic, chemical and biogenic rocks

Chapter 8. Metamorphic rocks - **8.1** Classification and textures **8.2** Index minerals

Laboratory practice:

- Crystal models and symmetry
- Hand-specimen mineral identification (4 sessions)
- Introduction to thin sections
- Rock identification (3 sessions)

Assessment: Lab tests 40%; midterm 20%; final 40%

Recommended textbooks

- **Mineralogy** - Dexter Perkins et al. (Univ. of North Dakota; LibreTexts). *CC BY-NC-SA; free web and PDF export.* PDF/online: https://geo.libretexts.org/Bookshelves/Geology/Mineralogy_(Perkins_et_al.). Table of contents: Appendix A, item A11.
- Klein and Philpotts, Earth Materials: Introduction to Mineralogy and Petrology (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.108 Structure of Earth Materials (OCW, Fall 2004).** Syllabus: Crystal structure, crystal chemistry and bonding of rock-forming minerals; structure and symmetry versus physical properties; silicate, oxide and metallic minerals in planetary interiors and surfaces. Two 3-hour labs per week. Link: https://ocw.mit.edu/courses/12-108-structure-of-earth-materials-fall-2004/pages/syllabus
- **Colorado School of Mines - GEGN212 The Rock Cycle (4).** Syllabus: Foundations of mineralogy and petrology: crystal chemistry, mineral classification, rock-forming processes and classification; lab identification of minerals and rocks in hand sample. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Colorado School of Mines - GEOL310 Earth Materials (3).** Syllabus: Structure, formation, distribution and engineering behaviour of minerals and rocks; stress/strain and rock mechanics; lab and field evaluation of natural materials for mining engineers. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Crystallography and Mineralogy (S3) + Crystal Optics. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S021 Electronic Engineering and Instrumentation

Used in: All ten courses - Sem 4 (3 cr)

Syllabus

Chapter 1. Semiconductor devices - **1.1** Diodes and rectifiers **1.2** Bipolar and field-effect transistors

Chapter 2. Amplifiers - **2.1** Operational amplifiers **2.2** Inverting/non-inverting amplifiers, filters, instrumentation amplifiers

Chapter 3. Digital electronics - **3.1** Logic gates and Boolean algebra **3.2** Flip-flops, counters

Chapter 4. Data conversion - **4.1** Sampling theorem **4.2** ADC and DAC

Chapter 5. Sensors - **5.1** Temperature, pressure, strain, magnetic and accelerometer sensors **5.2** Signal conditioning

Chapter 6. Microcontrollers and data acquisition - **6.1** Microcontroller basics: *Arduino-type* **6.2** Serial communication, data logging **6.3** Power supplies and batteries for field instruments

Laboratory practice:

- Diode rectifier
- Transistor switch
- Op-amp amplifier and filter
- ADC sampling

- Build a temperature/strain data logger

Assessment: Labs 40%; project 20%; final 40%

Recommended textbooks

- **Lessons in Electric Circuits (6 volumes)** - Tony R. Kuphaldt. *Design Science / Creative Commons licence; free PDF per volume.* PDF/online: https://ibiblio.org/kuphaldt/electricCircuits/index.htm. Table of contents: Appendix A, item A06.
- Hambley, Electrical Engineering: Principles and Applications (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Horowitz and Hill, The Art of Electronics (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed; related to CUGB Dept. of Measurement and Control Technology.

S022 Numerical Methods and Geodata Analysis

Used in: All ten courses - Sem 4 (3 cr)

Syllabus

Chapter 1. Errors and floating point - **1.1** Round-off and truncation error **1.2** Stability

Chapter 2. Roots of equations - **2.1** Bisection, Newton-Raphson, secant

Chapter 3. Linear systems - **3.1** Direct and iterative solvers **3.2** Sparse matrices

Chapter 4. Interpolation and curve fitting - **4.1** Polynomial and spline interpolation **4.2** Least-squares regression

Chapter 5. Numerical integration and differentiation - **5.1** Newton-Cotes and Gauss quadrature **5.2** Finite differences

Chapter 6. Ordinary differential equations - **6.1** Runge-Kutta methods **6.2** Boundary-value problems

Chapter 7. Partial differential equations - **7.1** Finite-difference schemes for diffusion and Laplace equations

Chapter 8. Optimisation - **8.1** Gradient methods **8.2** Linear programming introduction

Chapter 9. Data analysis - **9.1** Time series and Fourier transform: *FFT* **9.2** Introduction to machine learning: *regression, clustering*

Computer practice: Weekly Python notebooks: borehole temperature profile, gravity profile smoothing, 2D groundwater head solution

Assessment: Notebooks 40%; project 20%; final 40%

Recommended textbooks

- Chapra and Canale, Numerical Methods for Engineers (McGraw-Hill). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Trauth, Python/MATLAB Recipes for Earth Sciences (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.010 Computational Methods of Scientific Programming (OCW, Fall 2024).** Syllabus: Python, Julia and C/C++; code design; algorithm development and verification; Jupyter; machine learning; cloud and HPC workflows; data analysis and modelling; visualisation; numerical analysis; reproducibility and version control. Link: https://www.ocw.mit.edu/courses/12-010-computational-methods-of-scientific-programming-fall-2024/pages/syllabus
- **Colorado School of Mines - GEGN432 Geological Data Management (3).** Syllabus: Probability, distributions, estimation, hypothesis testing, data sequences, sampling, regression, uni- and multivariate statistics; programming with geological case-history data sets. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** TH 40250443 Numerical Analysis and Algorithms, 30030622 Math Modelling and Data Analysis. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S023 Petrology and Optical Petrography

Used in: Mineral Exploration - Sem 4, 4 cr; Geochemical Exploration - Sem 4, 4 cr; Mineral Engineering - Sem 4, 4 cr (as 'Optical Mineralogy and Petrography')

Versions: Mineral Engineering version 'Optical Mineralogy and Petrography' (4 cr) doubles the optical-mineralogy chapter and laboratory time.

Syllabus

Chapter 1. Optical mineralogy - **1.1** Polarised light and the petrographic microscope **1.2** Isotropic and anisotropic minerals **1.3** Interference colours, extinction, pleochroism, optic sign

Chapter 2. Igneous petrology - **2.1** Magma generation and differentiation **2.2** Phase diagrams: *binary, ternary* **2.3** Classification and textures **2.4** Igneous rocks and tectonic settings

Chapter 3. Sedimentary petrology - **3.1** Sandstones, mudrocks, carbonates **3.2** Diagenesis and porosity

Chapter 4. Metamorphic petrology - **4.1** Metamorphic reactions and facies **4.2** Pressure-temperature paths **4.3** Textures and fabrics

Chapter 5. Petrogenesis - **5.1** Geochemical diagrams for rock series **5.2** Linking rocks to tectonic environments

Laboratory practice:

- Optical properties of common minerals (4 sessions)
- Thin sections of igneous rocks (4 sessions)
- Sedimentary rocks (2 sessions)
- Metamorphic rocks (3 sessions)
- Petrographic report on an unknown suite

Assessment: Lab tests 40%; petrographic report 10%; final 50%

Recommended textbooks

- **Mineralogy** - Dexter Perkins et al. (Univ. of North Dakota; LibreTexts). *CC BY-NC-SA; free web and PDF export.* PDF/online: https://geo.libretexts.org/Bookshelves/Geology/Mineralogy_(Perkins_et_al.). Table of contents: Appendix A, item A11.
- Philpotts and Ague, Principles of Igneous and Metamorphic Petrology (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Nesse, Introduction to Optical Mineralogy (Oxford). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.108 Structure of Earth Materials (OCW, Fall 2004).** Syllabus: Crystal structure, crystal chemistry and bonding of rock-forming minerals; structure and symmetry versus physical properties; silicate, oxide and metallic minerals in planetary interiors and surfaces. Two 3-hour labs per week. Link: https://ocw.mit.edu/courses/12-108-structure-of-earth-materials-fall-2004/pages/syllabus
- **Colorado School of Mines - GEGN307 Petrology (4).** Syllabus: Igneous, sedimentary and metamorphic processes; megascopic and microscopic classification and description of rocks; fabric and physical properties. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Petrology (S4, 5.5 cr); CUGB Crystal Optics (S3) + Petrology (S4). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S024 Structural Geology

Used in: Mineral Exploration - Sem 4, 4 cr; Geochemical Exploration - Sem 4, 4 cr; Mineral Engineering - Sem 4, 2 cr; Geological Information - Sem 4, 3 cr; Petroleum Exploration and Engineering - Sem 4, 3 cr (as 'Structural Geology (Petroleum Applications)')

Versions: Petroleum course version (3 cr) emphasises Chapter 7 (trap geometry, fault seal) and seismic structural interpretation; Mineral Engineering (2 cr) and Geological Information (3 cr) versions omit Chapter 6.

Syllabus

Chapter 1. Stress and strain - **1.1** Stress tensor and Mohr circle **1.2** Strain, finite and incremental **1.3** Rheology of rocks

Chapter 2. Brittle deformation - **2.1** Fractures and joints **2.2** Faults: *types, kinematics, fault rocks*

Chapter 3. Ductile deformation - **3.1** Folds: *geometry, classification, mechanisms* **3.2** Foliation, lineation and shear zones

Chapter 4. Structural analysis - **4.1** Stereographic projection **4.2** Structural domains **4.3** Superposed deformation

Chapter 5. Maps and cross-sections - **5.1** Geological map interpretation **5.2** Balanced cross-sections

Chapter 6. Tectonic settings - **6.1** Extensional, contractional and strike-slip regimes

Chapter 7. Applications - **7.1** Structural control of ore bodies and hydrocarbon traps **7.2** Slope and tunnel stability

Laboratory practice:

- Apparent dip and three-point problems
- Stereonet exercises (3 sessions)
- Fold analysis
- Fault analysis
- Map interpretation and cross-sections (3 sessions)

Field practice: One-day structural mapping exercise

Assessment: Labs 30%; midterm 20%; final 50%

Recommended textbooks

- **Geological Structures: a Practical Introduction** - John W.F. Waldron, Morgan Snyder (Univ. of Alberta). *CC BY-NC; free web and PDF.* PDF/online: https://pressbooks.openeducationalberta.ca/introductorystructuralgeology/. Table of contents: Appendix A, item A12.
- Fossen, Structural Geology (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Lisle, Geological Structures and Maps (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.113 Structural Geology (OCW, Fall 2005).** Syllabus: Mechanics of rock deformation; faults and folds; structures of igneous and metamorphic rocks; superposed deformation; regional structural geology and tectonics. 10 labs (stereonets, Mohr circles, map interpretation, cross-sections). Link: https://ocw.mit.edu/courses/12-113-structural-geology-fall-2005/pages/syllabus/
- **Colorado School of Mines - GEOL309 Structural Geology and Tectonics (4).** Syllabus: Deformational structures, stress and strain, rock mechanics and microstructures; mountain belts, rifts, strike-slip and salt systems; stereographic projection,

structure contours, maps, cross-sections, seismic data. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

- **Colorado School of Mines - GEOL308 Introductory Applied Structural Geology (3).** Syllabus: Structural features with emphasis on hydrocarbon traps; plate-tectonic context; structural geometry, mapping, cross-sections and seismic interpretation. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Structural Geology (S4). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S025 Paleontology, Stratigraphy and Historical Geology

Used in: Mineral Exploration - Sem 4, 4 cr

Syllabus

Chapter 1. Palaeontology basics - **1.1** Fossilisation and preservation **1.2** Major fossil groups: *invertebrates, microfossils, plants*

Chapter 2. Stratigraphic principles - **2.1** Lithostratigraphy and stratigraphic codes **2.2** Biostratigraphy **2.3** Chronostratigraphy and the geological time scale

Chapter 3. Correlation - **3.1** Facies and Walther's law **3.2** Sequence stratigraphy basics

Chapter 4. Historical geology - **4.1** Precambrian Earth **4.2** Phanerozoic history **4.3** Mass extinctions

Chapter 5. Regional geology - **5.1** Geology of the home country **5.2** Stratigraphic columns used in exploration

Laboratory practice:

- Fossil identification (4 sessions)
- Stratigraphic sections and correlation
- Interpreting geological history from maps

Assessment: Labs 30%; midterm 25%; final 45%

Recommended textbooks

- **Physical Geology (2nd ed.)** - Steven Earle (BCcampus Open Education). *CC BY; free PDF and web.* PDF/online: https://yc-prod-openlibrary-repo.ecampusontario.ca/jspui/bitstream/123456789/713/3/Physical-Geology-2nd-Edition-1591817334.pdf. Table of contents: Appendix A, item A07.
- Stanley and Luczaj, Earth System History (W.H. Freeman). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Boggs, Principles of Sedimentology and Stratigraphy (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEOL314 Stratigraphy (4).** Syllabus: Stratigraphy and biostratigraphy; facies associations; sedimentary sequences and basin geometries; geohistory analysis; lab and field exercises. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Colorado School of Mines - GEOL444 Invertebrate Paleontology (3).** Syllabus: Fossil preservation, evolution, mass extinctions, biostratigraphy and correlation, invertebrate phyla and their geological history. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Paleontology + Historical Geology (S4). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S026 Geological Mapping Field School (4 weeks)

Used in: Mineral Exploration - Summer 2, 4 cr; Geochemical Exploration - Summer 2, 4 cr; Mineral Engineering - Summer 2, 4 cr; Geological Information - Summer 2, 4 cr; Petroleum Exploration and Engineering - Summer 2, 4 cr

Syllabus

Chapter 1. Preparation - **1.1** Base maps, air photos and satellite images **1.2** Safety plan and equipment

Chapter 2. Mapping methods - **2.1** Traversing and contact mapping **2.2** Measuring structural data **2.3** Measured stratigraphic sections **2.4** Sampling and photo-logging

Chapter 3. Data processing - **3.1** Fair-copy maps: *GIS* **3.2** Cross-sections **3.3** Stratigraphic columns

Chapter 4. Reporting - **4.1** Geological history **4.2** Economic geology and hazards of the area

Field practice:

- 4 weeks residential mapping of a 10-15 km2 area with mixed lithologies
- supervised mapping (week 1), independent mapping in pairs (weeks 2-3), map and report compilation (week 4)

Assessment: Field slips and notebook 30%; final map and cross-sections 40%; report 30%

Recommended textbooks

- **Geological Structures: a Practical Introduction** - John W.F. Waldron, Morgan Snyder (Univ. of Alberta). *CC BY-NC; free web and PDF.* PDF/online: https://pressbooks.openeducationalberta.ca/introductorystructuralgeology/. Table of contents: Appendix A, item A12.

- Lisle, Brabham and Barnes, Basic Geological Mapping (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN316 Field Geology (5).** Syllabus: Five weeks of field mapping in the Southern Rocky Mountains: igneous, metamorphic and sedimentary terrain; problems in environmental, energy, mining, engineering and structural geology; formal reports. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Zhoukoudian Mapping Field School (6 wks). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S027 Ore Deposit Geology

Used in: Mineral Exploration - Sem 5, 4 cr; Geophysical Exploration - Sem 6, 2 cr; Geochemical Exploration - Sem 5, 4 cr; Geological Information - Sem 5, 3 cr (as 'Mineral Deposits and Exploration for Information Engineers')

Versions: Shorter versions: Geophysical Exploration (2 cr) = Chapters 1-5 and 8; Geological Information (3 cr, 'Mineral Deposits and Exploration for Information Engineers') = Chapters 1-5, 8 plus exploration data flow.

Syllabus

Chapter 1. Introduction - **1.1** Ore, grade, tonnage, cut-off **1.2** Mineral systems concept

Chapter 2. Magmatic deposits - **2.1** Chromite and PGE **2.2** Ni-Cu sulfides **2.3** Pegmatites and REE

Chapter 3. Magmatic-hydrothermal deposits - **3.1** Porphyry Cu-Mo-Au **3.2** Skarns **3.3** Epithermal Au-Ag **3.4** Greisen Sn-W

Chapter 4. Volcanic and sedimentary-exhalative deposits - **4.1** VMS **4.2** SEDEX and MVT Pb-Zn

Chapter 5. Orogenic and other gold - **5.1** Orogenic gold **5.2** Carlin-type **5.3** IOCG

Chapter 6. Sedimentary and surficial deposits - **6.1** Banded iron formation **6.2** Placers **6.3** Laterites and bauxite **6.4** Supergene enrichment

Chapter 7. Ore fluids and alteration - **7.1** Fluid sources, inclusions, alteration zoning **7.2** Ore textures and paragenesis

Chapter 8. Exploration implications - **8.1** Deposit models and vectors to ore

Laboratory practice:

- Hand-specimen suites of major deposit types (8 sessions)
- Alteration minerals
- Ore-deposit case-study exercises

Field practice: Visit to an operating mine or deposit (2 days)

Assessment: Labs 30%; case-study report 15%; final 55%

Recommended textbooks

- **Mineral Deposit Models (USGS Bulletin 1693)** - D.P. Cox, D.A. Singer (eds), U.S. Geological Survey, 1986. *Public domain; free PDF.* PDF/online: https://pubs.usgs.gov/bul/b1693/. Table of contents: Appendix A, item A13.
- Robb, Introduction to Ore-Forming Processes (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Pohl, Economic Geology: Principles and Practice (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Moon, Whateley and Evans (Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN401 Mineral Deposits (4).** Syllabus: Magmatic, hydrothermal and sedimentary metallic ore deposits; ore-forming processes; classic deposits; exploration sequences; hand-specimen labs of ore suites and deposit evaluation problems. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Ore Deposit Geology (S5, 4 cr); CUG-W Ore Deposit Geology Field Training. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page; CUG (Wuhan) programme list.

S028 Geochemistry

Used in: Mineral Exploration - Sem 5, 3 cr; Geochemical Exploration - Sem 5, 4 cr

Syllabus

Chapter 1. Earth's chemical composition - **1.1** Cosmochemistry and element abundances **1.2** Goldschmidt classification

Chapter 2. Crystal chemistry and trace elements - **2.1** Partitioning and compatibility **2.2** REE patterns

Chapter 3. Thermodynamics - **3.1** Gibbs free energy and equilibrium **3.2** Phase equilibria **3.3** Activity

Chapter 4. Aqueous geochemistry - **4.1** Speciation **4.2** Mineral solubility **4.3** Redox and Eh-pH diagrams

Chapter 5. Isotope geochemistry introduction - **5.1** Radiogenic isotopes and dating **5.2** Stable isotopes: *O, H, C, S*

Chapter 6. Organic geochemistry introduction - **6.1** Organic matter and kerogen **6.2** Hydrocarbon generation

Chapter 7. Geochemical cycles - **7.1** Weathering and element mobility **7.2** Hydrothermal processes

Laboratory practice:

- Calculation exercises (thermodynamics, speciation)
- Plotting and interpreting geochemical data
- Use of PHREEQC

Assessment: Exercises 30%; midterm 20%; final 50%

Recommended textbooks

- White, Geochemistry (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Misra, Introduction to Geochemistry: Principles and Applications (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Geochemistry incl. organic (S5). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S029 Ore Microscopy and Ore Petrography

Used in: Mineral Exploration - Sem 5, 3 cr

Syllabus

Chapter 1. Reflected-light microscopy - **1.1** Microscope and polished sections **1.2** Reflectance, colour, bireflectance, anisotropy

Chapter 2. Ore mineral identification - **2.1** Sulfides, oxides, native metals **2.2** Systematic identification scheme

Chapter 3. Ore textures - **3.1** Primary textures: *open-space filling, replacement, exsolution* **3.2** Deformation and recrystallisation

Chapter 4. Paragenesis - **4.1** Mineral sequences and paragenetic diagrams

Chapter 5. Applied ore microscopy - **5.1** Grain size and liberation **5.2** Deleterious minerals **5.3** Automated mineralogy introduction

Laboratory practice:

- Polished section identification (10 sessions)
- Paragenetic study of a deposit suite (report)

Assessment: Lab tests 50%; paragenesis report 20%; final 30%

Recommended textbooks

- Craig and Vaughan, Ore Microscopy and Ore Petrography (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Ramdohr, The Ore Minerals and Their Intergrowths (Pergamon). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN404 Ore Microscopy (3).** Syllabus: Identification of ore minerals by reflected light, micro-hardness and reflectivity; ore textures (magmatic, open-space filling, replacement, exsolution, recrystallisation); research on classic deposits. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Ore Petrology and Microscopy (S5). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S030 Applied Geophysics for Exploration

Used in: Mineral Exploration - Sem 5, 3 cr; Geochemical Exploration - Sem 5, 3 cr

Syllabus

Chapter 1. Introduction - **1.1** Physical properties of rocks **1.2** Choosing methods for exploration targets

Chapter 2. Gravity method - **2.1** Instruments and corrections **2.2** Bouguer anomalies **2.3** Interpretation of simple bodies

Chapter 3. Magnetic method - **3.1** Earth's field and magnetisation **3.2** Survey and reduction **3.3** Anomaly shapes and depth estimation

Chapter 4. Electrical methods - **4.1** Resistivity and IP **4.2** Self-potential

Chapter 5. Electromagnetic methods - **5.1** FDEM and TDEM **5.2** Airborne EM

Chapter 6. Radiometric method - **6.1** Gamma-ray spectrometry: *K, U, Th*

Chapter 7. Seismic methods - **7.1** Refraction and reflection basics

Chapter 8. Integration - **8.1** Geophysical signatures of deposit types **8.2** Joint interpretation with geology

Laboratory practice:

- Anomaly modelling exercises (gravity, magnetics)
- Resistivity sounding interpretation
- Interpretation of airborne datasets

Field practice: One-day ground magnetic and resistivity survey

Assessment: Labs 30%; field report 10%; final 60%

Recommended textbooks

- **Geophysics for Practicing Geoscientists (GPG)** - UBC Geophysical Inversion Facility. *Open web resource (CC BY).* PDF/online: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm. Table of contents: Appendix A, item A14.
- Dentith and Mudge, Geophysics for the Mineral Exploration Geoscientist (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Kearey, Brooks and Hill, An Introduction to Geophysical Exploration (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN439 Petroleum Exploration Design (3).** Syllabus: Multidisciplinary design course integrating well and 3D seismic data, seismic interpretation (Petrel) and prospect development in exploration plays; team reports and presentations. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Exploration Geophysics Methods and Applications (S5). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S031 Remote Sensing and Digital Image Processing

Used in: Mineral Exploration - Sem 5, 2 cr (as 'Remote Sensing Geology'); Geological Information - Sem 5, 4 cr; Space - Sem 6, 2 cr (as 'Remote Sensing Image Processing')

Versions: Shorter versions: 'Remote Sensing Geology' (2 cr) = Chapters 1, 2, 4, 7 with alteration and lineament mapping; 'Remote Sensing Image Processing' (Space, 2 cr) = Chapters 3-6.

Syllabus

Chapter 1. Remote sensing principles - **1.1** EM radiation **1.2** Interactions with atmosphere and surface **1.3** Spectral signatures

Chapter 2. Sensors and platforms - **2.1** Optical multispectral: *Landsat, Sentinel-2* **2.2** Thermal **2.3** Radar **2.4** UAV

Chapter 3. Pre-processing - **3.1** Radiometric correction **3.2** Atmospheric correction **3.3** Geometric correction

Chapter 4. Enhancement - **4.1** Contrast stretch **4.2** Band ratios and indices **4.3** Principal components

Chapter 5. Classification - **5.1** Supervised and unsupervised **5.2** Accuracy assessment

Chapter 6. Change detection - **6.1** Image differencing **6.2** Post-classification comparison

Chapter 7. Applications - **7.1** Geological and alteration mapping **7.2** Lineament analysis **7.3** Land use and hazards

Computer practice: Labs with Sentinel-2 and Landsat imagery (QGIS/SNAP)

Assessment: Labs 40%; final 60%

Recommended textbooks

- **Fundamentals of Remote Sensing (tutorial)** - Canada Centre for Remote Sensing / Natural Resources Canada. *Free web tutorial.* PDF/online: https://natural-resources.canada.ca/science-data/science-research/geomatics/remote-sensing/tutorial-fundamentals-remote-sensing. Table of contents: Appendix A, item A15.
- Gupta, Remote Sensing Geology (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Lillesand, Kiefer and Chipman, Remote Sensing and Image Interpretation (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Jensen, Introductory Digital Image Processing (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

- Richards, Remote Sensing Digital Image Analysis (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEOL470 Applications of Satellite Remote Sensing (3).** Syllabus: Satellites, sensors and methods; geoscience applications from a systems perspective; guest case studies; independent term project. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Remote Sensing Geology (S5); TH 40040913; TH 40040913 Remote Sensing Principles and Methods; CUG-W Remote Sensing Science and Technology. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page; CUG (Wuhan) programme list.

S032 Sedimentology, Stratigraphy and Sedimentary Basins

Used in: Mineral Exploration - Sem 5, 3 cr (as 'Sedimentology and Sedimentary Basins'); Petroleum Exploration and Engineering - Sem 4, 4 cr (as 'Sedimentology and Stratigraphy')

Versions: Petroleum version 'Sedimentology and Stratigraphy' (4 cr) adds sequence stratigraphy and reservoir architecture case studies.

Syllabus

Chapter 1. Sediment transport - **1.1** Grain size and sorting **1.2** Fluid flow and bedforms

Chapter 2. Depositional environments - **2.1** Continental: *fluvial, aeolian, lacustrine* **2.2** Coastal and deltaic **2.3** Shallow and deep marine **2.4** Carbonate platforms

Chapter 3. Facies analysis - **3.1** Facies models **3.2** Sedimentary logging

Chapter 4. Diagenesis - **4.1** Compaction and cementation **4.2** Porosity evolution

Chapter 5. Sedimentary basins - **5.1** Basin types and plate tectonics **5.2** Subsidence history

Chapter 6. Sediment-hosted resources - **6.1** Coal, petroleum, placers, evaporites, sediment-hosted metals

Laboratory practice:

- Grain-size analysis
- Sedimentary structures
- Core and outcrop logging

Field practice: Two-day sedimentary logging excursion

Assessment: Labs 30%; field log 15%; final 55%

Recommended textbooks

- Nichols, Sedimentology and Stratigraphy (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Allen and Allen, Basin Analysis (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Boggs, Principles of Sedimentology and Stratigraphy (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.110 Sedimentary Geology (OCW, Spring 2007).** Syllabus: Sediment texture and transport; siliciclastic rocks; carbonates; cherts and evaporites; diagenesis; paleontology; depositional environments; stratigraphy; sedimentary basins; fossil fuels; laboratory sessions; 4-day field trip with field report. Link: https://ocw.mit.edu/courses/12-110-sedimentary-geology-spring-2007/pages/syllabus
- **Colorado School of Mines - GEOL314 Stratigraphy (4).** Syllabus: Stratigraphy and biostratigraphy; facies associations; sedimentary sequences and basin geometries; geohistory analysis; lab and field exercises. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Colorado School of Mines - GEOL315 Sedimentology and Stratigraphy (3).** Syllabus: Genesis of sedimentary rocks and reservoir porosity/permeability; diagenesis, facies analysis, correlation, sequence and seismic stratigraphy for petroleum engineers. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S033 Geographic Information Systems (GIS)

Used in: Mineral Exploration - Sem 5, 2 cr (as 'GIS for Geoscientists'); Geochemical Exploration - Sem 6, 2 cr (as 'GIS for Geoscientists'); Geological Information - Sem 5, 4 cr (as 'Geographic Information Systems'); Tourist District Development - Sem 5, 3 cr (as 'GIS for Planning')

Versions: Shorter versions: 'GIS for Geoscientists' (2 cr) = Chapters 1-5 and 7 with geological datasets; 'GIS for Planning' (3 cr) = Chapters 1-6 with land-suitability and accessibility projects.

Syllabus

Chapter 1. GIS fundamentals - **1.1** Spatial data and representation **1.2** GIS components

Chapter 2. Coordinate systems - **2.1** Datums and projections in GIS

Chapter 3. Data models - **3.1** Vector: *points, lines, polygons* **3.2** Raster **3.3** Topology

Chapter 4. Data capture and management - **4.1** Digitising and georeferencing **4.2** GNSS and field data **4.3** Geodatabases and metadata

Chapter 5. Spatial analysis - vector - **5.1** Queries **5.2** Buffers **5.3** Overlay **5.4** Network analysis

Chapter 6. Spatial analysis - raster - **6.1** Map algebra **6.2** Reclassification **6.3** Suitability analysis

Chapter 7. Terrain analysis - **7.1** DEMs **7.2** Slope, aspect, hillshade, watersheds

Chapter 8. Cartographic output - **8.1** Layout, symbology and labels

Chapter 9. GIS project management - **9.1** Workflow and documentation

Computer practice:

- Weekly QGIS labs
- final GIS project

Assessment: Labs 40%; project 20%; final 40%

Recommended textbooks

- **Essentials of Geographic Information Systems** - Jonathan Campbell, Michael Shin. *CC BY-NC-SA; free (Open Textbook Library).* PDF/online: https://open.umn.edu/opentextbooks/textbooks/67. Table of contents: Appendix A, item A16.
- Bonham-Carter, Geographic Information Systems for Geoscientists (Pergamon). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Longley et al., Geographic Information Science and Systems (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN475 Applications of GIS (3).** Syllabus: Principles of GIS, data structures, DEMs, data input and verification, spatial analysis and modelling, data quality, GIS project methods. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** TH 20030212 Fundamentals of Geospatial Information Technology; TH 20030212; CUG-W National Engineering Research Center of GIS. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Wuhan) programme list.

S034 Mineral Exploration Methods

Used in: Mineral Exploration - Sem 6, 4 cr; Geochemical Exploration - Sem 6, 2 cr

Syllabus

Chapter 1. Exploration framework - **1.1** Mineral exploration stages: *reconnaissance to feasibility* **1.2** Exploration strategy and budgets **1.3** Mineral systems and targeting

Chapter 2. Area selection - **2.1** Regional data compilation **2.2** Licensing and land access **2.3** Remote sensing and airborne surveys

Chapter 3. Ground exploration - **3.1** Geological mapping and prospecting **3.2** Geochemical surveys **3.3** Ground geophysics

Chapter 4. Trenching and drilling - **4.1** Trench and pit sampling **4.2** Drilling methods and drill-hole planning **4.3** Core logging standards

Chapter 5. Sampling - **5.1** Sampling theory (Gy) and errors **5.2** Sample preparation protocols **5.3** QA/QC

Chapter 6. Evaluation - **6.1** Interpretation and target ranking **6.2** Introduction to resource estimation **6.3** Reporting progress

Laboratory practice:

- Compiling a regional prospectivity map
- Planning a drilling programme on a target
- Core logging session (2 sessions)

Assessment: Exercises 30%; exploration plan 20%; final 50%

Recommended textbooks

- **Mineral Deposit Models (USGS Bulletin 1693)** - D.P. Cox, D.A. Singer (eds), U.S. Geological Survey, 1986. *Public domain; free PDF.* PDF/online: https://pubs.usgs.gov/bul/b1693/. Table of contents: Appendix A, item A13.
- Moon, Whateley and Evans (eds), Introduction to Mineral Exploration (Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Marjoribanks, Geological Methods in Mineral Exploration and Mining (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN403 Mineral Exploration Design (3).** Syllabus: Capstone: commodity and target selection, genetic and exploration models, property acquisition, costs; simulated exploration from planning to discovery, initial reserve calculation and preliminary economic evaluation. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Mineral Prospecting and Exploration (S6); CUG-W Resource Exploration Eng. programme. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page; CUG (Wuhan) programme list.

S035 Exploration Geochemistry

Used in: Mineral Exploration - Sem 6, 3 cr; Geochemical Exploration - Sem 6, 4 cr (as 'Exploration Geochemistry I (Survey Methods)')

Versions: Shorter version for Mineral Exploration (3 cr): Chapters 1-4 and 6 plus anomaly recognition from Exploration Geochemistry II.

Syllabus

Chapter 1. Principles - **1.1** Primary and secondary dispersion **1.2** Geochemical landscapes and regolith

Chapter 2. Stream sediments - **2.1** Sampling and orientation surveys

Chapter 3. Soils and regolith - **3.1** Soil horizons **3.2** Residual and transported cover

Chapter 4. Rocks - **4.1** Lithogeochemistry **4.2** Alteration indices

Chapter 5. Other media - **5.1** Till, lake sediments, heavy minerals

Chapter 6. Survey design - **6.1** Density, media and analytical choice

Laboratory practice:

- Survey planning exercise
- Orientation-survey data analysis

Field practice: Two-day stream and soil sampling exercise

Assessment: Labs 30%; field report 20%; final 50%

Recommended textbooks

- Rose, Hawkes and Webb, Geochemistry in Mineral Exploration (Academic Press). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Levinson, Introduction to Exploration Geochemistry (Applied Publishing). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Exploration Geochemistry (S6). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S036 Exploration Drilling and Sampling

Used in: Mineral Exploration - Sem 6, 2 cr

Syllabus

Chapter 1. Drilling methods - **1.1** Diamond core drilling **1.2** RC and RAB/aircore **1.3** Percussion and auger

Chapter 2. Drill planning - **2.1** Drill-hole design and orientation **2.2** Downhole surveying

Chapter 3. Core handling - **3.1** Core recovery and RQD **3.2** Geological and geotechnical logging **3.3** Core photography and storage

Chapter 4. Sampling - **4.1** Splitting and sampling intervals **4.2** Sample chain of custody **4.3** Density measurement

Field practice: Two days at a drill site: rig observation, core logging and sampling

Assessment: Logging exercise 50%; test 50%

Recommended textbooks

- Marjoribanks, Geological Methods in Mineral Exploration and Mining (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Heinz, Diamond Drill Handbook (Smith-Heinz). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN473 Geological Engineering Site Investigation (3).** Syllabus: Field investigation, testing and monitoring: drilling and sampling, logging, field tests, instrumentation, trench logging, foundation inspection; technical writing; simulated investigations. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUG-W Exploration Technology and Engineering programme. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S037 Geostatistics and Mineral Resource Estimation

Used in: Mineral Exploration - Sem 6, 3 cr (as 'Mineral Resource Estimation and Geostatistics'); Geochemical Exploration - Sem 7, 2 cr (as 'Mineral Resource Estimation and Geostatistics'); Geological Information - Sem 5, 3 cr (as 'Geostatistics')

Versions: Geological Information version 'Geostatistics' (3 cr) = Chapters 2-4 and 6 with environmental and geotechnical datasets; Geochemical course uses a 2 cr version.

Syllabus

Chapter 1. Data preparation - **1.1** Drill-hole databases and validation **1.2** Compositing and domaining **1.3** Top-cutting

Chapter 2. Exploratory statistics - **2.1** Distributions and outliers **2.2** Spatial continuity

Chapter 3. Variography - **3.1** Experimental variograms **3.2** Variogram modelling **3.3** Anisotropy

Chapter 4. Estimation methods - **4.1** Polygonal and inverse-distance **4.2** Ordinary kriging **4.3** Block models and support

Chapter 5. Validation and classification - **5.1** Swath plots and visual checks **5.2** Resource classification: *Measured, Indicated, Inferred*

Chapter 6. Introduction to simulation - **6.1** Conditional simulation and uncertainty

Computer practice: Resource estimation project with a public drill-hole dataset (Python/GSLIB-type tools)

Assessment: Project 50%; final 50%

Recommended textbooks

- **The JORC Code (2012 edition)** - Joint Ore Reserves Committee. *Free PDF (Chinese translation also free).* PDF/online: https://jorc.org/docs/jorc_code2012.pdf. Table of contents: Appendix A, item A17.
- Sinclair and Blackwell, Applied Mineral Inventory Estimation (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Rossi and Deutsch, Mineral Resource Estimation (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Isaaks and Srivastava, An Introduction to Applied Geostatistics (Oxford). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Chiles and Delfiner, Geostatistics (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN403 Mineral Exploration Design (3).** Syllabus: Capstone: commodity and target selection, genetic and exploration models, property acquisition, costs; simulated exploration from planning to discovery, initial reserve calculation and preliminary economic evaluation. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S038 Introduction to Mining and Mineral Processing

Used in: Mineral Exploration - Sem 6, 2 cr

Syllabus

Chapter 1. Mining methods - **1.1** Surface mining: *open pit, strip, quarry* **1.2** Underground methods: *room and pillar, stoping, caving*

Chapter 2. Mine planning basics - **2.1** Pit design and stripping ratio **2.2** Production scheduling

Chapter 3. Unit operations - **3.1** Drilling and blasting **3.2** Loading and haulage

Chapter 4. Mineral processing - **4.1** Comminution **4.2** Concentration: *gravity, flotation, magnetic* **4.3** Recovery and grade

Assessment: Assignments 40%; final 60%

Recommended textbooks

- Hartman and Mutmansky, Introductory Mining Engineering (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Wills and Finch, Wills' Mineral Processing Technology (Butterworth-Heinemann). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEOL311 Mining Geology (3).** Syllabus: Formation, distribution and engineering behaviour of ores; exploration and development geology; labs on recognition and evaluation of ores and host rocks. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Colorado School of Mines - MNGN210 Introductory Mining (3).** Syllabus: Mining law; exploration and sampling; reserve estimation and project evaluation; drilling, blasting, loading, hauling and support; shaft sinking; surface and underground methods. Link: https://facultyhandbook.mines.edu/undergraduate/programs/miningengineering/miningengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Intro to Mining and Mineral Processing (S7). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S039 Hydrogeology

Used in: Mineral Exploration - Sem 6, 2 cr

Syllabus

Chapter 1. Groundwater in the hydrological cycle - **1.1** Water balance **1.2** Aquifers and aquitards

Chapter 2. Physical properties - **2.1** Porosity, permeability, storage

Chapter 3. Groundwater flow - **3.1** Darcy's law **3.2** Flow equations **3.3** Flow nets

Chapter 4. Well hydraulics - **4.1** Steady and unsteady flow to wells **4.2** Pumping test analysis: *Theis, Cooper-Jacob*

Chapter 5. Groundwater chemistry - **5.1** Major ions and water types **5.2** Contamination and transport

Chapter 6. Applied hydrogeology - **6.1** Mine and construction dewatering **6.2** Groundwater resources

Laboratory practice:

- Permeameter test
- Flow-net construction
- Pumping test analysis
- Water-chemistry plotting

Assessment: Labs 30%; final 70%

Recommended textbooks

- **Groundwater** - R. Allen Freeze, John A. Cherry (1979). *Free PDF from The Groundwater Project (preserved book).* PDF/online: https://gw-project.org/books/groundwater/. Table of contents: Appendix A, item A18.
- Fetter, Applied Hydrogeology (Waveland). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Freeze and Cherry, Groundwater (Prentice Hall). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN466/466L Groundwater Engineering (3+1).** Syllabus: Groundwater occurrence and flow; relation to surface water; hydraulic head; aquifer tests; water chemistry and quality; contaminant transport; laboratory and field methods. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** TH 30040573 Hydrology (related). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S040 Metallogeny and Mineral Prospectivity

Used in: Mineral Exploration - Sem 6, 2 cr

Syllabus

Chapter 1. Metallogenic concepts - **1.1** Metallogenic provinces and epochs **1.2** Tectonic settings of ore deposits

Chapter 2. Mineral systems approach - **2.1** Source, pathway, trap, preservation

Chapter 3. Regional prospectivity - **3.1** Knowledge-driven and data-driven methods **3.2** Target ranking

Computer practice: Weights-of-evidence exercise in GIS

Assessment: Project 50%; final 50%

Recommended textbooks

- **Mineral Deposit Models (USGS Bulletin 1693)** - D.P. Cox, D.A. Singer (eds), U.S. Geological Survey, 1986. *Public domain; free PDF.* PDF/online: https://pubs.usgs.gov/bul/b1693/. Table of contents: Appendix A, item A13.
- Pirajno, Hydrothermal Processes and Mineral Systems (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Carranza, Geochemical Anomaly and Mineral Prospectivity Mapping in GIS (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed as a separate subject (part of CUGB Mineral Prospecting and Exploration).

S041 Mineral Economics and Project Evaluation

Used in: Mineral Exploration - Sem 6, 2 cr

Syllabus

Chapter 1. Commodity markets - **1.1** Supply, demand and prices **1.2** Critical minerals

Chapter 2. Cost estimation - **2.1** Capital and operating costs **2.2** Cut-off grade

Chapter 3. Project valuation - **3.1** Cash-flow models **3.2** NPV, IRR, payback **3.3** Taxes and royalties

Chapter 4. Risk - **4.1** Sensitivity and scenario analysis **4.2** Monte Carlo simulation

Chapter 5. Study stages - **5.1** Scoping, pre-feasibility and feasibility studies

Computer practice: Spreadsheet valuation of a small gold project

Assessment: Valuation project 40%; final 60%

Recommended textbooks

- Wellmer, Dalheimer and Wagner, Economic Evaluations in Exploration (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Mineral Resources Economics and Management (S6). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S042 Mineral Exploration Design (capstone)

Used in: Mineral Exploration - Sem 7, 3 cr

Syllabus

Chapter 1. Project brief - **1.1** Commodity and region selection **1.2** Deposit model

Chapter 2. Programme design - **2.1** Phased exploration plan **2.2** Methods and budget **2.3** Environmental and community aspects

Chapter 3. Evaluation - **3.1** Drilling results interpretation: *supplied data* **3.2** Preliminary resource **3.3** Scoping economics

Chapter 4. Reporting - **4.1** Written report to reporting-code style **4.2** Oral presentation to industry panel

Project work: Team capstone over the semester with weekly milestones

Assessment: Report 50%; presentations 30%; peer review 20%

Recommended textbooks

- Moon, Whateley and Evans (eds), Introduction to Mineral Exploration (Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN403 Mineral Exploration Design (3).** Syllabus: Capstone: commodity and target selection, genetic and exploration models, property acquisition, costs; simulated exploration from planning to discovery, initial reserve calculation and preliminary economic evaluation. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Mineral Exploration Course Design (S6). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S043 Reporting Codes, Exploration Standards and Mineral Law

Used in: Mineral Exploration - Sem 7, 2 cr

Syllabus

Chapter 1. Reporting codes - **1.1** CRIRSCO family: *JORC, NI 43-101, PERC* **1.2** Exploration results, resources and reserves **1.3** Competent/qualified person

Chapter 2. National standards - **2.1** National mineral exploration standards **2.2** Classification systems: *UNFC comparison*

Chapter 3. Mineral law - **3.1** Mineral rights and licensing **3.2** Environmental and community legislation

Assessment: Case-study critique 40%; final 60%

Recommended textbooks

- **The JORC Code (2012 edition)** - Joint Ore Reserves Committee. *Free PDF (Chinese translation also free).* PDF/online: https://jorc.org/docs/jorc_code2012.pdf. Table of contents: Appendix A, item A17.
- The JORC Code (Joint Ore Reserves Committee). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- national mineral resources law. *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Solid Mineral Exploration Standards + Environmental Law and Mineral Resources Law (S7). These universities publish course name, credits, hours and semester only; no subject

syllabus is published online. Source: CUG (Beijing) programme page.

S044 3D Geological Modelling

Used in: Mineral Exploration - Sem 7, 2 cr; Geological Information - Sem 6, 4 cr

Syllabus

Chapter 1. Data - **1.1** Drill-hole databases **1.2** Surface and section interpretation

Chapter 2. Modelling methods - **2.1** Explicit wireframing **2.2** Implicit modelling: *RBF* **2.3** Structural and stratigraphic models

Chapter 3. Block models - **3.1** Building and querying block models **3.2** Visualisation

Computer practice: 3D model of an ore body from supplied drill data (open or academic software)

Assessment: Modelling project 60%; test 40%

Recommended textbooks

- Mallet, Geomodeling (Oxford). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Pyrcz and Deutsch, Geostatistical Reservoir Modeling (Oxford). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S045 Coal and Industrial Minerals Geology

Used in: Mineral Exploration - Sem 7, 3 cr

Syllabus

Chapter 1. Coal geology - **1.1** Peat formation and coalification **1.2** Coal rank, type and grade **1.3** Coal-bearing basins

Chapter 2. Coal exploration and quality - **2.1** Drilling and logging **2.2** Proximate and ultimate analysis **2.3** Coalbed methane

Chapter 3. Industrial minerals - **3.1** Clays, kaolin, bentonite **3.2** Limestone and dolomite **3.3** Phosphate, potash, gypsum **3.4** Silica sand, feldspar, graphite

Chapter 4. Specifications and markets - **4.1** Product specifications **4.2** Exploration for industrial minerals

Laboratory practice:

- Coal macroscopic description
- Industrial mineral testing demonstrations

Assessment: Labs 30%; final 70%

Recommended textbooks

- Thomas, Coal Geology (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Kogel et al. (eds), Industrial Minerals and Rocks (SME). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Resource Exploration Eng. (new-energy class); CUGB School of Energy Resources. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page; CUG (Wuhan) programme list.

S046 Environmental Geology, Mine Wastes and Closure

Used in: Mineral Exploration - Sem 7, 2 cr (as 'Environmental Geology and Mine Closure'); Geochemical Exploration - Sem 7, 3 cr (as 'Geochemistry of the Mining Environment and Remediation')

Versions: Geochemical course version 'Geochemistry of the Mining Environment and Remediation' (3 cr) adds acid-drainage prediction tests, metal mobility and treatment design.

Syllabus

Chapter 1. Mine wastes - **1.1** Waste rock and tailings **1.2** Acid and metalliferous drainage

Chapter 2. Environmental baselines - **2.1** Water, soil and biological baselines

Chapter 3. Impacts and mitigation - **3.1** Dust, noise, water management

Chapter 4. Closure - **4.1** Rehabilitation planning **4.2** Closure costs and bonds

Assessment: Case-study report 40%; final 60%

Recommended textbooks

- Lottermoser, Mine Wastes: Characterization, Treatment and Environmental Impacts (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN466/466L Groundwater Engineering (3+1).** Syllabus: Groundwater occurrence and flow; relation to surface water; hydraulic head; aquifer tests; water chemistry and quality; contaminant transport; laboratory and field methods. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

- **Colorado School of Mines - GEGN470 Ground-water Engineering Design (3).** Syllabus: Hydrogeology applied to water supply, geotechnical and water-quality problems: design of well fields, drilling programmes and pump tests; engineering reports. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S047 Professional English and Technical Communication

Used in: All ten courses (or most) - Sem 7, 2 cr

Syllabus

Chapter 1. Reading technical literature - **1.1** Structure of papers and reports **1.2** Technical vocabulary of the discipline

Chapter 2. Writing - **2.1** Abstracts and summaries **2.2** Technical reports and proposals **2.3** Referencing

Chapter 3. Speaking - **3.1** Presentations **3.2** Meetings and discussions

Chapter 4. Standards and documents - **4.1** Reading international standards and specifications

Assessment: Written assignments 50%; presentation 30%; test 20%

Recommended textbooks

- Day and Gastel, How to Write and Publish a Scientific Paper (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Professional English for Resource Prospecting (S7). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S048 Elective (from any course of the faculty)

Used in: All ten courses (or most) - Sem 7, 2/3/4 cr

Syllabus

Chapter 1. Elective choice - **1.1** Selected with the course adviser from another course's subject list: *syllabus as per chosen subject*

Recommended textbooks

No textbook: practice-based subject (faculty guidelines, logbook and report templates).

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH category extension (6 cr). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S049 Structural Analysis I

Used in: Civil Engineering - Sem 4, 4 cr

Syllabus

Chapter 1. Introduction - **1.1** Structural forms and loads **1.2** Idealisation and supports

Chapter 2. Determinate structures - **2.1** Beams, frames and trusses **2.2** Shear, moment and axial diagrams

Chapter 3. Influence lines - **3.1** Moving loads on beams and trusses

Chapter 4. Deflections - **4.1** Double integration **4.2** Moment-area **4.3** Virtual work

Chapter 5. Indeterminate structures - **5.1** Force (flexibility) method **5.2** Slope-deflection method **5.3** Moment distribution

Chapter 6. Approximate analysis - **6.1** Portal and cantilever methods

Assessment: Problem sets 20%; midterm 30%; final 50%

Recommended textbooks

- Hibbeler, Structural Analysis (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 20030044 Structural Mechanics (1) (Y3 autumn, 4 cr). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S050 Construction Materials (with laboratory)

Used in: Civil Engineering - Sem 4, 3 cr

Syllabus

Chapter 1. Materials engineering basics - **1.1** Properties and testing **1.2** Standards

Chapter 2. Aggregates - **2.1** Sources, grading, properties

Chapter 3. Cement and concrete - **3.1** Cement chemistry and hydration **3.2** Concrete mix design **3.3** Fresh and hardened properties **3.4** Durability

Chapter 4. Steel - **4.1** Production, properties and heat treatment **4.2** Corrosion

Chapter 5. Other materials - **5.1** Masonry **5.2** Timber **5.3** Asphalt and bitumen **5.4** Polymers and composites

Laboratory practice:

- Aggregate grading and specific gravity
- Cement setting time
- Concrete mix, slump and cube strength
- Steel tension test
- Asphalt penetration/softening point

Assessment: Labs 30%; midterm 20%; final 50%

Recommended textbooks

- Mamlouk and Zaniewski, Materials for Civil and Construction Engineers (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Mehta and Monteiro, Concrete (McGraw-Hill). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 40030352 Building Materials + 40030361 Building Materials Lab. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S051 Soil Mechanics

Used in: Civil Engineering - Sem 4, 3 cr (as 'Soil Mechanics (with laboratory)'); Drill Engineering - Sem 5, 2 cr (as 'Geotechnical Engineering Basics')

Versions: Shorter version 'Geotechnical Engineering Basics' (Drill Engineering, 2 cr): Chapters 2, 5, 6 and 7 without the laboratory block.

Syllabus

Chapter 1. Soil formation and composition - **1.1** Soil types and clay minerals **1.2** Phase relations

Chapter 2. Classification - **2.1** Grain size and Atterberg limits **2.2** Unified/national classification

Chapter 3. Compaction - **3.1** Proctor tests **3.2** Field control

Chapter 4. Water in soil - **4.1** Permeability **4.2** Seepage and flow nets **4.3** Effective stress

Chapter 5. Compressibility - **5.1** One-dimensional consolidation **5.2** Settlement calculation

Chapter 6. Shear strength - **6.1** Mohr-Coulomb **6.2** Direct shear and triaxial tests **6.3** Drained and undrained behaviour

Chapter 7. Earth pressure - **7.1** Rankine and Coulomb theories

Laboratory practice:

- Sieve and hydrometer analysis
- Atterberg limits
- Proctor compaction
- Constant/falling head permeability
- Oedometer
- Direct shear
- Unconfined compression

Assessment: Labs 30%; midterm 20%; final 50%

Recommended textbooks

- Das and Sobhan, Principles of Geotechnical Engineering (Cengage). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Knappett and Craig, Craig's Soil Mechanics (CRC). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - CEEN312/312L Soil Mechanics + Laboratory (3+1).** Syllabus: Soil properties, phase relations and classification; effective stress; seepage and flow nets; compressibility, consolidation and settlement; shear strength; lab tests of classification, permeability, compressibility and shear strength. Link: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf

- **Tsinghua University / China University of Geosciences:** TH 30040513 Soil Mechanics. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S052 Engineering Surveying

Used in: Civil Engineering - Sem 4, 2 cr; Tourist District Development - Sem 4, 3 cr

Syllabus

Chapter 1. Surveying principles - **1.1** Errors and accuracy **1.2** Coordinate systems

Chapter 2. Levelling - **2.1** Differential and profile levelling **2.2** Contouring

Chapter 3. Angles and distances - **3.1** Theodolite and total station **3.2** EDM

Chapter 4. Traversing - **4.1** Traverse computation and adjustment

Chapter 5. GNSS - **5.1** RTK surveys

Chapter 6. Engineering surveys - **6.1** Topographic surveys **6.2** Setting out of buildings and roads

Field practice: Practical sessions on levelling, total station and GNSS

Assessment: Field books 40%; final 60%

Recommended textbooks

- Ghilani and Wolf, Elementary Surveying: An Introduction to Geomatics (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 30030611 Surveying + 40030402 Surveying Practice; CUGB Surveying A (S4). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page.

S053 Surveying and Engineering Geology Field Practice (4 weeks)

Used in: Civil Engineering - Summer 2, 4 cr

Syllabus

Chapter 1. Surveying camp - **1.1** Control network establishment **1.2** Detail and topographic survey **1.3** Map production

Chapter 2. Engineering geology mapping - **2.1** Site walkover **2.2** Mapping soils, rock and geohazards **2.3** Trial-pit logging

Field practice: 2 weeks surveying camp + 2 weeks engineering-geological mapping of a construction site

Assessment: Survey plans 40%; engineering-geology map and report 60%

Recommended textbooks

- Ghilani and Wolf, Elementary Surveying (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 40030402 Surveying Practice (2 wks); TH 30040551 Geology Practice. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S054 Structural Analysis II (Matrix and Finite Element Methods)

Used in: Civil Engineering - Sem 5, 3 cr

Syllabus

Chapter 1. Matrix methods - **1.1** Stiffness method for trusses **1.2** Beams and frames **1.3** Assembly and boundary conditions

Chapter 2. Finite element method - **2.1** Shape functions **2.2** Bar, beam and plane-stress elements

Chapter 3. Structural software - **3.1** Modelling, verification and interpretation of results

Computer practice:

- Coding a 2D frame stiffness solver
- FE analysis of a plate with opening

Assessment: Computer assignments 50%; final 50%

Recommended textbooks

- McGuire, Gallagher and Ziemian, Matrix Structural Analysis (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Logan, A First Course in the Finite Element Method (Cengage). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 40030322 Structural Matrix Analysis; TH 20030102 Elasticity and Finite Elements. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S055 Reinforced Concrete Design I

Used in: Civil Engineering - Sem 5, 4 cr

Syllabus

Chapter 1. Materials and design philosophy - **1.1** Concrete and reinforcement properties **1.2** Limit-state design

Chapter 2. Flexure - **2.1** Singly and doubly reinforced beams **2.2** T-beams

Chapter 3. Shear and torsion - **3.1** Shear design **3.2** Torsion

Chapter 4. Bond and anchorage - **4.1** Development length **4.2** Splices

Chapter 5. Serviceability - **5.1** Deflection and cracking

Chapter 6. Columns - **6.1** Short columns **6.2** Axial load and bending interaction

Laboratory practice: Reinforced-concrete beam test to failure

Assessment: Design assignments 30%; midterm 20%; final 50%

Recommended textbooks

- Wight, Reinforced Concrete: Mechanics and Design (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- McCormac and Brown, Design of Reinforced Concrete (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 20030153 Concrete Structures 1; TH 30030583 Reinforced Concrete. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S056 Steel Structures

Used in: Civil Engineering - Sem 5, 3 cr

Syllabus

Chapter 1. Steel and design basis - **1.1** Material properties **1.2** Limit-state design

Chapter 2. Members - **2.1** Tension members **2.2** Compression members and buckling **2.3** Beams and lateral-torsional buckling **2.4** Beam-columns

Chapter 3. Connections - **3.1** Bolted connections **3.2** Welded connections

Chapter 4. Systems - **4.1** Trusses and portal frames

Assessment: Design assignments 30%; final 70%

Recommended textbooks

- Segui, Steel Design (Cengage). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 30030113 Steel Structures (1). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S057 Foundation Engineering

Used in: Civil Engineering - Sem 5, 3 cr

Syllabus

Chapter 1. Site investigation - **1.1** Investigation planning **1.2** In-situ tests

Chapter 2. Shallow foundations - **2.1** Bearing capacity **2.2** Settlement **2.3** Raft foundations

Chapter 3. Deep foundations - **3.1** Pile types **3.2** Pile capacity and group effects

Chapter 4. Retaining structures - **4.1** Retaining walls **4.2** Sheet piles and anchored walls

Chapter 5. Ground improvement - **5.1** Compaction, preloading, grouting

Assessment: Design assignments 30%; final 70%

Recommended textbooks

- Das and Sivakugan, Principles of Foundation Engineering (Cengage). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 30040162 Foundation Engineering. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S058 Engineering Geology and Site Investigation

Used in: Civil Engineering - Sem 5, 3 cr

Syllabus

Chapter 1. Engineering geology basics - **1.1** Rock and soil as engineering materials **1.2** Weathering grades

Chapter 2. Site investigation - **2.1** Desk study and walkover **2.2** Drilling, sampling and in-situ testing **2.3** Geophysics for site investigation

Chapter 3. Geohazards - **3.1** Landslides **3.2** Karst and subsidence **3.3** Earthquakes and liquefaction

Chapter 4. Engineering applications - **4.1** Dams and reservoirs **4.2** Tunnels **4.3** Roads and foundations

Chapter 5. Reporting - **5.1** Geotechnical baseline reports

Laboratory practice:

- Engineering logging of core
- Interpreting site investigation data

Field practice: One-day geohazard excursion

Assessment: Labs 30%; final 70%

Recommended textbooks

- Gonzalez de Vallejo and Ferrer, Geological Engineering (CRC). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Clayton, Matthews and Simons, Site Investigation (Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN468 Engineering Geology and Geotechnics (4).** Syllabus: Geology in the evaluation of dams, waterways, tunnels, highways, bridges, buildings, mine design and waste facilities; design projects with field, lab and computer analysis. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Colorado School of Mines - GEGN473 Geological Engineering Site Investigation (3).** Syllabus: Field investigation, testing and monitoring: drilling and sampling, logging, field tests, instrumentation, trench logging, foundation inspection; technical writing; simulated investigations. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** TH 40040152 Engineering Geology; CUG-W Geological Engineering programme. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Wuhan) programme list.

S059 Engineering Hydraulics and Hydrology

Used in: Civil Engineering - Sem 5, 2 cr

Syllabus

Chapter 1. Hydrology - **1.1** Rainfall and runoff **1.2** Hydrographs and design floods

Chapter 2. Open channels - **2.1** Uniform and gradually varied flow **2.2** Culverts and bridges hydraulics

Chapter 3. Drainage - **3.1** Urban stormwater **3.2** Road drainage

Assessment: Assignments 30%; final 70%

Recommended textbooks

- Mays, Water Resources Engineering (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Chow, Open-Channel Hydraulics (McGraw-Hill). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 30040573 Hydrology, 20040143 River Hydraulics. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S060 Building Architecture and Design

Used in: Civil Engineering - Sem 5, 2 cr; Tourist District Development - Sem 5, 3 cr (as 'Architectural Design Basics')

Versions: Tourist course version 'Architectural Design Basics' (3 cr) replaces Chapter 4 with a small visitor-building design studio.

Syllabus

Chapter 1. Building planning - **1.1** Functional planning and space programmes **1.2** Building codes and regulations

Chapter 2. Design principles - **2.1** Form, space and order **2.2** Proportion and scale

Chapter 3. Building elements - **3.1** Foundations, walls, floors and roofs **3.2** Stairs, doors and windows

Chapter 4. Building envelope - **4.1** Insulation and moisture control **4.2** Fire safety and accessibility

Chapter 5. Architectural drawings - **5.1** Plans, sections, elevations **5.2** CAD drawing practice

Computer practice: Design and drawing of a small building (civil) or small visitor building (tourism)

Assessment: Drawings 50%; final 50%

Recommended textbooks

- Ching, Building Construction Illustrated (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Ching, Architecture: Form, Space and Order (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 30030522 Building Architecture; TH 40031102 Fundamentals of Architecture; TH 40030073 Architectural Design (1); TH 40030721 Intro to Architectural Design. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S061 Reinforced Concrete Design II (Buildings)

Used in: Civil Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. Slabs - **1.1** One-way and two-way slabs **1.2** Flat slabs

Chapter 2. Frames and walls - **2.1** Multi-storey frames **2.2** Shear walls

Chapter 3. Foundations - **3.1** Footings and pile caps

Chapter 4. Prestressed concrete - **4.1** Principles **4.2** Losses **4.3** Design of simple members

Project work: Design of a multi-storey RC building

Assessment: Project 40%; final 60%

Recommended textbooks

- Wight, Reinforced Concrete: Mechanics and Design (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 20030162 Concrete Structures (2). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S062 Structural Dynamics and Earthquake Engineering

Used in: Civil Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. SDOF systems - **1.1** Free and forced vibration **1.2** Damping

Chapter 2. Earthquake response - **2.1** Response spectra

Chapter 3. MDOF systems - **3.1** Modal analysis

Chapter 4. Seismic design - **4.1** Code provisions **4.2** Ductile detailing **4.3** Base isolation introduction

Computer practice: Response-spectrum analysis of a frame

Assessment: Assignments 30%; final 70%

Recommended textbooks

- Chopra, Dynamics of Structures (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.510 Introduction to Seismology (OCW, Spring 2010).** Syllabus: Elastic wave propagation; seismic ray theory; interpretation of travel times; finite-frequency effects; surface-wave dispersion; seismic tomography. Link: https://ocw.mit.edu/courses/12-510-introduction-to-seismology-spring-2010/pages/syllabus/
- **Tsinghua University / China University of Geosciences:** TH 30030601 Earthquake Engineering and Dynamics; TH 30040042 Structural Dynamics. These universities publish course name, credits, hours and semester only; no subject syllabus

is published online. Source: Tsinghua 2023 training plan PDF.

S063 Rock Mechanics

Used in: Civil Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. Introduction - **1.1** Rock as an engineering material **1.2** Role of rock mechanics in mining and civil engineering

Chapter 2. Intact rock - **2.1** Index properties **2.2** Laboratory strength tests: *UCS, triaxial, Brazilian* **2.3** Strength criteria: *Mohr-Coulomb, Hoek-Brown*

Chapter 3. Discontinuities - **3.1** Joint characterisation **3.2** Shear strength of discontinuities: *Barton-Bandis*

Chapter 4. Rock mass - **4.1** Rock mass classification: *RMR, Q, GSI* **4.2** Rock mass strength and deformability

Chapter 5. In-situ stress - **5.1** Stress measurement methods **5.2** Stresses around openings: *Kirsch solution*

Chapter 6. Design applications - **6.1** Slope stability: *kinematic analysis* **6.2** Underground openings and support

Laboratory practice:

- UCS test
- Point-load test
- Brazilian test
- Tilt and direct shear test
- Joint logging and RQD from core

Assessment: Labs 30%; assignments 20%; final 50%

Recommended textbooks

- **Practical Rock Engineering** - Evert Hoek (Rocscience). *Free PDF from publisher (author's course notes).* PDF/online: https://www.rocscience.com/assets/resources/learning/hoek/Practical-Rock-Engineering-Full-Text.pdf. Table of contents: Appendix A, item A19.
- Hudson and Harrison, Engineering Rock Mechanics (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Goodman, Introduction to Rock Mechanics (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - MNGN321 Introduction to Rock Mechanics (3).** Syllabus: Physical properties of rock; response of rock substance and rock mass to load; elastic analysis; basic design of underground openings and pit slopes. Link: https://facultyhandbook.mines.edu/undergraduate/programs/miningengineering/miningengineering.pdf
- **Tsinghua University / China University of Geosciences:** TH 30040142 Rock Mechanics. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S064 Construction Technology and Management

Used in: Civil Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. Construction methods - **1.1** Earthworks **1.2** Concrete and formwork **1.3** Steel erection

Chapter 2. Equipment - **2.1** Selection and productivity

Chapter 3. Planning and scheduling - **3.1** Work breakdown **3.2** CPM and PERT **3.3** Resource levelling

Chapter 4. Cost and quality - **4.1** Estimating **4.2** Cost control **4.3** Quality management

Chapter 5. Safety - **5.1** Site safety management

Computer practice: Scheduling software exercise

Assessment: Assignments 40%; final 60%

Recommended textbooks

- Halpin, Senior and Lucko, Construction Management (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Nunnally, Construction Methods and Management (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 30910032 Construction Technology, 30910042 Construction Organisation, 30910052 Project Management I. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S065 Road and Bridge Engineering

Used in: Civil Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. Highway engineering - **1.1** Geometric design **1.2** Pavement materials **1.3** Flexible and rigid pavement design

Chapter 2. Bridge engineering - **2.1** Bridge types **2.2** Loads **2.3** Design of short-span bridges **2.4** Bearings and substructures

Assessment: Design assignment 40%; final 60%

Recommended textbooks

- Garber and Hoel, Traffic and Highway Engineering (Cengage). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Barker and Puckett, Design of Highway Bridges (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 40030492 Road Engineering; TH 40030782 Bridge Engineering. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S066 Engineering Economics and Project Feasibility

Used in: Civil Engineering - Sem 6, 2 cr (as 'Engineering Economics'); Tourist District Development - Sem 7, 2 cr (as 'Project Finance and Feasibility Studies')

Versions: Tourist course version 'Project Finance and Feasibility Studies' (2 cr) = Chapters 1, 2 and 5.

Syllabus

Chapter 1. Time value of money - **1.1** Interest and equivalence **1.2** Cash-flow diagrams

Chapter 2. Evaluation methods - **2.1** Present and annual worth **2.2** Rate of return **2.3** Benefit-cost ratio

Chapter 3. Depreciation and taxes - **3.1** Methods **3.2** After-tax cash flow

Chapter 4. Decisions under uncertainty - **4.1** Sensitivity and scenario analysis **4.2** Replacement decisions

Chapter 5. Project finance and feasibility - **5.1** Market and financial feasibility **5.2** Financing structures (debt, equity, PPP) - used by the Tourist course version

Assessment: Assignments 30%; final 70%

Recommended textbooks

- Newnan, Eschenbach and Lavelle, Engineering Economic Analysis (Oxford). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 30030352 Engineering Economics. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S067 BIM and Computer-Aided Structural Design

Used in: Civil Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. BIM concepts - **1.1** BIM levels and standards **1.2** Information management

Chapter 2. Modelling - **2.1** Architectural and structural models **2.2** Clash detection

Chapter 3. Structural design software - **3.1** Analysis and design workflow **3.2** Drawing production

Computer practice: BIM model of a small building with structural analysis export

Assessment: Project 70%; test 30%

Recommended textbooks

- Sacks, Eastman, Lee and Teicholz, BIM Handbook (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 30030323 Civil Engineering CAD; TH 40031062 Intelligent Construction. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S068 Integrated Civil Engineering Design (capstone)

Used in: Civil Engineering - Sem 7, 4 cr

Syllabus

Chapter 1. Project definition - **1.1** Client brief and site data **1.2** Design criteria and codes

Chapter 2. Design - **2.1** Site investigation interpretation **2.2** Structural scheme and analysis **2.3** Foundation design

Chapter 3. Delivery - **3.1** Drawings and specifications **3.2** Cost estimate and construction plan

Project work: Team capstone design of a building or civil structure; design reviews at weeks 4, 8 and 12

Assessment: Design report 50%; drawings 20%; presentation 30%

Recommended textbooks

- National design codes. *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Ching, Building Construction Illustrated (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN469 Engineering Geology Design (3).** Syllabus: Capstone design of realistic engineering-geology/geotechnical projects; field trips; team and individual reports and presentations. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** TH 40031184 Integrated Course Design (Civil). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S069 Slope Stability and Geohazard Mitigation

Used in: Civil Engineering - Sem 7, 3 cr

Syllabus

Chapter 1. Slope failure mechanisms - **1.1** Soil and rock slope failures

Chapter 2. Analysis - **2.1** Limit equilibrium methods **2.2** Kinematic analysis of rock slopes **2.3** Probabilistic analysis

Chapter 3. Geohazards - **3.1** Landslides and debris flows **3.2** Rockfall

Chapter 4. Mitigation - **4.1** Drainage **4.2** Retaining and reinforcement **4.3** Monitoring

Computer practice: Slope stability analysis with software

Assessment: Assignments 40%; final 60%

Recommended textbooks

- **Practical Rock Engineering** - Evert Hoek (Rocscience). *Free PDF from publisher (author's course notes).* PDF/online: https://www.rocscience.com/assets/resources/learning/hoek/Practical-Rock-Engineering-Full-Text.pdf. Table of contents: Appendix A, item A19.
- **The Landslide Handbook - A Guide to Understanding Landslides (USGS Circular 1325)** - L.M. Highland, P. Bobrowsky (2008). *Public domain; free PDF.* PDF/online: https://pubs.usgs.gov/publication/cir1325. Table of contents: Appendix A, item A20.
- Duncan, Wright and Brandon, Soil Strength and Slope Stability (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Wyllie, Rock Slope Engineering (CRC). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN468 Engineering Geology and Geotechnics (4).** Syllabus: Geology in the evaluation of dams, waterways, tunnels, highways, bridges, buildings, mine design and waste facilities; design projects with field, lab and computer analysis. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Colorado School of Mines - GEGN469 Engineering Geology Design (3).** Syllabus: Capstone design of realistic engineering-geology/geotechnical projects; field trips; team and individual reports and presentations. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** TH 40040782 Engineering Disaster Science. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S070 Building Services and Green Building

Used in: Civil Engineering - Sem 7, 2 cr

Syllabus

Chapter 1. Building services - **1.1** Water supply and drainage **1.2** HVAC **1.3** Electrical services

Chapter 2. Sustainability - **2.1** Energy efficiency **2.2** Green-building rating systems **2.3** Life-cycle assessment

Assessment: Assignment 40%; final 60%

Recommended textbooks

- Kibert, Sustainable Construction: Green Building Design and Delivery (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 00990021 Green Building and Sustainable Development. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S071 Construction Contracts, Codes and Law

Used in: Civil Engineering - Sem 7, 2 cr

Syllabus

Chapter 1. Contracts - **1.1** Contract types **1.2** FIDIC-type conditions **1.3** Tendering

Chapter 2. Administration - **2.1** Variations and claims **2.2** Dispute resolution

Chapter 3. Regulation - **3.1** Building regulations **3.2** Professional liability and ethics

Assessment: Case studies 40%; final 60%

Recommended textbooks

- Fisk and Reynolds, Construction Project Administration (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 30030482 Engineering Contract Management; TH 40031141 Construction HSE Management. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S072 Tunnelling and Underground Space Engineering

Used in: Civil Engineering - Sem 7, 3 cr

Syllabus

Chapter 1. Underground space - **1.1** Planning and uses

Chapter 2. Ground investigation - **2.1** Investigation for tunnels

Chapter 3. Excavation methods - **3.1** Drill-and-blast **3.2** TBM **3.3** NATM/sequential excavation

Chapter 4. Support - **4.1** Rock bolts, shotcrete, segments **4.2** Support design

Chapter 5. Monitoring - **5.1** Convergence and settlement monitoring

Assessment: Design assignment 40%; final 60%

Recommended textbooks

- **Practical Rock Engineering** - Evert Hoek (Rocscience). *Free PDF from publisher (author's course notes).* PDF/online: https://www.rocscience.com/assets/resources/learning/hoek/Practical-Rock-Engineering-Full-Text.pdf. Table of contents: Appendix A, item A19.
- Chapman, Metje and Stark, Introduction to Tunnel Construction (CRC). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Hoek and Brown, Underground Excavations in Rock (CRC). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 40030152 Underground Structures, 40040042 Underground Cavern Eng.; CUG-W Urban Underground Space Eng. programme. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Wuhan) programme list.

S073 Fundamentals of Geophysics

Used in: Geophysical Exploration - Sem 4, 3 cr

Syllabus

Chapter 1. The Earth in the solar system - **1.1** Origin and composition of the Earth **1.2** Rotation, tides and the figure of the Earth

Chapter 2. Gravity and geodesy - **2.1** Gravitational potential, geoid and ellipsoid **2.2** Isostasy

Chapter 3. Geomagnetism - **3.1** Main field, secular variation, reversals **3.2** Palaeomagnetism and rock magnetism

Chapter 4. Seismology and Earth structure - **4.1** Seismic waves and ray paths **4.2** Earthquake location and magnitude **4.3** Earth's internal layering

Chapter 5. Geothermics - **5.1** Heat flow and heat sources **5.2** Thermal structure of the lithosphere

Chapter 6. Geodynamics - **6.1** Plate kinematics **6.2** Mantle convection **6.3** Geochronology overview

Assessment: Problem sets 30%; final 70%

Recommended textbooks

- Lowrie and Fichtner, Fundamentals of Geophysics (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.201 Essentials of Geophysics (OCW, Fall 2004).** Syllabus: The Earth in the solar system; the Earth's gravitational field; the magnetic field of the Earth; seismology; geodynamics. Link: https://ocw.mit.edu/courses/12-201-essentials-of-geophysics-fall-2004/pages/syllabus

S074 Signals, Systems and Digital Signal Processing

Used in: Geophysical Exploration - Sem 4, 3 cr; Space - Sem 4, 3 cr

Syllabus

Chapter 1. Signals and systems - **1.1** Continuous and discrete signals **1.2** Linear time-invariant systems

Chapter 2. Convolution - **2.1** Impulse response **2.2** Convolution sum and integral

Chapter 3. Fourier analysis - **3.1** Fourier series and transform **3.2** Spectra and filtering

Chapter 4. Sampling - **4.1** Sampling theorem **4.2** Aliasing **4.3** Discrete Fourier transform and FFT

Chapter 5. z-transform and digital filters - **5.1** FIR and IIR filters **5.2** Filter design

Chapter 6. Geophysical time series - **6.1** Correlation **6.2** Deconvolution **6.3** Spectral analysis of noise

Computer practice: Python labs on filtering seismic, magnetic and EM time series

Assessment: Labs 40%; final 60%

Recommended textbooks

- **Signals and Systems: Theory and Applications** - Fawwaz Ulaby, Andrew Yagle (Michigan Publishing). *Free PDF for students (Free Electrical Engineering Textbook Initiative).* PDF/online: https://ulabyweb.eecs.umich.edu/ftuc. Table of contents: Appendix A, item A21.
- Oppenheim, Willsky and Nawab, Signals and Systems (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Gubbins, Time Series Analysis and Inverse Theory for Geophysicists (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- MIT 12.012 Statistics, Regression, Signal Processing.

S075 Electromagnetic Field and Wave Theory

Used in: Geophysical Exploration - Sem 4, 3 cr (as 'Electromagnetic Field Theory'); Space - Sem 4, 3 cr (as 'Electromagnetism and Electromagnetic Waves')

Versions: Space course version 'Electromagnetism and EM Waves' (3 cr) emphasises Chapters 5-7 (propagation, antennas).

Syllabus

Chapter 1. Vector analysis - **1.1** Coordinate systems **1.2** Gradient, divergence, curl **1.3** Integral theorems

Chapter 2. Electrostatics - **2.1** Coulomb's and Gauss's laws **2.2** Potential and Laplace/Poisson equations **2.3** Boundary-value problems

Chapter 3. Steady currents - **3.1** Current density, conductivity, boundary conditions: *basis of DC resistivity*

Chapter 4. Magnetostatics - **4.1** Biot-Savart and Ampere's laws **4.2** Magnetic materials and magnetisation

Chapter 5. Time-varying fields - **5.1** Faraday's law **5.2** Maxwell's equations **5.3** Quasi-static (diffusion) regime

Chapter 6. Wave propagation - **6.1** Plane waves in lossless and conductive media **6.2** Skin depth **6.3** Reflection and transmission **6.4** Polarisation

Chapter 7. Radiation and antennas - **7.1** Dipole radiation **7.2** Antenna parameters: *basis for GPR, radar and satellite links*

Assessment: Problem sets 30%; midterm 20%; final 50%

Recommended textbooks

- **Electromagnetics, Volume 1** - Steven W. Ellingson (Virginia Tech Publishing). *CC BY-SA; free PDF.* PDF/online: https://vtechworks.lib.vt.edu/handle/10919/84164. Table of contents: Appendix A, item A22.
- Cheng, Field and Wave Electromagnetics (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Griffiths, Introduction to Electrodynamics (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed (physics-department subject).

S076 Applied Geology for Geophysicists and Drilling Engineers

Used in: Geophysical Exploration - Sem 4, 3 cr (as 'Petrology and Structural Geology for Geophysicists'); Drill Engineering - Sem 4, 2 cr (as 'Geology for Drilling Engineers')

Versions: Used by Geophysical Exploration (3 cr, emphasis on physical properties) and Drill Engineering (2 cr, emphasis on drillability and ground conditions).

Syllabus

Chapter 1. Rock-forming minerals and rocks - **1.1** Igneous, sedimentary and metamorphic rocks **1.2** Weathering and alteration

Chapter 2. Physical properties - **2.1** Density, magnetic susceptibility, resistivity, velocity and drillability of rock types

Chapter 3. Geological structures - **3.1** Folds, faults, joints and foliation **3.2** Stereonets basics **3.3** Structural settings producing anomalies or drilling problems

Chapter 4. Ground conditions - **4.1** Fractured and karstic ground **4.2** Groundwater and hazardous ground **4.3** Core description for engineers

Laboratory practice:

- Rock identification
- Measuring physical properties on samples
- Structural map exercises
- Core logging

Assessment: Labs 40%; final 60%

Recommended textbooks

- Klein and Philpotts, Earth Materials (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Fossen, Structural Geology (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.113 Structural Geology (OCW, Fall 2005).** Syllabus: Mechanics of rock deformation; faults and folds; structures of igneous and metamorphic rocks; superposed deformation; regional structural geology and tectonics. 10 labs (stereonets, Mohr circles, map interpretation, cross-sections). Link: https://ocw.mit.edu/courses/12-113-structural-geology-fall-2005/pages/syllabus/
- **Colorado School of Mines - GEOL308 Introductory Applied Structural Geology (3).** Syllabus: Structural features with emphasis on hydrocarbon traps; plate-tectonic context; structural geometry, mapping, cross-sections and seismic interpretation. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Colorado School of Mines - GEOL310 Earth Materials (3).** Syllabus: Structure, formation, distribution and engineering behaviour of minerals and rocks; stress/strain and rock mechanics; lab and field evaluation of natural materials for mining engineers. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S077 Geological and Geophysical Field Practice (4 weeks)

Used in: Geophysical Exploration - Summer 2, 4 cr

Syllabus

Chapter 1. Geological mapping - **1.1** Traverses and structural data: *2 weeks*

Chapter 2. Geophysical surveys - **2.1** Survey design and grids **2.2** Ground magnetics **2.3** Gravity with base-station loops **2.4** DC resistivity/IP profiling **2.5** Shallow seismic refraction

Chapter 3. Processing and reporting - **3.1** Data reduction **3.2** Joint geological-geophysical interpretation

Field practice: 4 weeks residential camp over a known target

Assessment: Field data quality 30%; maps and models 30%; report 40%

Recommended textbooks

- **Geophysics for Practicing Geoscientists (GPG)** - UBC Geophysical Inversion Facility. *Open web resource (CC BY).* PDF/online: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm. Table of contents: Appendix A, item A14.
- Milsom and Eriksen, Field Geophysics (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN316 Field Geology (5).** Syllabus: Five weeks of field mapping in the Southern Rocky Mountains: igneous, metamorphic and sedimentary terrain; problems in environmental, energy, mining, engineering and structural geology; formal reports. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S078 Gravity and Magnetic Prospecting

Used in: Geophysical Exploration - Sem 5, 4 cr

Syllabus

Chapter 1. Gravity theory - **1.1** Newtonian potential **1.2** Gravity of simple bodies

Chapter 2. Gravity instruments and surveys - **2.1** Relative and absolute gravimeters **2.2** Survey design

Chapter 3. Gravity reduction - **3.1** Drift, latitude, free-air, Bouguer and terrain corrections

Chapter 4. Magnetic theory - **4.1** Magnetisation, remanence, susceptibility **4.2** Fields of dipoles and prisms

Chapter 5. Magnetic surveys - **5.1** Magnetometers **5.2** Diurnal and IGRF corrections

Chapter 6. Data enhancement - **6.1** Upward continuation **6.2** Derivatives **6.3** Reduction to pole **6.4** Analytic signal

Chapter 7. Interpretation - **7.1** Depth estimation: *Euler, Werner* **7.2** 2D/3D forward modelling and inversion

Laboratory practice:

- Gravity reduction exercise
- Magnetic data processing
- 2.5D modelling
- Euler deconvolution

Assessment: Labs 40%; final 60%

Recommended textbooks

- **Geophysics for Practicing Geoscientists (GPG)** - UBC Geophysical Inversion Facility. *Open web resource (CC BY).* PDF/online: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm. Table of contents: Appendix A, item A14.
- Blakely, Potential Theory in Gravity and Magnetic Applications (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Hinze, von Frese and Saad, Gravity and Magnetic Exploration (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.201 Essentials of Geophysics (OCW, Fall 2004).** Syllabus: The Earth in the solar system; the Earth's gravitational field; the magnetic field of the Earth; seismology; geodynamics. Link: https://ocw.mit.edu/courses/12-201-essentials-of-geophysics-fall-2004/pages/syllabus
- **Tsinghua University / China University of Geosciences:** CUG-W Exploration Geophysics programme; CUGB Dept. of Geophysics. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page; CUG (Wuhan) programme list.

S079 Electrical Prospecting (Resistivity, IP, SP)

Used in: Geophysical Exploration - Sem 5, 4 cr

Syllabus

Chapter 1. Electrical properties of rocks - **1.1** Archie's law **1.2** Clay and mineral conduction

Chapter 2. DC resistivity theory - **2.1** Point source in half-space **2.2** Apparent resistivity **2.3** Electrode arrays

Chapter 3. Vertical electrical sounding - **3.1** Curve types **3.2** 1D inversion

Chapter 4. Electrical resistivity tomography - **4.1** 2D/3D surveys **4.2** Inversion

Chapter 5. Induced polarisation - **5.1** Time and frequency domain IP **5.2** Chargeability **5.3** Disseminated sulfide exploration

Chapter 6. Self-potential - **6.1** Sources **6.2** Surveys for sulfides and seepage

Laboratory practice:

- VES interpretation
- ERT inversion with open software
- IP pseudosection interpretation

Field practice: One-day ERT/IP survey

Assessment: Labs 30%; field report 10%; final 60%

Recommended textbooks

- **Geophysics for Practicing Geoscientists (GPG)** - UBC Geophysical Inversion Facility. *Open web resource (CC BY).* PDF/online: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm. Table of contents: Appendix A, item A14.
- Telford, Geldart and Sheriff, Applied Geophysics (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Reynolds, An Introduction to Applied and Environmental Geophysics (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Exploration Geophysics Methods (S5); CUG-W Exploration Geophysics programme. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page; CUG (Wuhan) programme list.

S080 Seismic Exploration

Used in: Geophysical Exploration - Sem 5, 4 cr; Petroleum Exploration and Engineering - Sem 5, 3 cr

Syllabus

Chapter 1. Elastic waves - **1.1** Stress-strain **1.2** P, S and surface waves **1.3** Velocities of rocks

Chapter 2. Wave behaviour - **2.1** Reflection, refraction, Snell's law **2.2** Attenuation

Chapter 3. Refraction method - **3.1** Travel-time curves **3.2** Layer interpretation **3.3** Delay-time methods

Chapter 4. Reflection method - **4.1** Common midpoint **4.2** Normal moveout **4.3** Velocity analysis

Chapter 5. Acquisition - **5.1** Sources and receivers **5.2** 2D and 3D survey design

Chapter 6. Interpretation basics - **6.1** Seismic sections and horizons

Laboratory practice:

- Refraction interpretation
- NMO and velocity exercise
- Picking reflections on sections

Assessment: Labs 30%; final 70%

Recommended textbooks

- **Geophysics for Practicing Geoscientists (GPG)** - UBC Geophysical Inversion Facility. *Open web resource (CC BY).* PDF/online: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm. Table of contents: Appendix A, item A14.
- **Seismic Data Analysis: Processing, Inversion, and Interpretation of Seismic Data** - Oz Yilmaz (SEG). *Open access on the SEG Wiki (web, not PDF).* PDF/online: https://wiki.seg.org/wiki/Seismic_Data_Analysis. Table of contents: Appendix A, item A23.
- Sheriff and Geldart, Exploration Seismology (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.510 Introduction to Seismology (OCW, Spring 2010).** Syllabus: Elastic wave propagation; seismic ray theory; interpretation of travel times; finite-frequency effects; surface-wave dispersion; seismic tomography. Link: https://ocw.mit.edu/courses/12-510-introduction-to-seismology-spring-2010/pages/syllabus/
- **Colorado School of Mines - GEGN439 Petroleum Exploration Design (3).** Syllabus: Multidisciplinary design course integrating well and 3D seismic data, seismic interpretation (Petrel) and prospect development in exploration plays; team reports and presentations. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S081 Petrophysics and Rock Physics

Used in: Geophysical Exploration - Sem 5, 2 cr

Syllabus

Chapter 1. Porosity and density - **1.1** Measurement methods

Chapter 2. Elastic properties - **2.1** Velocity-porosity relations **2.2** Gassmann fluid substitution

Chapter 3. Electrical and magnetic properties - **3.1** Archie and shaly-sand **3.2** Susceptibility and remanence

Chapter 4. Thermal and radioactive properties - **4.1** Conductivity **4.2** Gamma-ray emitters

Laboratory practice:

- Density, porosity and susceptibility measurements
- Velocity measurement on core

Assessment: Labs 50%; final 50%

Recommended textbooks

- Schon, Physical Properties of Rocks (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Mavko, Mukerji and Dvorkin, The Rock Physics Handbook (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- MIT 12.202 Flow, Deformation and Fracture; MIT Earth Resources Laboratory research.

S082 Geophysical Inverse Theory

Used in: Geophysical Exploration - Sem 5, 3 cr

Syllabus

Chapter 1. Forward and inverse problems - **1.1** Data, models, misfit

Chapter 2. Linear inversion - **2.1** Least squares **2.2** Damped least squares and regularisation: *Tikhonov*

Chapter 3. Model appraisal - **3.1** Resolution and covariance

Chapter 4. Non-linear inversion - **4.1** Gauss-Newton and gradient methods **4.2** Global methods: *simulated annealing, genetic algorithms*

Chapter 5. Probabilistic inversion - **5.1** Bayesian framework **5.2** Markov chain Monte Carlo

Computer practice: Inverting gravity profile and 1D resistivity data in Python

Assessment: Computer assignments 50%; final 50%

Recommended textbooks

- **Geophysics for Practicing Geoscientists (GPG)** - UBC Geophysical Inversion Facility. *Open web resource (CC BY).* PDF/online: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm. Table of contents: Appendix A, item A14.
- Menke, Geophysical Data Analysis: Discrete Inverse Theory (Academic Press). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Aster, Borchers and Thurber, Parameter Estimation and Inverse Problems (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- MIT 12.012 (part); MIT graduate geophysics.

S083 Geophysical Instrumentation

Used in: Geophysical Exploration - Sem 5, 3 cr

Syllabus

Chapter 1. Measurement principles - **1.1** Sensors and transducers **1.2** Noise and resolution

Chapter 2. Magnetometers - **2.1** Fluxgate, proton, optically pumped

Chapter 3. Gravimeters - **3.1** Spring and superconducting gravimeters

Chapter 4. Electrical and EM instruments - **4.1** Transmitters and receivers **4.2** Electrodes and coils

Chapter 5. Seismic instruments - **5.1** Geophones and seismographs **5.2** Data recorders

Chapter 6. Timing and positioning - **6.1** GNSS synchronisation **6.2** Calibration procedures

Laboratory practice:

- Calibrating a magnetometer
- Geophone response test
- Electrode contact resistance
- Instrument noise measurement

Assessment: Labs 50%; final 50%

Recommended textbooks

- Milsom and Eriksen, Field Geophysics (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Horowitz and Hill, The Art of Electronics (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Dept. of Measurement and Control Technology; CUG-W Mech. Eng. and Electronic Information (intelligent geo-instruments). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page; CUG (Wuhan) programme list.

S084 Electromagnetic Prospecting (TEM, MT, CSAMT, GPR)

Used in: Geophysical Exploration - Sem 6, 4 cr

Syllabus

Chapter 1. EM induction principles - **1.1** Primary and secondary fields **1.2** Diffusion and skin depth

Chapter 2. Frequency-domain EM - **2.1** Slingram and terrain conductivity

Chapter 3. Time-domain EM - **3.1** Loop configurations **3.2** Decay curves **3.3** Conductor modelling

Chapter 4. Airborne EM - **4.1** Systems and processing

Chapter 5. Magnetotellurics and CSAMT - **5.1** Natural sources **5.2** Impedance tensor **5.3** 1D/2D inversion

Chapter 6. Ground-penetrating radar - **6.1** Principles **6.2** Survey and processing

Laboratory practice:

- TEM decay interpretation
- MT data processing and inversion
- GPR processing

Field practice: One-day TEM/GPR survey

Assessment: Labs 30%; field 10%; final 60%

Recommended textbooks

- **Geophysics for Practicing Geoscientists (GPG)** - UBC Geophysical Inversion Facility. *Open web resource (CC BY).* PDF/online: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm. Table of contents: Appendix A, item A14.
- Nabighian (ed.), Electromagnetic Methods in Applied Geophysics (SEG). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Simpson and Bahr, Practical Magnetotellurics (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Exploration Geophysics programme; CUGB Dept. of Geophysics. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page; CUG (Wuhan) programme list.

S085 Seismic Data Processing

Used in: Geophysical Exploration - Sem 6, 4 cr

Syllabus

Chapter 1. Processing overview - **1.1** Data formats: *SEG-Y* **1.2** Processing sequence

Chapter 2. Pre-processing - **2.1** Geometry **2.2** Editing **2.3** Amplitude correction

Chapter 3. Deconvolution - **3.1** Spiking and predictive deconvolution

Chapter 4. Velocity analysis and statics - **4.1** Velocity spectra **4.2** Refraction and residual statics

Chapter 5. Stacking and migration - **5.1** NMO and stack **5.2** Time and depth migration

Chapter 6. Noise and multiples - **6.1** f-k filtering **6.2** Radon transform

Computer practice: Processing a 2D line with open-source software (Madagascar/Seismic Unix)

Assessment: Processing project 60%; final 40%

Recommended textbooks

- **Seismic Data Analysis: Processing, Inversion, and Interpretation of Seismic Data** - Oz Yilmaz (SEG). *Open access on the SEG Wiki (web, not PDF).* PDF/online: https://wiki.seg.org/wiki/Seismic_Data_Analysis. Table of contents: Appendix A, item A23.
- Yilmaz, Seismic Data Analysis (SEG). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- Mines Geophysical Engineering programme; MIT Earth Resources Laboratory.

S086 Well Logging and Formation Evaluation

Used in: Geophysical Exploration - Sem 6, 3 cr; Petroleum Exploration and Engineering - Sem 5, 4 cr

Syllabus

Chapter 1. Borehole environment - **1.1** Invasion and mud filtrate **1.2** Logging operations

Chapter 2. Electrical logs - **2.1** SP **2.2** Resistivity and induction logs

Chapter 3. Nuclear logs - **3.1** Gamma ray and spectral GR **3.2** Density and neutron

Chapter 4. Acoustic and imaging logs - **4.1** Sonic **4.2** Borehole images

Chapter 5. Interpretation - **5.1** Lithology and porosity **5.2** Water saturation: *Archie* **5.3** Cross-plots

Chapter 6. Mineral and engineering applications - **6.1** Logging in mineral and water boreholes

Computer practice: Log interpretation exercises (Python/spreadsheet)

Assessment: Exercises 40%; final 60%

Recommended textbooks

- Ellis and Singer, Well Logging for Earth Scientists (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Rider and Kennedy, The Geological Interpretation of Well Logs (Rider-French). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN438 Petroleum Geology (4).** Syllabus: Source and reservoir rocks, traps, reservoir conditions, origin and accumulation of petroleum, major petroleum provinces, exploration methods; labs in well-log analysis, correlation and mapping. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S087 Geophysical Equipment Design

Used in: Geophysical Exploration - Sem 6, 3 cr

Syllabus

Chapter 1. System design - **1.1** Requirements and specifications

Chapter 2. Sensors and front ends - **2.1** Analogue signal conditioning **2.2** Low-noise design

Chapter 3. Embedded systems - **3.1** Microcontrollers and FPGA **3.2** Real-time acquisition

Chapter 4. Power and environment - **4.1** Field power supplies **4.2** Ruggedisation

Chapter 5. Data and telemetry - **5.1** Logging, wireless links **5.2** Firmware and software

Laboratory practice:

- Design and test an analogue front end
- Embedded data logger

Assessment: Labs 50%; design report 50%

Recommended textbooks

- Fraden, Handbook of Modern Sensors (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Horowitz and Hill, The Art of Electronics (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Dept. of Measurement and Control Technology; CUG-W intelligent geological equipment. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page; CUG (Wuhan) programme list.

S088 Airborne and UAV Geophysics

Used in: Geophysical Exploration - Sem 6, 2 cr

Syllabus

Chapter 1. Platforms - **1.1** Fixed-wing, helicopter and UAV

Chapter 2. Airborne magnetics and radiometrics - **2.1** Compensation **2.2** Levelling **2.3** Gridding

Chapter 3. Airborne EM - **3.1** Systems and processing

Chapter 4. Survey planning - **4.1** Line spacing and flight height **4.2** Regulations and safety

Computer practice: Processing an airborne magnetic dataset

Assessment: Exercise 50%; final 50%

Recommended textbooks

- Reeves, Aeromagnetic Surveys: Principles, Practice and Interpretation (Geosoft). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Dentith and Mudge (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S089 Numerical Modelling in Geophysics

Used in: Geophysical Exploration - Sem 6, 2 cr

Syllabus

Chapter 1. Numerical methods - **1.1** Finite differences **1.2** Finite elements **1.3** Integral equations

Chapter 2. Forward modelling - **2.1** Potential fields **2.2** DC resistivity **2.3** EM diffusion **2.4** Seismic wave propagation

Computer practice: Coding 2D finite-difference forward models

Assessment: Projects 70%; test 30%

Recommended textbooks

- Igel, Computational Seismology: A Practical Introduction (Oxford). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.010 Computational Methods of Scientific Programming (OCW, Fall 2024).** Syllabus: Python, Julia and C/C++; code design; algorithm development and verification; Jupyter; machine learning; cloud and HPC workflows; data analysis and modelling; visualisation; numerical analysis; reproducibility and version control. Link: https://www.ocw.mit.edu/courses/12-010-computational-methods-of-scientific-programming-fall-2024/pages/syllabus

S090 Seismic Interpretation and Prospect Evaluation

Used in: Geophysical Exploration - Sem 7, 3 cr (as 'Seismic Interpretation'); Petroleum Exploration and Engineering - Sem 6, 3 cr (as 'Seismic Interpretation and Petroleum Exploration')

Versions: Petroleum version 'Seismic Interpretation and Petroleum Exploration' (3 cr) is the same subject with petroleum datasets.

Syllabus

Chapter 1. Seismic data and well ties - **1.1** Synthetic seismograms **1.2** Wavelet and polarity

Chapter 2. Structural interpretation - **2.1** Horizons and faults **2.2** Time-depth conversion

Chapter 3. Stratigraphic interpretation - **3.1** Seismic stratigraphy and facies **3.2** Sequence stratigraphy

Chapter 4. Attributes - **4.1** Amplitude and AVO **4.2** Direct hydrocarbon indicators

Chapter 5. Play and prospect evaluation - **5.1** Petroleum system mapping **5.2** Risk and volumetrics

Computer practice: 3D seismic interpretation project on a public dataset

Assessment: Project 60%; final 40%

Recommended textbooks

- **Seismic Data Analysis: Processing, Inversion, and Interpretation of Seismic Data** - Oz Yilmaz (SEG). *Open access on the SEG Wiki (web, not PDF).* PDF/online: https://wiki.seg.org/wiki/Seismic_Data_Analysis. Table of contents: Appendix A, item A23.
- Bacon, Simm and Redshaw, 3-D Seismic Interpretation (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Brown, Interpretation of Three-Dimensional Seismic Data (AAPG/SEG). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN439 Petroleum Exploration Design (3).** Syllabus: Multidisciplinary design course integrating well and 3D seismic data, seismic interpretation (Petrel) and prospect development in exploration plays; team reports and presentations. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S091 Integrated Geophysical Interpretation and Survey Design (capstone)

Used in: Geophysical Exploration - Sem 7, 3 cr

Syllabus

Chapter 1. Brief - **1.1** Exploration or engineering problem definition

Chapter 2. Design - **2.1** Method selection and survey design **2.2** Budget and logistics

Chapter 3. Integration - **3.1** Processing and joint inversion of supplied data **3.2** Geological interpretation

Project work: Team capstone with report and presentation

Assessment: Report 50%; presentation 30%; peer review 20%

Recommended textbooks

- Dentith and Mudge, Geophysics for the Mineral Exploration Geoscientist (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN439 Petroleum Exploration Design (3).** Syllabus: Multidisciplinary design course integrating well and 3D seismic data, seismic interpretation (Petrel) and prospect development in exploration plays; team reports and presentations. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S092 Engineering and Environmental Geophysics

Used in: Geophysical Exploration - Sem 7, 3 cr

Syllabus

Chapter 1. Near-surface methods - **1.1** Shallow seismic **1.2** ERT **1.3** GPR **1.4** EM

Chapter 2. Applications - **2.1** Site investigation **2.2** Groundwater and contamination **2.3** Cavities and karst **2.4** Archaeology and UXO

Field practice: Two-day near-surface survey

Assessment: Field report 40%; final 60%

Recommended textbooks

- **Geophysics for Practicing Geoscientists (GPG)** - UBC Geophysical Inversion Facility. *Open web resource (CC BY).* PDF/online: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm. Table of contents: Appendix A, item A14.
- Reynolds, An Introduction to Applied and Environmental Geophysics (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- MIT 12.214 Essentials of Field Geophysics.

S093 Radiometric and Nuclear Prospecting

Used in: Geophysical Exploration - Sem 7, 2 cr

Syllabus

Chapter 1. Radioactivity - **1.1** Decay series **1.2** Gamma-ray spectra

Chapter 2. Instruments - **2.1** Scintillation spectrometers **2.2** Calibration

Chapter 3. Surveys - **3.1** Ground and airborne surveys **3.2** Data reduction

Chapter 4. Applications - **4.1** Uranium exploration **4.2** Geological mapping **4.3** Environmental monitoring

Chapter 5. Radiation safety - **5.1** Dose and protection

Laboratory practice:

- Spectrometer calibration
- Survey data processing

Assessment: Labs 40%; final 60%

Recommended textbooks

- **Guidelines for Radioelement Mapping Using Gamma Ray Spectrometry Data (IAEA-TECDOC-1363)** - International Atomic Energy Agency (2003). *Free PDF.* PDF/online: https://www-pub.iaea.org/MTCD/Publications/PDF/te_1363_web/PDF/print%20version.pdf. Table of contents: Appendix A, item A24.
- IAEA, Guidelines for Radioelement Mapping Using Gamma Ray Spectrometry Data (IAEA-TECDOC-1363). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S094 Geophysical Equipment Development Project

Used in: Geophysical Exploration - Sem 7, 3 cr

Syllabus

Project work: Team design, build and field test of a prototype instrument or software module; design review, testing and report

Assessment: Prototype 40%; report 40%; demonstration 20%

Recommended textbooks

- Fraden, Handbook of Modern Sensors (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Engineering Research Centre of Rock-Soil Drilling and Equipment (research link). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S095 Analytical Chemistry (Quantitative and Instrumental)

Used in: Geochemical Exploration - Sem 4, 4 cr

Syllabus

Chapter 1. Fundamentals - **1.1** Analytical process **1.2** Units and stoichiometry

Chapter 2. Data evaluation - **2.1** Errors and statistics **2.2** Calibration and detection limits

Chapter 3. Classical methods - **3.1** Gravimetric analysis **3.2** Titrimetry

Chapter 4. Equilibrium in analysis - **4.1** Activity **4.2** Complexation and solubility

Chapter 5. Spectroscopy - **5.1** UV-Vis **5.2** Atomic absorption and emission

Chapter 6. Electrochemistry - **6.1** Potentiometry, ion-selective electrodes

Chapter 7. Chromatography - **7.1** Gas and ion chromatography

Chapter 8. Quality assurance - **8.1** Standard methods **8.2** Reference materials

Laboratory practice:

- Gravimetric sulfate
- Acid-base titration
- Complexometric titration of Ca/Mg
- AAS calibration
- Ion chromatography of water

Assessment: Labs 40%; final 60%

Recommended textbooks

- **Analytical Chemistry 2.1** - David Harvey (DePauw University; LibreTexts). *Free; full-book PDF download.* PDF/online: https://chem.libretexts.org/Bookshelves/Analytical_Chemistry/Analytical_Chemistry_2.1_by_David_Harvey. Table of contents: Appendix A, item A25.
- Harris, Quantitative Chemical Analysis (W.H. Freeman). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Skoog, Holler and Crouch, Principles of Instrumental Analysis (Cengage). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Physical Chemistry C (S3) (related). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S096 Instrumental Geoanalysis

Used in: Geochemical Exploration - Sem 5, 3 cr

Syllabus

Chapter 1. Sample preparation - **1.1** Crushing, splitting, pulverising **1.2** Digestion and fusion

Chapter 2. X-ray methods - **2.1** XRF **2.2** XRD: *for mineral phases*

Chapter 3. Plasma methods - **3.1** ICP-OES **3.2** ICP-MS

Chapter 4. Other techniques - **4.1** AAS **4.2** Fire assay for gold and PGE **4.3** LECO for C and S

Chapter 5. Microanalysis - **5.1** EPMA **5.2** SEM-EDS **5.3** LA-ICP-MS

Chapter 6. Method selection - **6.1** Detection limits and costs

Laboratory practice:

- Sample preparation
- XRF pellet analysis
- Fire assay demonstration
- SEM session

Assessment: Labs 40%; final 60%

Recommended textbooks

- **Analytical Chemistry 2.1** - David Harvey (DePauw University; LibreTexts). *Free; full-book PDF download.* PDF/online: https://chem.libretexts.org/Bookshelves/Analytical_Chemistry/Analytical_Chemistry_2.1_by_David_Harvey. Table of contents: Appendix A, item A25.
- Potts, A Handbook of Silicate Rock Analysis (Blackie). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Reed, Electron Microprobe Analysis and Scanning Electron Microscopy in Geology (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.335 Experimental Atmospheric Chemistry (OCW, Fall 2014).** Syllabus: Atmospheric chemistry of climate change, air pollution and biogeochemical cycles through laboratory work, field sampling of gases, aerosols and clouds, and simple computer models; three labs with reports. Link: https://ocw.mit.edu/courses/12-335-experimental-atmospheric-chemistry-fall-2014/pages/syllabus
- **Colorado School of Mines - GEOL321 Mineralogy and Mineral Characterization (3).** Syllabus: Crystallography, crystal chemistry, structure-property relations, mineral stability; labs in X-ray diffraction, SEM and optical microscopy. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S097 Isotope Geochemistry and Geochronology

Used in: Geochemical Exploration - Sem 5, 3 cr

Syllabus

Chapter 1. Radioactive decay - **1.1** Decay laws **1.2** Isochrons

Chapter 2. Geochronology - **2.1** Rb-Sr, Sm-Nd, U-Pb, Ar-Ar

Chapter 3. Radiogenic isotope tracers - **3.1** Mantle and crustal sources

Chapter 4. Stable isotopes - **4.1** Fractionation **4.2** O, H, C and S isotopes in hydrothermal systems

Chapter 5. Applications - **5.1** Dating ore deposits **5.2** Fluid sources

Assessment: Problem sets 40%; final 60%

Recommended textbooks

- Faure and Mensing, Isotopes: Principles and Applications (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Dickin, Radiogenic Isotope Geology (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- MIT 12.104 (part); MIT graduate geochemistry.

S098 Aqueous Geochemistry and Geochemical Thermodynamics

Used in: Geochemical Exploration - Sem 5, 3 cr

Syllabus

Chapter 1. Thermodynamics of solutions - **1.1** Activity models **1.2** Equilibrium constants

Chapter 2. Speciation - **2.1** Complexation **2.2** Saturation indices

Chapter 3. Carbonate system - **3.1** Alkalinity and pH

Chapter 4. Redox - **4.1** Eh-pH diagrams

Chapter 5. Surfaces - **5.1** Sorption and ion exchange

Chapter 6. Modelling - **6.1** PHREEQC speciation and reaction paths

Computer practice: PHREEQC exercises

Assessment: Exercises 40%; final 60%

Recommended textbooks

- Langmuir, Aqueous Environmental Geochemistry (Prentice Hall). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Anderson, Thermodynamics of Natural Systems (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN330 Geoscientists Thermodynamics (3).** Syllabus: Thermodynamic properties as functions of P, T and composition; heat and volume change of reactions; phase stability by Gibbs energy minimisation; solutions, activities, equilibrium constants; GEM-Selektor software. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S099 Geochemical Data Processing and Multivariate Statistics

Used in: Geochemical Exploration - Sem 6, 3 cr

Syllabus

Chapter 1. Data quality - **1.1** Detection limits and censored data

Chapter 2. Distributions - **2.1** Univariate statistics **2.2** Background and threshold

Chapter 3. Compositional data - **3.1** Closure **3.2** Log-ratio transforms

Chapter 4. Multivariate methods - **4.1** PCA **4.2** Factor and cluster analysis **4.3** Discriminant analysis

Chapter 5. Spatial analysis - **5.1** Geochemical maps **5.2** Interpolation

Computer practice: R/Python workflow on a regional geochemical dataset

Assessment: Project 60%; final 40%

Recommended textbooks

- Reimann, Filzmoser, Garrett and Dutter, Statistical Data Analysis Explained (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Davis, Statistics and Data Analysis in Geology (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN432 Geological Data Management (3).** Syllabus: Probability, distributions, estimation, hypothesis testing, data sequences, sampling, regression, uni- and multivariate statistics; programming with geological case-history data sets. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S100 Biogeochemical and Gas Methods

Used in: Geochemical Exploration - Sem 6, 2 cr

Syllabus

Chapter 1. Biogeochemistry - **1.1** Plant uptake **1.2** Plant and humus sampling

Chapter 2. Geobotany - **2.1** Indicator plants

Chapter 3. Gas methods - **3.1** Soil gas **3.2** Hg and radon surveys

Chapter 4. Partial leaches - **4.1** Selective extraction methods

Assessment: Assignment 40%; final 60%

Recommended textbooks

- Dunn, Biogeochemistry in Mineral Exploration (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- MIT 12.107 Geobiology (related).

S101 Hydrogeochemistry

Used in: Geochemical Exploration - Sem 6, 2 cr

Syllabus

Chapter 1. Water chemistry - **1.1** Major ions **1.2** Water types and diagrams

Chapter 2. Water-rock interaction - **2.1** Weathering reactions **2.2** Mixing

Chapter 3. Hydrogeochemical exploration - **3.1** Sampling of springs and wells

Chapter 4. Water quality - **4.1** Standards and pollution

Laboratory practice: Water analysis and Piper diagrams

Assessment: Lab 40%; final 60%

Recommended textbooks

- **Groundwater** - R. Allen Freeze, John A. Cherry (1979). *Free PDF from The Groundwater Project (preserved book).* PDF/online: https://gw-project.org/books/groundwater/. Table of contents: Appendix A, item A18.
- Appelo and Postma, Geochemistry, Groundwater and Pollution (CRC). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN466/466L Groundwater Engineering (3+1).** Syllabus: Groundwater occurrence and flow; relation to surface water; hydraulic head; aquifer tests; water chemistry and quality; contaminant transport; laboratory and field methods. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S102 Environmental Geochemistry

Used in: Geochemical Exploration - Sem 6, 3 cr

Syllabus

Chapter 1. Potentially toxic elements - **1.1** Sources and behaviour

Chapter 2. Soils and sediments - **2.1** Contamination and baselines

Chapter 3. Risk assessment - **3.1** Bioavailability **3.2** Human and ecological risk

Chapter 4. Remediation - **4.1** Options overview

Assessment: Case study 40%; final 60%

Recommended textbooks

- Siegel, Environmental Geochemistry of Potentially Toxic Metals (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Lottermoser, Mine Wastes (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- MIT 12.104; Mines Environmental Geosciences track.

S103 Organic Geochemistry

Used in: Geochemical Exploration - Sem 6, 2 cr; Petroleum Exploration and Engineering - Sem 5, 2 cr

Syllabus

Chapter 1. Organic matter - **1.1** Production and preservation

Chapter 2. Kerogen - **2.1** Types **2.2** Maturation and hydrocarbon generation

Chapter 3. Biomarkers - **3.1** Source and maturity indicators

Chapter 4. Source-rock evaluation - **4.1** Rock-Eval **4.2** Vitrinite reflectance **4.3** TOC

Laboratory practice: Interpretation of Rock-Eval and biomarker data

Assessment: Lab 40%; final 60%

Recommended textbooks

- Tissot and Welte, Petroleum Formation and Occurrence (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Killops and Killops, Introduction to Organic Geochemistry (Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Geochemistry incl. organic geochemistry (S5). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S104 Exploration Geochemistry II (Anomaly Interpretation and Regional Mapping)

Used in: Geochemical Exploration - Sem 7, 3 cr

Syllabus

Chapter 1. Regional geochemical mapping - **1.1** National and global programmes

Chapter 2. Anomaly interpretation - **2.1** Ranking and follow-up

Chapter 3. Integration - **3.1** Geochemistry with geology and geophysics

Chapter 4. Case histories - **4.1** By deposit type and climate

Assessment: Case-study report 40%; final 60%

Recommended textbooks

- Govett (ed.), Handbook of Exploration Geochemistry (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Carranza, Geochemical Anomaly and Mineral Prospectivity Mapping in GIS (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Exploration Geochemistry (S6). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S105 Geochemical Survey Design (capstone)

Used in: Geochemical Exploration - Sem 7, 3 cr

Syllabus

Project work: Team capstone: objectives, media, sampling plan, QA/QC, budget; interpretation of a supplied dataset; report and presentation

Assessment: Report 50%; presentation 30%; peer review 20%

Recommended textbooks

- Rose, Hawkes and Webb, Geochemistry in Mineral Exploration (Academic Press). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN403 Mineral Exploration Design (3).** Syllabus: Capstone: commodity and target selection, genetic and exploration models, property acquisition, costs; simulated exploration from planning to discovery, initial reserve calculation and preliminary economic evaluation. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Mineral Exploration Course Design (S6). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S106 QA/QC and Laboratory Management

Used in: Geochemical Exploration - Sem 7, 2 cr

Syllabus

Chapter 1. Quality concepts - **1.1** Accuracy and precision

Chapter 2. QA/QC programme - **2.1** Certified reference materials **2.2** Blanks **2.3** Duplicates

Chapter 3. Control charts - **3.1** Monitoring and failures

Chapter 4. Laboratory management - **4.1** Accreditation: *ISO/IEC 17025* **4.2** Data management

Computer practice: QA/QC analysis of an assay dataset

Assessment: Exercise 50%; final 50%

Recommended textbooks

- **Analytical Chemistry 2.1** - David Harvey (DePauw University; LibreTexts). *Free; full-book PDF download.* PDF/online: https://chem.libretexts.org/Bookshelves/Analytical_Chemistry/Analytical_Chemistry_2.1_by_David_Harvey. Table of contents: Appendix A, item A25.
- Abzalov, Applied Mining Geology (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S107 Rock Mechanics and Rock Breaking

Used in: Drill Engineering - Sem 4, 3 cr

Syllabus

Chapter 1. Rock properties for drilling - **1.1** Strength and hardness: *UCS, Shore, Cerchar abrasivity* **1.2** Drillability indices: *DRI, bit wear index*

Chapter 2. Fracture mechanics - **2.1** Crack initiation and propagation: *Griffith theory* **2.2** Fracture toughness: *mode I tests*

Chapter 3. Breakage mechanisms - **3.1** Indentation and crushing: *crater formation* **3.2** Cutting and scraping: *PDC cutter-rock interaction* **3.3** Percussive breakage: *stress waves, energy transfer*

Chapter 4. Drilling and boring - **4.1** Rock-bit interaction: *specific energy* **4.2** Rate of penetration models: *Bingham, Warren*

Chapter 5. Blasting basics - **5.1** Explosives and detonation **5.2** Blast design for excavation

Laboratory practice:

- Cerchar abrasivity and Schmidt hammer tests
- Indentation test on rock cores
- Drillability test with a laboratory drill

Assessment: Labs 30%; final 70%

Recommended textbooks

- **Practical Rock Engineering** - Evert Hoek (Rocscience). *Free PDF from publisher (author's course notes).* PDF/online: https://www.rocscience.com/assets/resources/learning/hoek/Practical-Rock-Engineering-Full-Text.pdf. Table of contents: Appendix A, item A19.
- Jaeger, Cook and Zimmerman, Fundamentals of Rock Mechanics (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Hudson and Harrison, Engineering Rock Mechanics (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - MNGN321 Introduction to Rock Mechanics (3).** Syllabus: Physical properties of rock; response of rock substance and rock mass to load; elastic analysis; basic design of underground openings and pit slopes. Link: https://facultyhandbook.mines.edu/undergraduate/programs/miningengineering/miningengineering.pdf
- **Colorado School of Mines - MNGN407 Rock Fragmentation (3).** Syllabus: Rock drilling and boring; explosives and blasting; mechanical rock breakage; blast round design for surface and underground excavation. Link: https://facultyhandbook.mines.edu/undergraduate/programs/miningengineering/miningengineering.pdf
- **Colorado School of Mines - MNGN333 Explosives Engineering I (3).** Syllabus: Explosives terminology and fundamental explosives science and engineering; guest lectures from government and industry. Link: https://facultyhandbook.mines.edu/undergraduate/programs/miningengineering/miningengineering.pdf
- **Tsinghua University / China University of Geosciences:** TH 30040142 Rock Mechanics. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S108 Machine Elements and Mechanical Design

Used in: Drill Engineering - Sem 4, 4 cr

Syllabus

Chapter 1. Design process - **1.1** Requirements, codes and safety factors **1.2** Materials selection

Chapter 2. Stress and failure - **2.1** Static failure theories: *von Mises, Tresca* **2.2** Fatigue: *S-N curves, Goodman diagram*

Chapter 3. Shafts and couplings - **3.1** Shaft design for torsion and bending **3.2** Keys and couplings

Chapter 4. Bearings - **4.1** Rolling-element bearing selection and life **4.2** Journal bearings

Chapter 5. Gears - **5.1** Spur and helical gears **5.2** Gear trains and gearboxes

Chapter 6. Fasteners and springs - **6.1** Threaded fasteners and preload **6.2** Helical springs

Chapter 7. Clutches and brakes - **7.1** Friction devices for hoists

Chapter 8. Drill-rig components - **8.1** Rotary head, mast and rod-handling design examples

Project work: Design of a drill-rig rotary head gearbox (calculations and drawings)

Assessment: Project 40%; final 60%

Recommended textbooks

- Budynas and Nisbett, Shigley's Mechanical Engineering Design (McGraw-Hill). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Dept. of Mechanical Design and Automation. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S109 Hydraulic and Pneumatic Drives

Used in: Drill Engineering - Sem 4, 3 cr

Syllabus

Chapter 1. Hydraulic fundamentals - **1.1** Pascal's law, hydraulic fluids and viscosity

Chapter 2. Pumps and motors - **2.1** Gear, vane and piston pumps **2.2** Hydraulic motors **2.3** Efficiency

Chapter 3. Actuators - **3.1** Cylinders and cushioning

Chapter 4. Valves and control - **4.1** Directional, pressure and flow control valves **4.2** Proportional and servo valves

Chapter 5. Circuits - **5.1** Basic and regenerative circuits **5.2** Load-sensing systems on rigs

Chapter 6. Pneumatics - **6.1** Compressors: *screw, piston* **6.2** Air treatment **6.3** DTH hammer air requirements

Chapter 7. Maintenance - **7.1** Contamination control **7.2** Troubleshooting

Laboratory practice:

- Hydraulic trainer circuits (4 sessions)
- Compressor performance test

Assessment: Labs 40%; final 60%

Recommended textbooks

- Esposito, Fluid Power with Applications (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W School of Mechanical Engineering and Electronic Information. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S110 Drilling Equipment and Geology Field Practice (4 weeks)

Used in: Drill Engineering - Summer 2, 4 cr

Syllabus

Chapter 1. Geological mapping - **1.1** Mapping and core description: *2 weeks*

Chapter 2. Drill-site practice - **2.1** Rig components and safety induction **2.2** Rig set-up, levelling and operation **2.3** Core handling, recovery and boxes **2.4** Daily drilling reports and consumables

Field practice: 2 weeks geological mapping + 2 weeks at a teaching drill site

Assessment: Logbook 40%; report 60%

Recommended textbooks

- Australian Drilling Industry Training Committee, Drilling: The Manual of Methods, Applications and Management (CRC). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Exploration Technology and Engineering practice. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S111 Drilling Engineering I (Methods)

Used in: Drill Engineering - Sem 5, 4 cr

Syllabus

Chapter 1. Drilling overview - **1.1** Purposes: *exploration, water, geotechnical, oil, geothermal* **1.2** Classification of methods

Chapter 2. Rotary drilling - **2.1** Rotary-table and top-drive rigs **2.2** Tricone and drag bits

Chapter 3. Diamond core drilling - **3.1** Conventional and wireline coring **3.2** Core barrels and bits

Chapter 4. Percussion drilling - **4.1** Top-hammer **4.2** Down-the-hole hammer **4.3** Cable-tool

Chapter 5. Reverse circulation and air drilling - **5.1** RC dual-tube **5.2** Air and foam drilling

Chapter 6. Auger and sonic drilling - **6.1** Hollow-stem auger **6.2** Sonic drilling for unconsolidated ground

Chapter 7. Drilling parameters - **7.1** Weight on bit, rotary speed, flushing **7.2** Rate of penetration and specific energy

Chapter 8. Method selection - **8.1** Ground, depth, sample quality and cost criteria

Laboratory practice:

- Drilling simulator sessions (4)
- Method-selection case exercises

Assessment: Labs 30%; final 70%

Recommended textbooks

- Australian Drilling Industry Training Committee, Drilling: The Manual of Methods, Applications and Management (CRC). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - MNGN407 Rock Fragmentation (3).** Syllabus: Rock drilling and boring; explosives and blasting; mechanical rock breakage; blast round design for surface and underground excavation. Link: https://facultyhandbook.mines.edu/undergraduate/programs/miningengineering/miningengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUG-W Exploration Technology and Engineering programme. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S112 Drilling Fluids and Hydraulics

Used in: Drill Engineering - Sem 5, 3 cr

Syllabus

Chapter 1. Drilling fluid functions - **1.1** Cooling, cleaning, borehole stability, pressure control

Chapter 2. Fluid types - **2.1** Water-based bentonite and polymer muds **2.2** Oil-based fluids **2.3** Air, mist and foam

Chapter 3. Properties and testing - **3.1** Density, rheology: *Bingham, power law* **3.2** Filtration and filter cake **3.3** Sand content and pH

Chapter 4. Hydraulics - **4.1** Pressure losses in pipe and annulus **4.2** Hole cleaning and cuttings transport **4.3** Pump selection

Chapter 5. Borehole stability - **5.1** Shale inhibition **5.2** Lost circulation materials

Chapter 6. Solids control - **6.1** Shakers, desanders, desilters, centrifuges

Chapter 7. Environmental aspects - **7.1** Fluid disposal and recycling

Laboratory practice:

- Mud balance, Marsh funnel, rotational viscometer, API filter press
- Polymer mud mixing
- Sand content

Assessment: Labs 40%; final 60%

Recommended textbooks

- Caenn, Darley and Gray, Composition and Properties of Drilling and Completion Fluids (Gulf). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Exploration Technology and Engineering programme. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S113 Drilling Machinery and Equipment

Used in: Drill Engineering - Sem 5, 4 cr

Syllabus

Chapter 1. Rig types - **1.1** Exploration, water-well, geotechnical and oil rigs **1.2** Truck, crawler and skid rigs

Chapter 2. Hoisting system - **2.1** Mast and derrick **2.2** Drawworks **2.3** Wire rope selection and inspection

Chapter 3. Rotary system - **3.1** Rotary heads, chucks and tables **3.2** Torque-speed characteristics

Chapter 4. Circulation system - **4.1** Triplex mud pumps **4.2** Air compressors and boosters

Chapter 5. Power systems - **5.1** Diesel engines **5.2** Electric and hydraulic drives

Chapter 6. Drill string - **6.1** Rods, collars, casing **6.2** Thread types and wear

Chapter 7. Maintenance and reliability - **7.1** Preventive maintenance **7.2** Failure analysis

Laboratory practice:

- Rig component disassembly and inspection
- Wire-rope inspection
- Pump performance test

Assessment: Labs 30%; final 70%

Recommended textbooks

- Bourgoyne, Millheim, Chenevert and Young, Applied Drilling Engineering (SPE). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Mech. Eng. (geological drilling equipment); CUGB Cooperation Base for Deep Drilling Equipment. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page; CUG (Wuhan) programme list.

S114 Drill Bits and Diamond Tools

Used in: Drill Engineering - Sem 5, 3 cr

Syllabus

Chapter 1. Bit types - **1.1** Roller-cone: *milled tooth, insert* **1.2** PDC **1.3** Surface-set and impregnated diamond bits **1.4** Percussive button bits

Chapter 2. Superhard materials - **2.1** Natural and synthetic diamond **2.2** PCD and PDC cutters

Chapter 3. Powder metallurgy of matrices - **3.1** Matrix powders **3.2** Hot pressing and sintering

Chapter 4. Bit design - **4.1** Waterways, profile, cutter layout

Chapter 5. Bit selection - **5.1** Matching bit to rock properties

Chapter 6. Wear and dull grading - **6.1** IADC dull grading **6.2** Bit performance records

Laboratory practice:

- Dull grading of used bits
- Matrix sample preparation demonstration
- Bit performance analysis

Assessment: Labs 30%; final 70%

Recommended textbooks

- Konstanty, Powder Metallurgy Diamond Tools (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Heinz, Diamond Drill Handbook (Smith-Heinz). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Exploration Technology and Engineering programme. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S115 Engineering Thermodynamics and Heat Transfer

Used in: Drill Engineering - Sem 5, 2 cr; Petroleum Exploration and Engineering - Sem 4, 3 cr

Syllabus

Chapter 1. Basic concepts - **1.1** Systems, properties, states, processes

Chapter 2. Properties of pure substances - **2.1** Phase diagrams and property tables **2.2** Ideal and real gases

Chapter 3. First law - **3.1** Closed systems: *work, heat* **3.2** Control volumes: *nozzles, turbines, compressors, pumps*

Chapter 4. Second law and entropy - **4.1** Heat engines and refrigerators **4.2** Entropy and isentropic efficiency

Chapter 5. Power and compression cycles - **5.1** Diesel cycle **5.2** Compressor work

Chapter 6. Heat transfer - **6.1** Conduction **6.2** Convection **6.3** Radiation

Chapter 7. Applications - **7.1** Downhole temperature **7.2** Reservoir fluid expansion

Assessment: Problem sets 30%; final 70%

Recommended textbooks

- **Introduction to Engineering Thermodynamics** - Claire Yu Yan (Univ. of British Columbia). *CC BY; free (Open Textbook Library).* PDF/online: https://open.umn.edu/opentextbooks/textbooks/1252. Table of contents: Appendix A, item A26.
- Cengel and Boles, Thermodynamics: An Engineering Approach (McGraw-Hill). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN330 Geoscientists Thermodynamics (3).** Syllabus: Thermodynamic properties as functions of P, T and composition; heat and volume change of reactions; phase stability by Gibbs energy minimisation; solutions, activities, equilibrium constants; GEM-Selektor software. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S116 Electrical Drives and Automation

Used in: Drill Engineering - Sem 5, 2 cr

Syllabus

Chapter 1. Electric motors - **1.1** DC, induction and permanent-magnet motors **1.2** Torque-speed curves

Chapter 2. Power electronics and VFDs - **2.1** Rectifiers, inverters **2.2** Variable-frequency drives

Chapter 3. Control - **3.1** Feedback control and PID **3.2** Stability

Chapter 4. Automation - **4.1** PLC programming basics **4.2** Sensors and actuators on rigs

Laboratory practice:

- VFD motor control
- PLC exercise

Assessment: Labs 40%; final 60%

Recommended textbooks

- Bolton, Mechatronics (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W School of Automation. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S117 Drilling Engineering II (Core Drilling for Exploration)

Used in: Drill Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. Wireline coring - **1.1** Wireline system components **1.2** Inner-tube operations

Chapter 2. Core quality - **2.1** Core recovery and RQD **2.2** Orientation of core

Chapter 3. Difficult ground - **3.1** Broken and swelling ground **3.2** Lost circulation and cementing

Chapter 4. Deviation control - **4.1** Causes of deviation **4.2** Downhole survey tools **4.3** Wedging and navigational drilling

Chapter 5. Deep core drilling - **5.1** Deep exploration hole design **5.2** Rod string selection

Field practice: Site visit to an exploration drilling project

Assessment: Report 30%; final 70%

Recommended textbooks

- Heinz, Diamond Drill Handbook (Smith-Heinz). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Marjoribanks, Geological Methods in Mineral Exploration and Mining (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN473 Geological Engineering Site Investigation (3).** Syllabus: Field investigation, testing and monitoring: drilling and sampling, logging, field tests, instrumentation, trench logging, foundation inspection; technical writing; simulated investigations. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Laboratory of Deep Geological Drilling Technology (research link). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S118 Directional and Horizontal Drilling

Used in: Drill Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. Well planning - **1.1** Trajectory types: *J, S, horizontal* **1.2** Survey calculations: *minimum curvature*

Chapter 2. Deflection tools - **2.1** Whipstocks **2.2** Bent-housing mud motors

Chapter 3. Steerable systems - **3.1** Rotary steerable systems

Chapter 4. Surveying - **4.1** Magnetic and gyro tools **4.2** MWD surveys **4.3** Anti-collision

Chapter 5. Horizontal and multilateral wells - **5.1** Design and torque-drag

Computer practice: Well-path planning and torque-drag exercise

Assessment: Exercise 40%; final 60%

Recommended textbooks

- Inglis, Directional Drilling (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Mitchell and Miska, Fundamentals of Drilling Engineering (SPE). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- Mines Petroleum Engineering programme.

S119 Well Design: Casing, Cementing and Completion

Used in: Drill Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. Casing design - **1.1** Loads: *burst, collapse, tension* **1.2** Casing setting depths

Chapter 2. Cementing - **2.1** Cement chemistry and additives **2.2** Primary cementing operations **2.3** Squeeze cementing

Chapter 3. Completion - **3.1** Open-hole and cased completions **3.2** Perforating

Chapter 4. Well integrity - **4.1** Barriers and testing **4.2** Abandonment

Laboratory practice: Cement slurry density, thickening time and strength

Assessment: Labs 30%; final 70%

Recommended textbooks

- Nelson and Guillot (eds), Well Cementing (Schlumberger). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Rabia, Well Engineering and Construction (Entrac). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- Mines Petroleum Engineering programme.

S120 Water Well Drilling and Groundwater

Used in: Drill Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. Hydrogeology for wells - **1.1** Aquifer types **1.2** Well siting: *geology, geophysics*

Chapter 2. Well design - **2.1** Casing and screen selection **2.2** Gravel pack design

Chapter 3. Construction - **3.1** Rotary, cable-tool and DTH drilling for water wells

Chapter 4. Development and testing - **4.1** Well development methods **4.2** Step and constant-rate pumping tests

Chapter 5. Water quality - **5.1** Sampling and protection: *sanitary seals*

Chapter 6. Maintenance - **6.1** Well rehabilitation **6.2** Encrustation and corrosion

Field practice: Visit to a water-well construction site

Assessment: Design assignment 40%; final 60%

Recommended textbooks

- **Groundwater** - R. Allen Freeze, John A. Cherry (1979). *Free PDF from The Groundwater Project (preserved book).* PDF/online: https://gw-project.org/books/groundwater/. Table of contents: Appendix A, item A18.
- Driscoll, Groundwater and Wells (Johnson Screens). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Misstear, Banks and Clark, Water Wells and Boreholes (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN466/466L Groundwater Engineering (3+1).** Syllabus: Groundwater occurrence and flow; relation to surface water; hydraulic head; aquifer tests; water chemistry and quality; contaminant transport; laboratory and field methods. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Colorado School of Mines - GEGN470 Ground-water Engineering Design (3).** Syllabus: Hydrogeology applied to water supply, geotechnical and water-quality problems: design of well fields, drilling programmes and pump tests; engineering reports. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S121 Geotechnical Drilling, In-situ Testing and Piling

Used in: Drill Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. Investigation drilling - **1.1** Borehole methods **1.2** Undisturbed sampling

Chapter 2. In-situ testing - **2.1** SPT **2.2** CPT/CPTu **2.3** Pressuremeter **2.4** Packer (Lugeon) tests

Chapter 3. Piling - **3.1** Bored piles **3.2** Driven piles **3.3** Continuous flight auger

Chapter 4. Ground engineering - **4.1** Anchors and soil nails **4.2** Grouting and jet grouting

Laboratory practice: Interpretation of SPT/CPT data

Assessment: Lab 30%; final 70%

Recommended textbooks

- Clayton, Matthews and Simons, Site Investigation (Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Tomlinson and Woodward, Pile Design and Construction Practice (CRC). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN473 Geological Engineering Site Investigation (3).** Syllabus: Field investigation, testing and monitoring: drilling and sampling, logging, field tests, instrumentation, trench logging, foundation inspection; technical writing; simulated investigations. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** TH 30040162 Foundation Engineering. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S122 Drilling Instrumentation and Monitoring (MWD/LWD)

Used in: Drill Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. Surface monitoring - **1.1** Drilling parameter sensors: *hookload, torque, pressure* **1.2** Data acquisition

Chapter 2. Downhole sensors - **2.1** Inclination, azimuth, gamma, pressure, vibration

Chapter 3. MWD/LWD - **3.1** Tools and mud-pulse/EM telemetry

Chapter 4. Real-time interpretation - **4.1** Drilling optimisation **4.2** Geosteering

Assessment: Exercise 40%; final 60%

Recommended textbooks

- Mitchell and Miska, Fundamentals of Drilling Engineering (SPE). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Ellis and Singer, Well Logging for Earth Scientists (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- Mines Petroleum Engineering programme.

S123 Drilling Project Management and Cost Engineering

Used in: Drill Engineering - Sem 6, 2 cr

Syllabus

Chapter 1. Project planning - **1.1** Scope and work breakdown **1.2** Scheduling

Chapter 2. Contracts and tendering - **2.1** Metre-rate and day-rate contracts

Chapter 3. Cost engineering - **3.1** Cost per metre **3.2** Budget control

Chapter 4. HSE planning - **4.1** Risk assessment **4.2** Emergency response plans

Assessment: Assignment 40%; final 60%

Recommended textbooks

- Halpin, Senior and Lucko, Construction Management (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 30910052 Engineering Project Management I. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S124 Drilling Accidents, Well Control and Safety

Used in: Drill Engineering - Sem 7, 3 cr

Syllabus

Chapter 1. Drilling problems - **1.1** Stuck pipe: *differential, mechanical* **1.2** Lost circulation **1.3** Fishing operations

Chapter 2. Well control - **2.1** Formation pressure and kicks **2.2** Blowout preventers **2.3** Kill methods

Chapter 3. Safety - **3.1** H2S **3.2** Rig safety management systems

Computer practice: Well-control simulator exercises

Assessment: Simulator 40%; final 60%

Recommended textbooks

- Grace, Blowout and Well Control Handbook (Gulf). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- Mines Petroleum Engineering programme.

S125 Drilling Project Design (capstone)

Used in: Drill Engineering - Sem 7, 3 cr

Syllabus

Project work: Team capstone: complete drilling programme for an exploration core hole, water well or geothermal well (rig, bits, fluids, casing, schedule, budget); report and presentation

Assessment: Report 50%; presentation 30%; peer review 20%

Recommended textbooks

- Bourgoyne et al., Applied Drilling Engineering (SPE). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN470 Ground-water Engineering Design (3).** Syllabus: Hydrogeology applied to water supply, geotechnical and water-quality problems: design of well fields, drilling programmes and pump tests; engineering reports. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Colorado School of Mines - GEGN439 Petroleum Exploration Design (3).** Syllabus: Multidisciplinary design course integrating well and 3D seismic data, seismic interpretation (Petrel) and prospect development in exploration plays; team reports and presentations. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S126 Trenchless Technology and Mechanised Tunnelling

Used in: Drill Engineering - Sem 7, 3 cr

Syllabus

Chapter 1. Trenchless methods - **1.1** Horizontal directional drilling **1.2** Microtunnelling **1.3** Pipe jacking and ramming

Chapter 2. Mechanised excavation - **2.1** Raise boring **2.2** Tunnel boring machines

Chapter 3. Design - **3.1** Bore path planning **3.2** Ground risks and frac-out

Assessment: Assignment 40%; final 60%

Recommended textbooks

- Najafi, Trenchless Technology: Planning, Equipment, and Methods (McGraw-Hill). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 40040042 Underground Cavern Engineering; CUG-W Urban Underground Space Eng.. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Wuhan) programme list.

S127 Deep, Scientific and Geothermal Drilling

Used in: Drill Engineering - Sem 7, 2 cr

Syllabus

Chapter 1. Deep and scientific drilling - **1.1** Continental scientific drilling programmes **1.2** Deep-hole challenges

Chapter 2. Geothermal drilling - **2.1** Geothermal well design **2.2** High-temperature bits, fluids and cements **2.3** Lost circulation in geothermal wells

Assessment: Report 40%; final 60%

Recommended textbooks

- **Handbook of Best Practices for Geothermal Drilling** - J. Finger, D. Blankenship (Sandia National Laboratories). *Free PDF (U.S. DOE).* PDF/online: https://www1.eere.energy.gov/geothermal/pdfs/drillinghandbook.pdf. Table of contents: Appendix A, item A27.
- Finger and Blankenship, Handbook of Best Practices for Geothermal Drilling (Sandia National Laboratories). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Laboratory of Deep Geological Drilling Technology. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S128 Drilling Automation and Intelligent Drilling

Used in: Drill Engineering - Sem 7, 2 cr

Syllabus

Chapter 1. Drilling data - **1.1** Acquisition and data quality

Chapter 2. Optimisation - **2.1** ROP and mechanical specific energy optimisation

Chapter 3. Machine learning - **3.1** Models on drilling data: *prediction of ROP, stuck pipe*

Chapter 4. Automation - **4.1** Automated rigs and digital twins

Computer practice: ML project on a public drilling dataset

Assessment: Project 60%; test 40%

Recommended textbooks

- Mitchell and Miska, Fundamentals of Drilling Engineering (SPE). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S129 Systematic Mineralogy

Used in: Mineral Engineering - Sem 4, 3 cr

Syllabus

Chapter 1. Classification - **1.1** Chemical classes of minerals: *Dana/Strunz*

Chapter 2. Silicates - **2.1** Nesosilicates and sorosilicates: *olivine, garnet, epidote* **2.2** Inosilicates: *pyroxenes, amphiboles* **2.3** Phyllosilicates and clay minerals **2.4** Tectosilicates: *quartz, feldspars, zeolites*

Chapter 3. Non-silicates - **3.1** Oxides and hydroxides **3.2** Sulfides and sulfosalts **3.3** Carbonates **3.4** Sulfates and phosphates **3.5** Halides **3.6** Native elements

Chapter 4. Occurrence and uses - **4.1** Economic minerals and industrial applications

Laboratory practice: Hand-specimen identification of about 150 minerals (8 sessions)

Assessment: Lab tests 50%; final 50%

Recommended textbooks

- **Mineralogy** - Dexter Perkins et al. (Univ. of North Dakota; LibreTexts). *CC BY-NC-SA; free web and PDF export.* PDF/online: https://geo.libretexts.org/Bookshelves/Geology/Mineralogy_(Perkins_et_al.). Table of contents: Appendix A, item A11.
- Nesse, Introduction to Mineralogy (Oxford). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Deer, Howie and Zussman, An Introduction to the Rock-Forming Minerals (Mineralogical Society). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEOL321 Mineralogy and Mineral Characterization (3).** Syllabus: Crystallography, crystal chemistry, structure-property relations, mineral stability; labs in X-ray diffraction, SEM and optical microscopy. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Crystallography and Mineralogy (S3). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S130 Physical Chemistry

Used in: Mineral Engineering - Sem 4, 3 cr

Syllabus

Chapter 1. Thermodynamics - **1.1** First and second laws **1.2** Gibbs energy and equilibrium

Chapter 2. Phase equilibria - **2.1** Phase rule **2.2** Binary and ternary diagrams

Chapter 3. Solutions - **3.1** Activity **3.2** Electrolyte solutions

Chapter 4. Kinetics - **4.1** Rate laws **4.2** Arrhenius equation

Chapter 5. Surface chemistry - **5.1** Adsorption isotherms **5.2** Colloids and zeta potential

Laboratory practice:

- Binary phase diagram
- Adsorption isotherm
- Reaction kinetics

Assessment: Labs 30%; final 70%

Recommended textbooks

- Atkins and de Paula, Atkins' Physical Chemistry (Oxford). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Physical Chemistry C (S3). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S131 Mineral Physics and Crystal Chemistry

Used in: Mineral Engineering - Sem 5, 3 cr

Syllabus

Chapter 1. Crystal structures - **1.1** Close packing and coordination polyhedra **1.2** Pauling's rules

Chapter 2. Defects - **2.1** Point and line defects **2.2** Non-stoichiometry

Chapter 3. Solid solution and exsolution - **3.1** Substitution **3.2** Exsolution textures

Chapter 4. Phase transformations - **4.1** Polymorphism **4.2** Order-disorder

Chapter 5. Physical properties - **5.1** Optical, mechanical, thermal and magnetic properties

Assessment: Assignments 30%; final 70%

Recommended textbooks

- Putnis, An Introduction to Mineral Sciences (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEOL321 Mineralogy and Mineral Characterization (3).** Syllabus: Crystallography, crystal chemistry, structure-property relations, mineral stability; labs in X-ray diffraction, SEM and optical microscopy. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S132 Analytical Mineralogy

Used in: Mineral Engineering - Sem 5, 4 cr

Syllabus

Chapter 1. X-ray diffraction - **1.1** Diffraction theory **1.2** Qualitative phase analysis **1.3** Rietveld quantification

Chapter 2. Electron microscopy - **2.1** SEM imaging and EDS **2.2** Electron microprobe: *WDS*

Chapter 3. Spectroscopy - **3.1** Raman **3.2** Infrared: *FTIR*

Chapter 4. Thermal analysis - **4.1** TG-DTA, DSC

Chapter 5. Automated mineralogy - **5.1** MLA/QEMSCAN principles

Laboratory practice:

- XRD phase identification
- SEM-EDS session
- Raman spectra
- TG-DTA of clays

Assessment: Labs 50%; final 50%

Recommended textbooks

- Cullity and Stock, Elements of X-Ray Diffraction (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Reed, Electron Microprobe Analysis and SEM in Geology (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEOL321 Mineralogy and Mineral Characterization (3).** Syllabus: Crystallography, crystal chemistry, structure-property relations, mineral stability; labs in X-ray diffraction, SEM and optical microscopy. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S133 Mineral Processing I (Comminution and Physical Separation)

Used in: Mineral Engineering - Sem 5, 4 cr

Syllabus

Chapter 1. Introduction - **1.1** Ore, concentrate, tailings **1.2** Recovery and grade

Chapter 2. Sampling and mass balance - **2.1** Sampling **2.2** Mass balance calculations

Chapter 3. Comminution - **3.1** Crushers **3.2** Grinding mills **3.3** Energy laws: *Bond*

Chapter 4. Sizing - **4.1** Screening **4.2** Hydrocyclones and classification

Chapter 5. Gravity separation - **5.1** Jigs, spirals, shaking tables, dense-medium separation

Laboratory practice:

- Crushing and grinding
- Sieve analysis

- Bond work index
- Shaking-table separation

Assessment: Labs 40%; final 60%

Recommended textbooks

- Wills and Finch, Wills' Mineral Processing Technology (Butterworth-Heinemann). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - MNGN322 Introduction to Mineral Processing and Laboratory (3).** Syllabus: Crushing, grinding, size classification; magnetic, electrostatic, gravity and flotation concentration; sedimentation, thickening, filtration, drying; tailings disposal. Link: https://facultyhandbook.mines.edu/undergraduate/programs/miningengineering/miningengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUGB Intro to Mining and Mineral Processing (S7). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S134 Materials Science and Engineering

Used in: Mineral Engineering - Sem 5, 3 cr

Syllabus

Chapter 1. Atomic structure and bonding - **1.1** Bonding types

Chapter 2. Crystal structures - **2.1** Metals and ceramics structures

Chapter 3. Defects and diffusion - **3.1** Imperfections **3.2** Fick's laws

Chapter 4. Mechanical properties - **4.1** Stress-strain **4.2** Hardness **4.3** Fracture

Chapter 5. Phase diagrams - **5.1** Binary systems **5.2** Transformations and microstructures

Chapter 6. Material classes - **6.1** Metals, ceramics, polymers, composites

Chapter 7. Functional properties - **7.1** Thermal, electrical, magnetic

Laboratory practice:

- Hardness tests
- Metallography
- Thermal expansion

Assessment: Labs 30%; final 70%

Recommended textbooks

- Callister and Rethwisch, Materials Science and Engineering: An Introduction (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W / CUGB Materials Science and Engineering programmes. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page; CUG (Wuhan) programme list.

S135 Industrial Minerals and Rocks

Used in: Mineral Engineering - Sem 5, 3 cr

Syllabus

Chapter 1. Overview - **1.1** Definitions, markets and specifications

Chapter 2. Clays - **2.1** Kaolin **2.2** Bentonite **2.3** Ball clay

Chapter 3. Carbonates - **3.1** Limestone and dolomite: *lime, cement*

Chapter 4. Silica and feldspar - **4.1** Glass and ceramic raw materials

Chapter 5. Other industrial minerals - **5.1** Phosphate **5.2** Potash **5.3** Gypsum **5.4** Graphite **5.5** Talc **5.6** Barite

Chapter 6. Evaluation - **6.1** Testing for product specifications

Assessment: Assignment 30%; final 70%

Recommended textbooks

- Kogel et al. (eds), Industrial Minerals and Rocks (SME). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Harben and Kuzvart, Industrial Minerals: A Global Geology (IMIL). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Materials Science and Chemistry. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S136 Process Mineralogy

Used in: Mineral Engineering - Sem 5, 3 cr

Syllabus

Chapter 1. Liberation - **1.1** Liberation analysis and locking

Chapter 2. Textures - **2.1** Grade-recovery relations

Chapter 3. Automated mineralogy - **3.1** Application to process diagnosis

Chapter 4. Geometallurgy - **4.1** Ore variability and domain models

Assessment: Exercise 40%; final 60%

Recommended textbooks

- Becker, Wightman and Evans (eds), Process Mineralogy (JKMRC). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN404 Ore Microscopy (3).** Syllabus: Identification of ore minerals by reflected light, micro-hardness and reflectivity; ore textures (magmatic, open-space filling, replacement, exsolution, recrystallisation); research on classic deposits. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S137 Mineral Processing II (Flotation and Separation)

Used in: Mineral Engineering - Sem 6, 4 cr

Syllabus

Chapter 1. Surface chemistry - **1.1** Interfaces **1.2** Zeta potential

Chapter 2. Flotation - **2.1** Collectors, frothers, modifiers **2.2** Flotation cells **2.3** Circuit design

Chapter 3. Magnetic separation - **3.1** Low and high intensity

Chapter 4. Electrostatic separation - **4.1** Principles

Chapter 5. Dewatering - **5.1** Thickening **5.2** Filtration **5.3** Drying

Chapter 6. Tailings - **6.1** Disposal and storage

Laboratory practice:

- Batch flotation tests
- Magnetic separation
- Settling tests

Assessment: Labs 40%; final 60%

Recommended textbooks

- Wills and Finch (Butterworth-Heinemann). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Fuerstenau and Han (eds), Principles of Mineral Processing (SME). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - MNGN422 Flotation (3).** Syllabus: Interfacial phenomena, reagents and zeta potential; flotation circuit design and evaluation; tailings handling. Link: https://facultyhandbook.mines.edu/undergraduate/programs/miningengineering/miningengineering.pdf

S138 Non-metallic Mineral Materials (Ceramics, Glass, Cement)

Used in: Mineral Engineering - Sem 6, 4 cr

Syllabus

Chapter 1. Ceramics - **1.1** Raw materials **1.2** Forming methods **1.3** Drying and firing

Chapter 2. Refractories - **2.1** Types and properties

Chapter 3. Glass - **3.1** Glass chemistry **3.2** Production

Chapter 4. Cement and lime - **4.1** Clinker manufacture **4.2** Hydration **4.3** Lime production

Chapter 5. Quality control - **5.1** Testing standards

Laboratory practice:

- Ceramic tile making and firing
- Cement setting and strength tests

Assessment: Labs 40%; final 60%

Recommended textbooks

- Carter and Norton, Ceramic Materials: Science and Engineering (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Taylor, Cement Chemistry (Thomas Telford). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB School of Materials Science and Technology. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

S139 Rock Mechanics and Dimension Stone

Used in: Mineral Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. Rock properties - **1.1** Strength **1.2** Porosity and water absorption **1.3** Durability

Chapter 2. Dimension stone - **2.1** Quarrying methods **2.2** Processing: *sawing, polishing*

Chapter 3. Stone decay - **3.1** Weathering mechanisms **3.2** Conservation and cleaning

Laboratory practice: Water absorption, freeze-thaw and salt crystallisation tests

Assessment: Labs 30%; final 70%

Recommended textbooks

- **Practical Rock Engineering** - Evert Hoek (Rocscience). *Free PDF from publisher (author's course notes).* PDF/online: https://www.rocscience.com/assets/resources/learning/hoek/Practical-Rock-Engineering-Full-Text.pdf. Table of contents: Appendix A, item A19.
- Siegesmund and Snethlage (eds), Stone in Architecture (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Hudson and Harrison (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - MNGN321 Introduction to Rock Mechanics (3).** Syllabus: Physical properties of rock; response of rock substance and rock mass to load; elastic analysis; basic design of underground openings and pit slopes. Link: https://facultyhandbook.mines.edu/undergraduate/programs/miningengineering/miningengineering.pdf
- **Tsinghua University / China University of Geosciences:** TH 30040142 Rock Mechanics. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S140 Construction Aggregates and Building Stone Technology

Used in: Mineral Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. Resources - **1.1** Sand, gravel and crushed rock deposits

Chapter 2. Processing - **2.1** Crushing and screening plants

Chapter 3. Specifications - **3.1** Grading and shape **3.2** Strength and durability standards

Chapter 4. Alkali-aggregate reaction - **4.1** Mechanisms **4.2** Testing

Laboratory practice:

- Los Angeles abrasion
- Grading
- Flakiness index
- Aggregate crushing value

Assessment: Labs 40%; final 60%

Recommended textbooks

- Smith and Collis (eds), Aggregates: Sand, Gravel and Crushed Rock Aggregates for Construction Purposes (Geological Society). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 40030352 Building Materials. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S141 Gemmology

Used in: Mineral Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. Gem properties - **1.1** Hardness, density, cleavage **1.2** Optical properties

Chapter 2. Instruments - **2.1** Refractometer **2.2** Polariscope **2.3** Spectroscope **2.4** Microscope

Chapter 3. Gem identification - **3.1** Diamonds and grading **3.2** Coloured stones

Chapter 4. Synthetics and treatments - **4.1** Detection methods

Chapter 5. Gem deposits - **5.1** Geology of gem deposits

Laboratory practice: Gem identification (6 sessions)

Assessment: Labs 50%; final 50%

Recommended textbooks

- O'Donoghue (ed.), Gems (Butterworth-Heinemann). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Read, Gemmology (Butterworth-Heinemann). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Gemological Institute; CUGB School of Gemology. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page; CUG (Wuhan) programme list.

S142 Introduction to Extractive Metallurgy

Used in: Mineral Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. Pyrometallurgy - **1.1** Roasting **1.2** Smelting **1.3** Refining

Chapter 2. Hydrometallurgy - **2.1** Leaching **2.2** Solvent extraction **2.3** Ion exchange

Chapter 3. Electrometallurgy - **3.1** Electrowinning **3.2** Electrorefining

Chapter 4. Flowsheets - **4.1** Process selection for Cu, Au, Zn, Al

Laboratory practice: Agitation leaching test

Assessment: Lab 30%; final 70%

Recommended textbooks

- Havlik, Hydrometallurgy: Principles and Applications (Woodhead). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Rosenqvist, Principles of Extractive Metallurgy (Tapir). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- Mines Kroll Institute; Mines Metallurgical and Materials Eng..

S143 Mineral Materials Development Project (capstone)

Used in: Mineral Engineering - Sem 7, 3 cr

Syllabus

Project work: Team capstone: develop a product from a local mineral or rock (characterisation, processing tests, properties, cost); report and presentation

Assessment: Report 50%; presentation 30%; peer review 20%

Recommended textbooks

- Gupta and Yan, Mineral Processing Design and Operations (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S144 Functional Mineral Materials

Used in: Mineral Engineering - Sem 7, 3 cr

Syllabus

Chapter 1. Clays and nanoclays - **1.1** Properties **1.2** Organic modification

Chapter 2. Zeolites - **2.1** Structure **2.2** Ion exchange and catalysis

Chapter 3. Carbon minerals - **3.1** Graphite **3.2** Graphene production

Chapter 4. Other functional minerals - **4.1** Diatomite **4.2** Mineral adsorbents **4.3** Fillers and pigments

Laboratory practice:

- Clay modification
- Adsorption test with zeolite

Assessment: Labs 40%; final 60%

Recommended textbooks

- Bergaya and Lagaly (eds), Handbook of Clay Science (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S145 Testing and Standards for Mineral Products

Used in: Mineral Engineering - Sem 7, 2 cr

Syllabus

Chapter 1. Sampling and testing - **1.1** Sampling plans

Chapter 2. Physical and chemical testing - **2.1** Test methods

Chapter 3. Standards - **3.1** ISO **3.2** ASTM **3.3** EN **3.4** National standards

Chapter 4. Certification - **4.1** Quality management systems

Assessment: Assignment 40%; final 60%

Recommended textbooks

- Relevant ISO / ASTM / national standards. *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S146 Environmental Mineralogy and Tailings Utilisation

Used in: Mineral Engineering - Sem 7, 3 cr

Syllabus

Chapter 1. Mineral-water interactions - **1.1** Dissolution and precipitation **1.2** Surface reactions and sorption on minerals

Chapter 2. Secondary minerals - **2.1** Oxidation of sulfides **2.2** Efflorescent salts and iron oxyhydroxides

Chapter 3. Mine wastes as resources - **3.1** Characterisation of tailings and waste rock **3.2** Re-use in bricks, cement and aggregates **3.3** Recovery of residual metals

Chapter 4. Environmental assessment - **4.1** Leaching tests **4.2** Stability of waste products

Laboratory practice:

- Leaching test on tailings
- Making test bricks from tailings

Assessment: Labs 40%; final 60%

Recommended textbooks

- Vaughan and Wogelius (eds), Environmental Mineralogy (EMU Notes in Mineralogy). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Lottermoser, Mine Wastes (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- MIT 12.104 (related).

S147 Plant Design and Process Economics

Used in: Mineral Engineering - Sem 7, 3 cr

Syllabus

Chapter 1. Flowsheet development - **1.1** Process selection **1.2** Testwork

Chapter 2. Equipment sizing - **2.1** Crushers, mills, flotation cells

Chapter 3. Plant layout - **3.1** General arrangement

Chapter 4. Cost estimation - **4.1** Capital and operating costs

Project work: Design of a small processing plant

Assessment: Project 50%; final 50%

Recommended textbooks

- Mular, Halbe and Barratt (eds), Mineral Processing Plant Design, Practice, and Control (SME). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- Mines EBGN321 Engineering Economics.

S148 Database Systems

Used in: Geological Information - Sem 4, 3 cr

Syllabus

Chapter 1. Database concepts - **1.1** Files versus databases **1.2** DBMS architecture and users

Chapter 2. Data models - **2.1** Hierarchical, network, relational and NoSQL models

Chapter 3. Relational model - **3.1** Tables, keys and integrity constraints

Chapter 4. ER modelling - **4.1** Entities, attributes, relationships **4.2** ER to relational mapping

Chapter 5. Normalisation - **5.1** Functional dependencies **5.2** 1NF to BCNF

Chapter 6. SQL - **6.1** DDL: *create, alter* **6.2** DML: *select, insert, update, delete* **6.3** Joins, subqueries, views

Chapter 7. Transactions and indexing - **7.1** ACID and concurrency **7.2** Indexes and performance

Chapter 8. Application - **8.1** Drill-hole, sample and assay database design

Computer practice: Build and query a drill-hole database in PostgreSQL (6 labs)

Assessment: Labs 40%; final 60%

Recommended textbooks

- **Database Design (2nd ed.)** - Adrienne Watt, Nelson Eng (BCcampus). *CC BY; free PDF.* PDF/online: https://yc-prod-openlibrary-repo.ecampusontario.ca/jspui/bitstream/123456789/247/16/Database-Design-2nd-Edition-1540829576.pdf. Table of contents: Appendix A, item A28.
- Silberschatz, Korth and Sudarshan, Database System Concepts (McGraw-Hill). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 20740063 Database Technology and Applications. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S149 Data Structures and Algorithms

Used in: Geological Information - Sem 4, 3 cr

Syllabus

Chapter 1. Algorithm analysis - **1.1** Big-O notation **1.2** Recursion

Chapter 2. Linear structures - **2.1** Stacks, queues, deques **2.2** Linked lists

Chapter 3. Trees - **3.1** Binary trees and traversal **3.2** Binary search trees **3.3** Heaps and priority queues

Chapter 4. Hashing - **4.1** Hash functions and tables

Chapter 5. Graphs - **5.1** Representations **5.2** BFS and DFS **5.3** Shortest paths: *Dijkstra*

Chapter 6. Sorting and searching - **6.1** Merge, quick, heap sort **6.2** Binary search

Chapter 7. Spatial indexing - **7.1** Quad-trees **7.2** k-d trees **7.3** R-trees

Computer practice: Weekly Python programming labs

Assessment: Labs 40%; final 60%

Recommended textbooks

- Goodrich, Tamassia and Goldwasser, Data Structures and Algorithms in Python (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 20740112 Data Structures and Algorithms. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S150 Surveying and Geodesy

Used in: Geological Information - Sem 4, 3 cr

Syllabus

Chapter 1. Geodesy - **1.1** Figure of the Earth **1.2** Ellipsoids and datums

Chapter 2. Map projections - **2.1** Projection properties **2.2** UTM and national grids

Chapter 3. Heights - **3.1** Geoid, ellipsoidal and orthometric heights **3.2** Levelling

Chapter 4. Total station surveys - **4.1** Angles, distances **4.2** Traverse adjustment

Chapter 5. GNSS surveys - **5.1** Static and RTK methods

Chapter 6. Coordinate transformations - **6.1** Helmert transformation **6.2** Datum conversion

Field practice: Survey practicals (levelling, traverse, GNSS)

Assessment: Practicals 40%; final 60%

Recommended textbooks

- Ghilani and Wolf, Elementary Surveying (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Torge and Muller, Geodesy (De Gruyter). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 30030611 Surveying; CUGB Surveying A (S4); CUG-W Surveying and Mapping Eng.. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page; CUG (Wuhan) programme list.

S151 Cartography and Geological Map Compilation

Used in: Geological Information - Sem 5, 3 cr

Syllabus

Chapter 1. Cartographic principles - **1.1** Map scale **1.2** Generalisation

Chapter 2. Symbolisation - **2.1** Visual variables **2.2** Colour schemes **2.3** Typography

Chapter 3. Thematic mapping - **3.1** Choropleth **3.2** Proportional symbols **3.3** Isolines

Chapter 4. Geological maps - **4.1** Map standards **4.2** Legend and stratigraphic column **4.3** Cross-sections

Chapter 5. Digital publishing - **5.1** Web maps **5.2** Print production

Computer practice: Compile a geological map sheet

Assessment: Map project 60%; final 40%

Recommended textbooks

- **Essentials of Geographic Information Systems** - Jonathan Campbell, Michael Shin. *CC BY-NC-SA; free (Open Textbook Library).* PDF/online: https://open.umn.edu/opentextbooks/textbooks/67. Table of contents: Appendix A, item A16.
- Slocum et al., Thematic Cartography and Geovisualization (CRC). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Lisle, Geological Structures and Maps (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Geography and Information Eng.. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S152 Software Engineering

Used in: Geological Information - Sem 5, 3 cr

Syllabus

Chapter 1. Software process - **1.1** Waterfall, agile and Scrum

Chapter 2. Requirements - **2.1** Elicitation **2.2** Use cases and user stories

Chapter 3. Design - **3.1** Architecture **3.2** Design patterns **3.3** UML

Chapter 4. Implementation - **4.1** Version control: *Git* **4.2** Code review **4.3** Coding standards

Chapter 5. Testing - **5.1** Unit, integration and system tests **5.2** Continuous integration

Chapter 6. Maintenance and documentation - **6.1** Refactoring **6.2** Documentation

Project work: Team software project (geodata tool)

Assessment: Project 60%; final 40%

Recommended textbooks

- Sommerville, Software Engineering (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Software Engineering programme (Mech. Eng. and Electronic Information). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S153 Spatial Databases and Geological Data Models

Used in: Geological Information - Sem 6, 3 cr

Syllabus

Chapter 1. Spatial data types - **1.1** Geometry and geography types

Chapter 2. Spatial SQL - **2.1** PostGIS functions **2.2** Spatial joins

Chapter 3. Spatial indexing - **3.1** GiST and R-tree indexes

Chapter 4. Geological data models - **4.1** GeoSciML **4.2** Drill-hole standards **4.3** Metadata: *ISO 19115*

Chapter 5. Data governance - **5.1** Data management plans **5.2** Quality control

Computer practice: PostGIS labs

Assessment: Labs 50%; final 50%

Recommended textbooks

- **Database Design (2nd ed.)** - Adrienne Watt, Nelson Eng (BCcampus). *CC BY; free PDF.* PDF/online: https://yc-prod-openlibrary-repo.ecampusontario.ca/jspui/bitstream/123456789/247/16/Database-Design-2nd-Edition-1540829576.pdf. Table of contents: Appendix A, item A28.
- Shekhar and Chawla, Spatial Databases: A Tour (Prentice Hall). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Obe and Hsu, PostGIS in Action (Manning). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN432 Geological Data Management (3).** Syllabus: Probability, distributions, estimation, hypothesis testing, data sequences, sampling, regression, uni- and multivariate statistics; programming with geological case-history data sets. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S154 Photogrammetry and UAV Mapping

Used in: Geological Information - Sem 6, 3 cr

Syllabus

Chapter 1. Photogrammetry principles - **1.1** Camera geometry **1.2** Stereoscopy and parallax

Chapter 2. Aerial triangulation - **2.1** Ground control **2.2** Bundle adjustment

Chapter 3. Structure from motion - **3.1** Feature matching **3.2** Dense point clouds

Chapter 4. UAV mapping - **4.1** Mission planning **4.2** Regulations and safety

Chapter 5. Products - **5.1** DEM and DSM **5.2** Orthophotos **5.3** Volume calculations

Field practice: UAV mapping survey of a quarry or outcrop

Assessment: Field project 40%; final 60%

Recommended textbooks

- Wolf, Dewitt and Wilkinson, Elements of Photogrammetry with Applications in GIS (McGraw-Hill). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Surveying and Mapping / Remote Sensing programmes. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S155 GNSS Positioning and Satellite Geodesy

Used in: Geological Information - Sem 6, 2 cr (as 'GNSS Positioning'); Space - Sem 6, 2 cr (as 'Satellite Geodesy and GNSS')

Versions: Space course version 'Satellite Geodesy and GNSS' (2 cr) focuses on Chapters 2 and 6.

Syllabus

Chapter 1. GNSS systems - **1.1** GPS, GLONASS, Galileo, BeiDou

Chapter 2. Orbits and signals - **2.1** Satellite orbits **2.2** Code and carrier signals

Chapter 3. Positioning - **3.1** Code positioning **3.2** Carrier-phase positioning

Chapter 4. Error sources - **4.1** Ionosphere, troposphere **4.2** Multipath

Chapter 5. Methods - **5.1** Static **5.2** RTK **5.3** PPP

Chapter 6. Satellite geodesy - **6.1** Reference frames **6.2** Satellite gravimetry: *GRACE* **6.3** Altimetry

Field practice: GNSS static and RTK survey

Assessment: Field 40%; final 60%

Recommended textbooks

- Hofmann-Wellenhof, Lichtenegger and Wasle, GNSS - Global Navigation Satellite Systems (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

- Seeber, Satellite Geodesy (De Gruyter). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Hofmann-Wellenhof et al., GNSS (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Surveying and Mapping Eng.. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S156 Spatial Analysis and Mineral Prospectivity Mapping

Used in: Geological Information - Sem 6, 3 cr

Syllabus

Chapter 1. Evidence layers - **1.1** Data preparation and proxies

Chapter 2. Knowledge-driven methods - **2.1** Index overlay **2.2** Fuzzy logic

Chapter 3. Data-driven methods - **3.1** Weights of evidence **3.2** Logistic regression **3.3** Random forests

Chapter 4. Validation - **4.1** Success-rate curves **4.2** Uncertainty

Computer practice: Prospectivity mapping project

Assessment: Project 60%; final 40%

Recommended textbooks

- Bonham-Carter, GIS for Geoscientists (Pergamon). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Carranza, Geochemical Anomaly and Mineral Prospectivity Mapping in GIS (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN475 Applications of GIS (3).** Syllabus: Principles of GIS, data structures, DEMs, data input and verification, spatial analysis and modelling, data quality, GIS project methods. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** CUG-W Resource and Environment Big Data Eng.. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S157 Machine Learning for Geoscience

Used in: Geological Information - Sem 6, 3 cr

Syllabus

Chapter 1. ML fundamentals - **1.1** Learning paradigms **1.2** Bias-variance

Chapter 2. Supervised learning - **2.1** Linear and logistic regression **2.2** Decision trees and random forests **2.3** Support vector machines

Chapter 3. Unsupervised learning - **3.1** k-means and hierarchical clustering **3.2** Dimensionality reduction

Chapter 4. Neural networks - **4.1** Multilayer perceptrons **4.2** Convolutional networks

Chapter 5. Model validation - **5.1** Cross-validation **5.2** Metrics

Chapter 6. Geoscience applications - **6.1** Well logs **6.2** Satellite imagery **6.3** Geochemical maps

Computer practice: Python (scikit-learn, PyTorch) labs

Assessment: Labs 40%; project 30%; final 30%

Recommended textbooks

- **Dive into Deep Learning** - Zhang, Lipton, Li, Smola. *Free online book and PDF (arXiv).* PDF/online: https://arxiv.org/abs/2106.11342. Table of contents: Appendix A, item A29.
- Geron, Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow (O'Reilly). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 30030592 AI and Engineering Applications, 40240532 Intro to Machine Learning; CUG-W Data Science and Big Data Technology. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Wuhan) programme list.

S158 WebGIS and Geospatial Application Development

Used in: Geological Information - Sem 6, 2 cr

Syllabus

Chapter 1. Web mapping - **1.1** OGC services: *WMS, WFS, WMTS*

Chapter 2. Web map libraries - **2.1** Leaflet **2.2** OpenLayers

Chapter 3. Servers - **3.1** GeoServer

Chapter 4. Mobile data collection - **4.1** Field apps: *QField*

Chapter 5. APIs - **5.1** REST services

Computer practice: Build a web map application

Assessment: Project 70%; test 30%

Recommended textbooks

- Fu, Web GIS: Principles and Applications (Esri Press). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S159 Geological Information System Design (capstone)

Used in: Geological Information - Sem 7, 3 cr

Syllabus

Project work: Team capstone: design and implement an information system for a mine, survey or geohazard area (database, 3D model, web portal); report and demonstration

Assessment: System 40%; report 40%; demonstration 20%

Recommended textbooks

- Sommerville, Software Engineering (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S160 Big Data and Cloud Computing for Geoscience

Used in: Geological Information - Sem 7, 3 cr

Syllabus

Chapter 1. Big data concepts - **1.1** Volume, velocity, variety

Chapter 2. Distributed processing - **2.1** Hadoop and Spark

Chapter 3. Cloud platforms - **3.1** Cloud storage and computing

Chapter 4. Earth-observation data cubes - **4.1** Data cubes and analysis-ready data

Chapter 5. Parallel computing - **5.1** HPC basics

Computer practice: Cloud processing of satellite time series

Assessment: Project 60%; final 40%

Recommended textbooks

- Kleppmann, Designing Data-Intensive Applications (O'Reilly). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Resource and Environment Big Data Engineering programme; TH 10030022 Data Science. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Wuhan) programme list.

S161 Hyperspectral, Microwave and SAR Remote Sensing

Used in: Geological Information - Sem 7, 3 cr (as 'Hyperspectral and Radar (SAR) Remote Sensing'); Space - Sem 7, 3 cr (as 'Microwave and SAR Remote Sensing')

Versions: Space course version 'Microwave and SAR Remote Sensing' (3 cr) = Chapters 2-4 plus microwave radiometry.

Syllabus

Chapter 1. Imaging spectroscopy - **1.1** Spectral libraries **1.2** Mineral mapping: *SAM, MTMF*

Chapter 2. SAR principles - **2.1** Radar imaging geometry **2.2** Speckle **2.3** SAR sensors

Chapter 3. Polarimetry - **3.1** Polarimetric decomposition

Chapter 4. InSAR - **4.1** Interferometry **4.2** Persistent scatterers **4.3** Ground deformation

Computer practice: Hyperspectral and InSAR labs

Assessment: Labs 40%; final 60%

Recommended textbooks

- **Fundamentals of Remote Sensing (tutorial)** - Canada Centre for Remote Sensing / Natural Resources Canada. *Free web tutorial.* PDF/online: https://natural-resources.canada.ca/science-data/science-research/geomatics/remote-sensing/tutorial-fundamentals-remote-sensing. Table of contents: Appendix A, item A15.
- Elachi and van Zyl, Introduction to the Physics and Techniques of Remote Sensing (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Woodhouse, Introduction to Microwave Remote Sensing (CRC). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Ulaby and Long, Microwave Radar and Radiometric Remote Sensing (Univ. of Michigan Press). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEOL470 Applications of Satellite Remote Sensing (3).** Syllabus: Satellites, sensors and methods; geoscience applications from a systems perspective; guest case studies; independent term project. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S162 Digital Mine and Smart Mining

Used in: Geological Information - Sem 7, 2 cr

Syllabus

Chapter 1. Mine planning software - **1.1** Block models to schedules

Chapter 2. Fleet and sensors - **2.1** Fleet management **2.2** IoT sensors

Chapter 3. Digital twins - **3.1** Concepts and examples

Chapter 4. Automation - **4.1** Autonomous haulage and drilling

Assessment: Assignment 40%; final 60%

Recommended textbooks

- Darling (ed.), SME Mining Engineering Handbook (SME). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S163 Geohazard Monitoring and Early-Warning Systems

Used in: Geological Information - Sem 7, 3 cr

Syllabus

Chapter 1. Geohazards - **1.1** Landslides **1.2** Subsidence **1.3** Debris flows

Chapter 2. Monitoring technologies - **2.1** GNSS **2.2** InSAR **2.3** Extensometers, piezometers, inclinometers

Chapter 3. Early warning - **3.1** Thresholds **3.2** Warning systems and communication

Computer practice: Analysis of monitoring data

Assessment: Assignment 40%; final 60%

Recommended textbooks

- **The Landslide Handbook - A Guide to Understanding Landslides (USGS Circular 1325)** - L.M. Highland, P. Bobrowsky (2008). *Public domain; free PDF.* PDF/online: https://pubs.usgs.gov/publication/cir1325. Table of contents: Appendix A, item A20.
- Highland and Bobrowsky, The Landslide Handbook (USGS Circular 1325). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 40040782 Engineering Disaster Science. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training

plan PDF.

S164 Introduction to Space Science and Astronomy

Used in: Space - Sem 4, 3 cr

Syllabus

Chapter 1. The sky - **1.1** Celestial sphere **1.2** Coordinate systems **1.3** Time

Chapter 2. Solar system - **2.1** Planets **2.2** Small bodies

Chapter 3. Stars - **3.1** Stellar properties **3.2** Stellar evolution

Chapter 4. Galaxies and cosmology - **4.1** Overview

Chapter 5. Telescopes - **5.1** Optics **5.2** Detectors: *CCD*

Chapter 6. Sun-Earth system - **6.1** Solar activity **6.2** Space weather overview

Chapter 7. Space exploration - **7.1** History **7.2** Missions and agencies

Laboratory practice: Telescope observations (4 sessions)

Assessment: Labs 30%; final 70%

Recommended textbooks

- Chaisson and McMillan, Astronomy Today (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- MIT 12.410J Observational Techniques of Optical Astronomy; MIT Astronomy minor.

S165 Atmospheric Physics

Used in: Space - Sem 4, 3 cr

Syllabus

Chapter 1. Atmosphere - **1.1** Composition **1.2** Vertical structure

Chapter 2. Thermodynamics - **2.1** Gas laws **2.2** Adiabatic processes **2.3** Stability

Chapter 3. Water vapour - **3.1** Humidity **3.2** Cloud formation

Chapter 4. Radiation - **4.1** Solar and terrestrial radiation **4.2** Energy balance

Chapter 5. Boundary layer - **5.1** Turbulence **5.2** Diurnal cycle

Assessment: Problem sets 30%; final 70%

Recommended textbooks

- **Practical Meteorology: An Algebra-based Survey of Atmospheric Science** - Roland Stull (Univ. of British Columbia). *CC BY-NC-SA; free whole-book and chapter PDFs.* PDF/online: https://www.eoas.ubc.ca/books/Practical_Meteorology. Table of contents: Appendix A, item A30.
- Wallace and Hobbs, Atmospheric Science: An Introductory Survey (Academic Press). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.003 Atmosphere, Ocean and Climate Dynamics (OCW, Fall 2008).** Syllabus: Physics of the large-scale circulation of the atmosphere and ocean and the processes controlling climate. Three lectures and one recitation per week. Link: https://ocw.mit.edu/courses/12-003-atmosphere-ocean-and-climate-dynamics-fall-2008/pages/syllabus
- **Colorado School of Mines - GEOC407 Atmosphere, Weather and Climate (3).** Syllabus: Origin and evolution of the atmosphere, heat budget, global circulation, climate zones, climate change, ozone, volcanic effects, ENSO, storms. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S166 Observatory and Meteorological Station Practice (4 weeks)

Used in: Space - Summer 2, 4 cr

Syllabus

Chapter 1. Astronomical observation - **1.1** Telescope operation **1.2** Image reduction

Chapter 2. Meteorological observation - **2.1** Surface observations **2.2** Radiosonde launches

Chapter 3. GNSS station - **3.1** Operation and data

Field practice: 4 weeks at an observatory and meteorological station

Assessment: Logbook 40%; report 60%

Recommended textbooks

No textbook: practice-based subject (faculty guidelines, logbook and report templates).

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.307 Weather and Climate Laboratory (OCW, Spring 2009).** Syllabus: Four 3-week projects (mass and wind; fronts; convection; general circulation), each combining real-time atmospheric data analysis with rotating-tank fluid experiments; reports and oral presentations. Link: https://ocw.mit.edu/courses/12-307-weather-and-climate-laboratory-spring-2009/pages/syllabus/

S167 Orbital Mechanics

Used in: Space - Sem 5, 4 cr

Syllabus

Chapter 1. Two-body problem - **1.1** Equations of motion **1.2** Conservation laws

Chapter 2. Orbits - **2.1** Conic sections **2.2** Kepler's equation **2.3** Orbital elements

Chapter 3. Manoeuvres - **3.1** Hohmann transfer **3.2** Plane changes **3.3** Rendezvous

Chapter 4. Perturbations - **4.1** J2 effect **4.2** Atmospheric drag

Chapter 5. Earth-observation orbits - **5.1** Sun-synchronous **5.2** Repeat ground tracks **5.3** Geostationary

Computer practice: Orbit propagation in Python

Assessment: Assignments 30%; final 70%

Recommended textbooks

- **Basics of Space Flight** - NASA Jet Propulsion Laboratory. *Free (U.S. Government work); web tutorial.* PDF/online: https://science.nasa.gov/learn/basics-of-space-flight/participants-guide. Table of contents: Appendix A, item A31.
- Curtis, Orbital Mechanics for Engineering Students (Butterworth-Heinemann). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Vallado, Fundamentals of Astrodynamics and Applications (Microcosm). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed (aeronautics departments).

S168 Atmospheric Dynamics

Used in: Space - Sem 5, 4 cr

Syllabus

Chapter 1. Equations of motion - **1.1** Rotating frame **1.2** Scale analysis

Chapter 2. Balanced flows - **2.1** Geostrophic, gradient wind **2.2** Thermal wind

Chapter 3. Vorticity - **3.1** Circulation and vorticity equation

Chapter 4. Waves - **4.1** Rossby waves **4.2** Gravity waves

Chapter 5. General circulation - **5.1** Hadley cell **5.2** Jet streams

Assessment: Problem sets 30%; final 70%

Recommended textbooks

- **Practical Meteorology: An Algebra-based Survey of Atmospheric Science** - Roland Stull (Univ. of British Columbia). *CC BY-NC-SA; free whole-book and chapter PDFs.* PDF/online: https://www.eoas.ubc.ca/books/Practical_Meteorology. Table of contents: Appendix A, item A30.
- Holton and Hakim, An Introduction to Dynamic Meteorology (Academic Press). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.003 Atmosphere, Ocean and Climate Dynamics (OCW, Fall 2008).** Syllabus: Physics of the large-scale circulation of the atmosphere and ocean and the processes controlling climate. Three lectures and one recitation per week. Link: https://ocw.mit.edu/courses/12-003-atmosphere-ocean-and-climate-dynamics-fall-2008/pages/syllabus
- **Massachusetts Institute of Technology (MIT) - 12.307 Weather and Climate Laboratory (OCW, Spring 2009).** Syllabus: Four 3-week projects (mass and wind; fronts; convection; general circulation), each combining real-time atmospheric data analysis with rotating-tank fluid experiments; reports and oral presentations. Link: https://ocw.mit.edu/courses/12-307-weather-and-climate-laboratory-spring-2009/pages/syllabus/

S169 Physics of Remote Sensing

Used in: Space - Sem 5, 4 cr

Syllabus

Chapter 1. Radiative transfer - **1.1** Radiative transfer equation

Chapter 2. Spectral signatures - **2.1** Surfaces **2.2** Atmospheric gases

Chapter 3. Sensors - **3.1** Optical **3.2** Thermal **3.3** Microwave

Chapter 4. Calibration - **4.1** Radiometric calibration

Chapter 5. Atmospheric correction - **5.1** Methods

Assessment: Assignments 30%; final 70%

Recommended textbooks

- **Fundamentals of Remote Sensing (tutorial)** - Canada Centre for Remote Sensing / Natural Resources Canada. *Free web tutorial.* PDF/online: https://natural-resources.canada.ca/science-data/science-research/geomatics/remote-sensing/tutorial-fundamentals-remote-sensing. Table of contents: Appendix A, item A15.
- Elachi and van Zyl, Introduction to the Physics and Techniques of Remote Sensing (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Petty, A First Course in Atmospheric Radiation (Sundog). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- MIT 12.421 Remote Sensing: Physical Principles and Techniques.

S170 Space Physics (Sun, Magnetosphere, Ionosphere)

Used in: Space - Sem 5, 4 cr

Syllabus

Chapter 1. The Sun - **1.1** Solar interior **1.2** Solar wind

Chapter 2. Magnetosphere - **2.1** Structure and dynamics

Chapter 3. Radiation belts - **3.1** Trapped particles

Chapter 4. Ionosphere - **4.1** Layers **4.2** Effects on radio propagation

Chapter 5. Geomagnetic storms - **5.1** Storms and substorms

Chapter 6. Plasma basics - **6.1** Plasma physics

Assessment: Problem sets 30%; final 70%

Recommended textbooks

- Kivelson and Russell (eds), Introduction to Space Physics (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- MIT Planetary Science programme (related).

S171 Atmospheric Chemistry

Used in: Space - Sem 5, 2 cr

Syllabus

Chapter 1. Ozone - **1.1** Stratospheric chemistry

Chapter 2. Aerosols - **2.1** Sources and properties

Chapter 3. Greenhouse gases - **3.1** Cycles

Chapter 4. Air pollution - **4.1** Tropospheric chemistry

Chapter 5. Satellite measurement - **5.1** Retrievals

Assessment: Assignment 30%; final 70%

Recommended textbooks

- Jacob, Introduction to Atmospheric Chemistry (Princeton). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Seinfeld and Pandis, Atmospheric Chemistry and Physics (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.335 Experimental Atmospheric Chemistry (OCW, Fall 2014).** Syllabus: Atmospheric chemistry of climate change, air pollution and biogeochemical cycles through laboratory work, field sampling of gases, aerosols and clouds, and simple computer models; three labs with reports. Link: https://ocw.mit.edu/courses/12-335-experimental-atmospheric-chemistry-fall-2014/pages/syllabus

S172 Planetary Science

Used in: Space - Sem 5, 2 cr

Syllabus

Chapter 1. Solar system formation - **1.1** Solar nebula

Chapter 2. Planetary interiors - **2.1** Structure

Chapter 3. Planetary surfaces - **3.1** Processes

Chapter 4. Planetary atmospheres - **4.1** Comparison

Chapter 5. Small bodies - **5.1** Asteroids **5.2** Comets

Chapter 6. Exoplanets - **6.1** Detection methods

Assessment: Assignment 30%; final 70%

Recommended textbooks

- de Pater and Lissauer, Planetary Sciences (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEOL410 Planetary Geology (3).** Syllabus: Origin and composition of the solar system; planetary surface and interior processes; solar-system exploration and resources; comparative planetology. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S173 Spacecraft Systems Engineering

Used in: Space - Sem 6, 4 cr

Syllabus

Chapter 1. Systems engineering - **1.1** Mission design **1.2** Requirements

Chapter 2. Power - **2.1** Solar arrays **2.2** Batteries

Chapter 3. Thermal control - **3.1** Passive and active

Chapter 4. Attitude determination and control - **4.1** Sensors and actuators

Chapter 5. Structures - **5.1** Launch loads

Chapter 6. Payloads - **6.1** Earth-observation instruments

Chapter 7. On-board computers - **7.1** Data handling

Assessment: Design assignment 40%; final 60%

Recommended textbooks

- **Basics of Space Flight** - NASA Jet Propulsion Laboratory. *Free (U.S. Government work); web tutorial.* PDF/online: https://science.nasa.gov/learn/basics-of-space-flight/participants-guide. Table of contents: Appendix A, item A31.
- **CubeSat 101: Basic Concepts and Processes for First-Time CubeSat Developers** - NASA CubeSat Launch Initiative (2017). *Free PDF (U.S. Government work).* PDF/online: https://www.nasa.gov/wp-content/uploads/2017/03/nasa_csli_cubesat_101_508.pdf. Table of contents: Appendix A, item A32.
- Fortescue, Swinerd and Stark (eds), Spacecraft Systems Engineering (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Wertz, Everett and Puschell (eds), Space Mission Engineering: The New SMAD (Microcosm). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed (aeronautics departments).

S174 Satellite Communications and Telemetry

Used in: Space - Sem 6, 3 cr

Syllabus

Chapter 1. Link budgets - **1.1** Signal **1.2** Noise

Chapter 2. Modulation and coding - **2.1** Schemes

Chapter 3. Ground stations - **3.1** Antennas **3.2** Tracking

Chapter 4. TT&C - **4.1** Telemetry and command

Assessment: Assignment 40%; final 60%

Recommended textbooks

- **Basics of Space Flight** - NASA Jet Propulsion Laboratory. *Free (U.S. Government work); web tutorial.* PDF/online: https://science.nasa.gov/learn/basics-of-space-flight/participants-guide. Table of contents: Appendix A, item A31.
- Maral, Bousquet and Sun, Satellite Communications Systems (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S175 Weather Analysis and Numerical Weather Prediction

Used in: Space - Sem 6, 3 cr

Syllabus

Chapter 1. Synoptic analysis - **1.1** Weather maps **1.2** Fronts

Chapter 2. Numerical models - **2.1** Equations **2.2** Grids

Chapter 3. Parameterisation - **3.1** Physical processes

Chapter 4. Ensemble forecasting - **4.1** Uncertainty

Chapter 5. Verification - **5.1** Skill scores

Computer practice: Running a simple numerical model

Assessment: Lab 40%; final 60%

Recommended textbooks

- **Practical Meteorology: An Algebra-based Survey of Atmospheric Science** - Roland Stull (Univ. of British Columbia). *CC BY-NC-SA; free whole-book and chapter PDFs.* PDF/online: https://www.eoas.ubc.ca/books/Practical_Meteorology. Table of contents: Appendix A, item A30.
- Warner, Numerical Weather and Climate Prediction (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.307 Weather and Climate Laboratory (OCW, Spring 2009).** Syllabus: Four 3-week projects (mass and wind; fronts; convection; general circulation), each combining real-time atmospheric data analysis with rotating-tank fluid experiments; reports and oral presentations. Link: https://ocw.mit.edu/courses/12-307-weather-and-climate-laboratory-spring-2009/pages/syllabus/

S176 Climate System Science

Used in: Space - Sem 6, 3 cr

Syllabus

Chapter 1. Energy balance - **1.1** Radiation budget **1.2** Greenhouse effect

Chapter 2. Climate system - **2.1** Atmosphere, ocean, land, cryosphere

Chapter 3. Carbon cycle - **3.1** Fluxes and reservoirs

Chapter 4. Climate variability - **4.1** ENSO **4.2** Monsoons

Chapter 5. Climate change - **5.1** Models **5.2** Projections

Assessment: Assignment 30%; final 70%

Recommended textbooks

- **Practical Meteorology: An Algebra-based Survey of Atmospheric Science** - Roland Stull (Univ. of British Columbia). *CC BY-NC-SA; free whole-book and chapter PDFs.* PDF/online: https://www.eoas.ubc.ca/books/Practical_Meteorology. Table of contents: Appendix A, item A30.
- Hartmann, Global Physical Climatology (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.003 Atmosphere, Ocean and Climate Dynamics (OCW, Fall 2008).** Syllabus: Physics of the large-scale circulation of the atmosphere and ocean and the processes controlling climate. Three lectures and one recitation per week. Link: https://ocw.mit.edu/courses/12-003-atmosphere-ocean-and-climate-dynamics-fall-2008/pages/syllabus
- **Tsinghua University / China University of Geosciences:** TH Dept. of Earth System Science. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S177 Satellite Meteorology

Used in: Space - Sem 6, 3 cr

Syllabus

Chapter 1. Meteorological satellites - **1.1** Orbits **1.2** Instruments

Chapter 2. Imagery - **2.1** Visible, infrared, water vapour

Chapter 3. Retrievals - **3.1** Soundings

Chapter 4. Products - **4.1** Precipitation **4.2** Cloud

Computer practice: Satellite imagery analysis

Assessment: Lab 40%; final 60%

Recommended textbooks

- **Practical Meteorology: An Algebra-based Survey of Atmospheric Science** - Roland Stull (Univ. of British Columbia). *CC BY-NC-SA; free whole-book and chapter PDFs.* PDF/online: https://www.eoas.ubc.ca/books/Practical_Meteorology. Table of contents: Appendix A, item A30.
- Kidder and Vonder Haar, Satellite Meteorology: An Introduction (Academic Press). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- MIT 12.421 (related).

S178 Space Mission Design (CubeSat capstone)

Used in: Space - Sem 7, 4 cr

Syllabus

Project work: Team capstone: small-satellite Earth-observation mission (requirements, orbit, payload, subsystems, budget, design reviews)

Assessment: Report 50%; reviews 30%; peer review 20%

Recommended textbooks

- **CubeSat 101: Basic Concepts and Processes for First-Time CubeSat Developers** - NASA CubeSat Launch Initiative (2017). *Free PDF (U.S. Government work).* PDF/online: https://www.nasa.gov/wp-content/uploads/2017/03/nasa_csli_cubesat_101_508.pdf. Table of contents: Appendix A, item A32.
- NASA, CubeSat 101: Basic Concepts and Processes for First-Time CubeSat Developers. *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Wertz et al., The New SMAD (Microcosm). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH Dept. of Earth System Science satellite (launched 2024) - research link. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S179 Planetary Geology and Space Resources

Used in: Space - Sem 7, 3 cr

Syllabus

Chapter 1. Planetary surface processes - **1.1** Impact cratering **1.2** Volcanism **1.3** Tectonics

Chapter 2. Lunar geology - **2.1** Moon

Chapter 3. Mars geology - **3.1** Mars

Chapter 4. Space resources - **4.1** In-situ resource utilisation: *ISRU*

Assessment: Assignment 40%; final 60%

Recommended textbooks

- Melosh, Planetary Surface Processes (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Badescu (ed.), Moon: Prospective Energy and Material Resources (Springer). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEOL410 Planetary Geology (3).** Syllabus: Origin and composition of the solar system; planetary surface and interior processes; solar-system exploration and resources; comparative planetology. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S180 Space Weather

Used in: Space - Sem 7, 2 cr

Syllabus

Chapter 1. Solar drivers - **1.1** Flares **1.2** Coronal mass ejections

Chapter 2. Effects - **2.1** Satellites **2.2** GNSS **2.3** Power grids

Chapter 3. Forecasting - **3.1** Models and services

Assessment: Assignment 40%; final 60%

Recommended textbooks

- Moldwin, An Introduction to Space Weather (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S181 Data Assimilation

Used in: Space - Sem 7, 2 cr

Syllabus

Chapter 1. Estimation theory - **1.1** Bayesian estimation

Chapter 2. Methods - **2.1** Optimal interpolation **2.2** 3D-Var **2.3** 4D-Var

Chapter 3. Kalman filters - **3.1** Ensemble Kalman filter

Chapter 4. Satellite data - **4.1** Assimilation of satellite observations

Computer practice: Simple assimilation exercise

Assessment: Lab 40%; final 60%

Recommended textbooks

- Kalnay, Atmospheric Modeling, Data Assimilation and Predictability (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- MIT Atmospheres, Oceans and Climate graduate programme.

S182 Geomorphology and Landscape Geology

Used in: Tourist District Development - Sem 4, 3 cr

Syllabus

Chapter 1. Landforms and processes - **1.1** Geomorphic systems and time scales: *equilibrium, thresholds* **1.2** Weathering and regolith: *physical, chemical, karstic weathering*

Chapter 2. Hillslopes and mass movement - **2.1** Hillslope forms and processes: *creep, landslides, rockfall* **2.2** Slope stability factors

Chapter 3. Fluvial landscapes - **3.1** River channels and floodplains: *channel patterns, terraces* **3.2** Waterfalls and gorges as scenic features

Chapter 4. Karst landscapes - **4.1** Caves, dolines and towers: *show caves, tourism value*

Chapter 5. Glacial, coastal and aeolian landscapes - **5.1** Glacial landforms: *cirques, moraines* **5.2** Coasts: *cliffs, beaches, sea stacks* **5.3** Deserts and dunes

Chapter 6. Volcanic and tectonic landscapes - **6.1** Volcanic landforms: *cones, calderas, lava fields* **6.2** Fault scarps and rift landscapes

Chapter 7. Landscape evaluation - **7.1** Scenic quality and geomorphosites: *assessment methods*

Laboratory practice: Interpreting landforms on maps, air photos and DEMs

Field practice: Two-day geomorphological excursion

Assessment: Labs 30%; field report 20%; final 50%

Recommended textbooks

- Huggett, Fundamentals of Geomorphology (Routledge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Massachusetts Institute of Technology (MIT) - 12.163 Surface Processes and Landscape Evolution (OCW, Fall 2004).** Syllabus: Fluvial, hillslope and glacial process mechanics; weathering, soil formation, runoff, erosion, slope stability, sediment transport; landscape response to climate and tectonics. MATLAB modelling, GIS analysis of DEMs; 5 labs and field trips. Link: https://ocw.mit.edu/courses/12-163-surface-processes-and-landscape-evolution-fall-2004/pages/syllabus/
- **Colorado School of Mines - GEGN203 Engineering Terrain Analysis (3).** Syllabus: Landforms and processes (weathering, hillslopes, drainage, rivers, glaciers); geomorphic data and maps; application to geological engineering problems; fieldwork in Colorado. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S183 Principles of Tourism and Geotourism

Used in: Tourist District Development - Sem 4, 3 cr

Syllabus

Chapter 1. Tourism system - **1.1** Definitions and components: *tourist, generating and destination regions* **1.2** History and trends of tourism

Chapter 2. Demand and behaviour - **2.1** Motivation and segmentation: *push-pull factors* **2.2** Travel decision process

Chapter 3. Supply - **3.1** Transport, accommodation, food and beverage, attractions **3.2** Travel trade and services

Chapter 4. Nature-based tourism - **4.1** Ecotourism and adventure tourism **4.2** Geotourism concepts: *geosites, geoparks, interpretation*

Chapter 5. Impacts - **5.1** Economic, socio-cultural and environmental impacts **5.2** Sustainability principles

Assessment: Case study 30%; test 20%; final 50%

Recommended textbooks

- **Introduction to Tourism and Hospitality in BC** - Morgan Westcott (ed.) (BCcampus). *CC BY; free PDF.* PDF/online: https://opentextbc.ca/introtourism/open/download?type=pdf. Table of contents: Appendix A, item A33.
- Goeldner and Ritchie, Tourism: Principles, Practices, Philosophies (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Dowling and Newsome (eds), Geotourism (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S184 Urban and Regional Planning Principles

Used in: Tourist District Development - Sem 4, 3 cr

Syllabus

Chapter 1. Planning history and theory - **1.1** Evolution of planning: *garden city, modernism, new urbanism* **1.2** Planning theories: *rational, participatory*

Chapter 2. Land-use planning - **2.1** Zoning and development control **2.2** Density and mixed use

Chapter 3. Regional planning - **3.1** Regional development and growth poles **3.2** Rural and resort-region planning

Chapter 4. Infrastructure planning - **4.1** Transport and utilities planning **4.2** Green infrastructure

Chapter 5. Plan making - **5.1** Planning process and surveys: *land use, population* **5.2** Public participation **5.3** Plan documents and regulations

Laboratory practice: Land-use survey and plan drafting exercise

Assessment: Studio exercise 40%; final 60%

Recommended textbooks

- Levy, Contemporary Urban Planning (Routledge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 30020572 Principles of Urban Planning; TH 40030832 Urban Planning and Transportation. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S185 Landscape and Geoheritage Field Survey (4 weeks)

Used in: Tourist District Development - Summer 2, 4 cr

Syllabus

Chapter 1. Surveying camp - **1.1** Control and topographic survey of a candidate site: *2 weeks*

Chapter 2. Geoheritage and tourism inventory - **2.1** Geosite inventory forms **2.2** Landscape character assessment **2.3** Visitor facility and access audit **2.4** Stakeholder interviews

Field practice: 4 weeks residential field school in a candidate tourist district

Assessment: Survey plans 30%; inventory database 30%; report 40%

Recommended textbooks

- Reynard and Brilha (eds), Geoheritage: Assessment, Protection, and Management (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 40030402 Surveying Practice. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S186 Tourism Planning

Used in: Tourist District Development - Sem 5, 4 cr

Syllabus

Chapter 1. Planning approaches - **1.1** Levels of tourism planning: *national, regional, site* **1.2** Integrated and sustainable planning

Chapter 2. Destination analysis - **2.1** Resource inventory and evaluation **2.2** Market and competitor analysis **2.3** Demand forecasting: *trend, gravity models*

Chapter 3. Zoning and land use - **3.1** Tourism zoning and carrying capacity **3.2** Circulation and access

Chapter 4. Product development - **4.1** Attractions and activity planning **4.2** Accommodation and facility needs

Chapter 5. Plan implementation - **5.1** Institutional arrangements **5.2** Phasing and investment programme **5.3** Monitoring indicators

Project work: Tourism plan outline for a real district (groups)

Assessment: Project 40%; final 60%

Recommended textbooks

- Gunn and Var, Tourism Planning: Basics, Concepts, Cases (Routledge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Inskeep, Tourism Planning: An Integrated and Sustainable Development Approach (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S187 Geoheritage Assessment and Geopark Management

Used in: Tourist District Development - Sem 5, 3 cr

Syllabus

Chapter 1. Geodiversity - **1.1** Concept and values: *intrinsic, cultural, aesthetic, economic, scientific*

Chapter 2. Geoheritage inventory - **2.1** Geosite selection criteria **2.2** Inventory methods and databases

Chapter 3. Quantitative assessment - **3.1** Scientific, educational and tourism value methods: *e.g. Brilha method* **3.2** Risk of degradation

Chapter 4. Protection and management - **4.1** Legal protection **4.2** Site management plans

Chapter 5. UNESCO Global Geoparks - **5.1** Criteria and evaluation **5.2** Management structure and networking **5.3** Geoeducation and community involvement

Laboratory practice: Quantitative assessment of 10 geosites

Assessment: Assessment report 40%; final 60%

Recommended textbooks

- **Operational Guidelines for UNESCO Global Geoparks** - UNESCO. *Free PDF.* PDF/online: https://unesco.org.uk/site/assets/files/2129/operational_guidelines_for_unesco_global_geoparks_2015.pdf. Table of contents: Appendix A, item A34.
- Gray, Geodiversity: Valuing and Conserving Abiotic Nature (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Reynard and Brilha (eds), Geoheritage (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W campus Shaw Museum is a national 4A tourist site (practical example). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S188 Landscape Architecture and Site Design

Used in: Tourist District Development - Sem 5, 4 cr

Syllabus

Chapter 1. Landscape design principles - **1.1** Elements and principles of design **1.2** Landscape character

Chapter 2. Site analysis - **2.1** Topography, climate, vegetation, views **2.2** Opportunities and constraints mapping

Chapter 3. Grading and drainage - **3.1** Contours and earthworks **3.2** Surface drainage and erosion control

Chapter 4. Planting design - **4.1** Native species and ecology

Chapter 5. Trails and viewpoints - **5.1** Trail design and construction **5.2** Viewpoint and rest-area design **5.3** Universal access

Design studio: Landscape design of a visitor node (studio, 4 h/week)

Assessment: Studio projects 60%; final 40%

Recommended textbooks

- Booth, Basic Elements of Landscape Architectural Design (Waveland). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Motloch, Introduction to Landscape Design (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 40030862 Civil and Ecological Engineering. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S189 Environmental Impact Assessment

Used in: Tourist District Development - Sem 5, 3 cr

Syllabus

Chapter 1. EIA framework - **1.1** Legal basis and EIA process **1.2** Screening and scoping

Chapter 2. Baseline studies - **2.1** Physical, biological and socio-economic baseline

Chapter 3. Impact prediction - **3.1** Methods: *checklists, matrices, modelling* **3.2** Significance evaluation

Chapter 4. Mitigation and monitoring - **4.1** Mitigation hierarchy **4.2** Environmental management plans

Chapter 5. Participation and review - **5.1** Public consultation **5.2** Review and decision-making

Chapter 6. Strategic environmental assessment - **6.1** SEA of plans and programmes

Project work: EIA scoping report for a tourism or mining project

Assessment: Report 40%; final 60%

Recommended textbooks

- Glasson and Therivel, Introduction to Environmental Impact Assessment (Routledge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- Mines Environmental Geosciences track.

S190 Tourism Infrastructure Engineering

Used in: Tourist District Development - Sem 6, 4 cr

Syllabus

Chapter 1. Access and transport - **1.1** Road geometric design for scenic roads **1.2** Parking and traffic management **1.3** Cable cars and trails: *overview*

Chapter 2. Bridges and boardwalks - **2.1** Footbridges and boardwalk design basics

Chapter 3. Water supply and sanitation - **3.1** Demand estimation **3.2** Small water supply systems **3.3** Wastewater treatment for remote sites

Chapter 4. Energy and telecom - **4.1** Power supply options and renewables **4.2** Telecom

Chapter 5. Solid waste - **5.1** Collection and disposal at tourist sites

Project work: Infrastructure layout for a tourist node

Assessment: Design project 40%; final 60%

Recommended textbooks

- Garber and Hoel, Traffic and Highway Engineering (Cengage). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Mays, Water Resources Engineering (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 40030492 Road Engineering, 40030782 Bridge Engineering. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S191 Tourism Economics and Market Analysis

Used in: Tourist District Development - Sem 6, 3 cr

Syllabus

Chapter 1. Tourism demand - **1.1** Determinants and elasticity **1.2** Demand forecasting

Chapter 2. Tourism supply and markets - **2.1** Market structures **2.2** Pricing

Chapter 3. Economic impact - **3.1** Tourism satellite accounts **3.2** Multipliers and input-output

Chapter 4. Market research - **4.1** Surveys and segmentation **4.2** Visitor statistics

Assessment: Market study 40%; final 60%

Recommended textbooks

- Stabler, Papatheodorou and Sinclair, The Economics of Tourism (Routledge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 40910052 Urban and Real Estate Economics. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S192 Tourist Facility Design

Used in: Tourist District Development - Sem 6, 3 cr

Syllabus

Chapter 1. Facility types - **1.1** Visitor centres **1.2** Accommodation **1.3** Viewing platforms and amenities

Chapter 2. Design standards - **2.1** Space standards: *Neufert data* **2.2** Accessibility and fire codes

Chapter 3. Sustainable building - **3.1** Local materials and passive design

Design studio: Design of a visitor centre

Assessment: Studio project 60%; final 40%

Recommended textbooks

- Neufert, Architects' Data (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Ching, Building Construction Illustrated (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 30030522 Building Architecture. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S193 Geohazards and Visitor Safety

Used in: Tourist District Development - Sem 6, 3 cr

Syllabus

Chapter 1. Geohazards at tourist sites - **1.1** Rockfall and landslides **1.2** Floods and flash floods **1.3** Karst collapse and caves **1.4** Volcanic and coastal hazards

Chapter 2. Risk assessment - **2.1** Hazard mapping **2.2** Risk matrices and acceptable risk

Chapter 3. Safety management - **3.1** Warning signage and closures **3.2** Emergency planning

Field practice: Hazard assessment at a tourist site

Assessment: Field report 40%; final 60%

Recommended textbooks

- **The Landslide Handbook - A Guide to Understanding Landslides (USGS Circular 1325)** - L.M. Highland, P. Bobrowsky (2008). *Public domain; free PDF.* PDF/online: https://pubs.usgs.gov/publication/cir1325. Table of contents: Appendix A, item A20.
- Keller, Environmental Geology (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Highland and Bobrowsky, The Landslide Handbook (USGS). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN468 Engineering Geology and Geotechnics (4).** Syllabus: Geology in the evaluation of dams, waterways, tunnels, highways, bridges, buildings, mine design and waste facilities; design projects with field, lab and computer analysis. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** TH 00030052 Disasters and Countermeasures. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S194 Natural and Cultural Heritage Conservation

Used in: Tourist District Development - Sem 6, 3 cr

Syllabus

Chapter 1. Heritage concepts - **1.1** Natural and cultural heritage **1.2** Values and significance

Chapter 2. World Heritage system - **2.1** Convention, criteria and Operational Guidelines **2.2** Nomination and management

Chapter 3. Conservation planning - **3.1** Management plans **3.2** Buffer zones

Chapter 4. Protected areas - **4.1** Categories and management

Assessment: Case study 40%; final 60%

Recommended textbooks

- UNESCO, Operational Guidelines for the Implementation of the World Heritage Convention. *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S195 Destination Management and Marketing

Used in: Tourist District Development - Sem 6, 2 cr

Syllabus

Chapter 1. Destination management - **1.1** DMOs and governance **1.2** Stakeholder coordination

Chapter 2. Marketing - **2.1** Branding and positioning **2.2** Digital marketing

Chapter 3. Visitor management - **3.1** Flows and capacity **3.2** Quality and satisfaction

Assessment: Marketing plan 50%; final 50%

Recommended textbooks

- **Introduction to Tourism and Hospitality in BC** - Morgan Westcott (ed.) (BCcampus). *CC BY; free PDF.* PDF/online: https://opentextbc.ca/introtourism/open/download?type=pdf. Table of contents: Appendix A, item A33.
- Morrison, Marketing and Managing Tourism Destinations (Routledge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S196 Structures for Leisure Facilities

Used in: Tourist District Development - Sem 6, 2 cr

Syllabus

Chapter 1. Structural concepts - **1.1** Load paths and stability

Chapter 2. Structural types - **2.1** Pavilions and canopies **2.2** Footbridges **2.3** Viewing towers

Chapter 3. Materials - **3.1** Timber, steel, concrete for leisure structures

Assessment: Concept design 50%; final 50%

Recommended textbooks

- Ching, Onouye and Zuberbuhler, Building Structures Illustrated (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 00030112 Structural Aesthetics. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S197 Tourist District Master Plan Studio (capstone)

Used in: Tourist District Development - Sem 7, 5 cr

Syllabus

Project work: Team capstone master plan for a real district: resource inventory, zoning, infrastructure, facilities, phasing, financial and environmental appraisal; presentation to a stakeholder panel

Assessment: Master plan 50%; presentations 30%; peer review 20%

Recommended textbooks

- Gunn and Var, Tourism Planning (Routledge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Inskeep, Tourism Planning (Wiley). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 40031184 Integrated Course Design (analogue). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S198 Sustainable Tourism and Carrying Capacity

Used in: Tourist District Development - Sem 7, 3 cr

Syllabus

Chapter 1. Sustainability principles - **1.1** Sustainable development goals and tourism

Chapter 2. Carrying capacity - **2.1** Physical, ecological, social capacity **2.2** Limits of acceptable change

Chapter 3. Monitoring - **3.1** Indicators and visitor impact monitoring

Chapter 4. Certification - **4.1** Eco-labels and standards

Assessment: Assignment 40%; final 60%

Recommended textbooks

- Weaver, Sustainable Tourism (Routledge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S199 Tourism Law, Policy and Investment

Used in: Tourist District Development - Sem 7, 2 cr

Syllabus

Chapter 1. Tourism legislation - **1.1** Tourism laws and regulations

Chapter 2. Land and investment - **2.1** Land tenure **2.2** Investment incentives **2.3** Public-private partnerships

Chapter 3. Policy - **3.1** Tourism policy process

Assessment: Case study 40%; final 60%

Recommended textbooks

- Hall, Tourism Planning: Policies, Processes and Relationships (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S200 Heritage Interpretation and Visitor-Centre Design

Used in: Tourist District Development - Sem 7, 2 cr

Syllabus

Chapter 1. Interpretation principles - **1.1** Tilden's principles **1.2** Thematic interpretation

Chapter 2. Media - **2.1** Signs and panels **2.2** Trails **2.3** Guided tours

Chapter 3. Visitor centres - **3.1** Exhibition design

Project work: Interpretive plan for a geosite

Assessment: Project 60%; final 40%

Recommended textbooks

- Ham, Interpretation: Making a Difference on Purpose (Fulcrum). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed.

S201 Sedimentary Petrology

Used in: Petroleum Exploration and Engineering - Sem 4, 2 cr

Syllabus

Chapter 1. Siliciclastic rocks - **1.1** Sandstone composition and classification **1.2** Mudrocks

Chapter 2. Carbonate rocks - **2.1** Carbonate grains and classification: *Folk, Dunham*

Chapter 3. Diagenesis - **3.1** Compaction, cementation, dissolution **3.2** Porosity evolution

Chapter 4. Reservoir quality - **4.1** Controls on porosity and permeability

Laboratory practice:

- Sandstone and carbonate thin sections (6 sessions)
- Point counting

Assessment: Labs 50%; final 50%

Recommended textbooks

- Boggs, Petrology of Sedimentary Rocks (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN307 Petrology (4).** Syllabus: Igneous, sedimentary and metamorphic processes; megascopic and microscopic classification and description of rocks; fabric and physical properties. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S202 Petroleum Geology

Used in: Petroleum Exploration and Engineering - Sem 5, 4 cr

Syllabus

Chapter 1. Petroleum - **1.1** Composition of oil and gas **1.2** Physical properties

Chapter 2. Source rocks - **2.1** Organic matter types **2.2** Maturation and generation windows

Chapter 3. Migration - **3.1** Primary and secondary migration **3.2** Accumulation

Chapter 4. Reservoirs - **4.1** Clastic and carbonate reservoir properties

Chapter 5. Seals and traps - **5.1** Seal capacity **5.2** Structural, stratigraphic and combination traps

Chapter 6. Petroleum systems - **6.1** Elements, processes and timing **6.2** Play and prospect concepts

Chapter 7. Exploration methods - **7.1** Seismic, wells and basin analysis

Chapter 8. Petroleum provinces - **8.1** Major basins of the world

Laboratory practice:

- Structure-contour mapping
- Trap identification
- Well-log correlation

Assessment: Labs 30%; final 70%

Recommended textbooks

- Gluyas and Swarbrick, Petroleum Geoscience (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Selley and Sonnenberg, Elements of Petroleum Geology (Academic Press). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN438 Petroleum Geology (4).** Syllabus: Source and reservoir rocks, traps, reservoir conditions, origin and accumulation of petroleum, major petroleum provinces, exploration methods; labs in well-log analysis, correlation and mapping. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- **Tsinghua University / China University of Geosciences:** TH 40040892 Marine Resources and Energy; CUG-W Petroleum Engineering programme. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Wuhan) programme list.

S203 Reservoir Rock and Fluid Properties

Used in: Petroleum Exploration and Engineering - Sem 5, 4 cr

Syllabus

Chapter 1. Porosity - **1.1** Types **1.2** Laboratory measurement

Chapter 2. Permeability - **2.1** Darcy's law **2.2** Klinkenberg effect

Chapter 3. Capillary pressure and wettability - **3.1** Saturation-height functions

Chapter 4. Relative permeability - **4.1** Two-phase flow concepts

Chapter 5. Fluid properties - **5.1** Phase behaviour of hydrocarbons **5.2** Black-oil PVT: *Bo, Rs, viscosity* **5.3** Gas properties: *Z-factor*

Laboratory practice:

- Core porosity and permeability
- PVT data interpretation

Assessment: Labs 40%; final 60%

Recommended textbooks

- Tiab and Donaldson, Petrophysics (Gulf). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- McCain, The Properties of Petroleum Fluids (PennWell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- Mines Petroleum Engineering programme.

S204 Sedimentary Basin Analysis

Used in: Petroleum Exploration and Engineering - Sem 5, 3 cr

Syllabus

Chapter 1. Basin formation - **1.1** Rifts, passive margins, foreland and strike-slip basins

Chapter 2. Thermal history - **2.1** Heat flow **2.2** Maturation modelling

Chapter 3. Burial history - **3.1** Backstripping and subsidence curves

Chapter 4. Basin modelling - **4.1** 1D petroleum-system models

Computer practice: 1D basin model of a well

Assessment: Project 40%; final 60%

Recommended textbooks

- Allen and Allen, Basin Analysis: Principles and Application to Petroleum Play Assessment (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEOL314 Stratigraphy (4).** Syllabus: Stratigraphy and biostratigraphy; facies associations; sedimentary sequences and basin geometries; geohistory analysis; lab and field exercises. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S205 Petroleum Drilling Engineering

Used in: Petroleum Exploration and Engineering - Sem 6, 4 cr

Syllabus

Chapter 1. Rig systems - **1.1** Hoisting, rotating, circulating, power

Chapter 2. Drilling fluids and hydraulics - **2.1** Mud design **2.2** Bit hydraulics optimisation

Chapter 3. Bits and drill string - **3.1** Bit selection **3.2** Drill-string design

Chapter 4. Formation pressure - **4.1** Pore and fracture pressure prediction

Chapter 5. Casing and cementing - **5.1** Casing design **5.2** Cementing

Chapter 6. Well control basics - **6.1** Kick detection **6.2** Kill methods

Laboratory practice: Mud property testing

Assessment: Labs 30%; final 70%

Recommended textbooks

- Bourgoyne, Millheim, Chenevert and Young, Applied Drilling Engineering (SPE). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Mitchell and Miska, Fundamentals of Drilling Engineering (SPE). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Petroleum Engineering programme. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Wuhan) programme list.

S206 Reservoir Engineering

Used in: Petroleum Exploration and Engineering - Sem 6, 4 cr

Syllabus

Chapter 1. Reservoir fundamentals - **1.1** Volumetric reserves estimation

Chapter 2. Material balance - **2.1** Oil and gas material balance **2.2** Havlena-Odeh

Chapter 3. Drive mechanisms - **3.1** Solution gas, gas cap, water drive

Chapter 4. Decline curves - **4.1** Arps decline analysis

Chapter 5. Water influx - **5.1** Aquifer models

Chapter 6. Displacement - **6.1** Fractional flow **6.2** Buckley-Leverett

Assessment: Assignments 30%; final 70%

Recommended textbooks

- Craft and Hawkins, Applied Petroleum Reservoir Engineering (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Dake, Fundamentals of Reservoir Engineering (Elsevier). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- Mines Petroleum Engineering programme.

S207 Production Engineering

Used in: Petroleum Exploration and Engineering - Sem 6, 4 cr

Syllabus

Chapter 1. Inflow performance - **1.1** Productivity index **1.2** IPR curves: *Vogel*

Chapter 2. Nodal analysis - **2.1** Tubing performance **2.2** System analysis

Chapter 3. Artificial lift - **3.1** Rod pumps **3.2** ESP **3.3** Gas lift

Chapter 4. Stimulation - **4.1** Acidising **4.2** Hydraulic fracturing

Chapter 5. Surface facilities - **5.1** Separation and treatment

Assessment: Assignments 30%; final 70%

Recommended textbooks

- Economides, Hill, Ehlig-Economides and Zhu, Petroleum Production Systems (Pearson). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Guo, Lyons and Ghalambor, Petroleum Production Engineering (Gulf). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- Mines Petroleum Engineering programme.

S208 Petroleum Machinery and Equipment

Used in: Petroleum Exploration and Engineering - Sem 6, 3 cr

Syllabus

Chapter 1. Drilling equipment - **1.1** Rigs, mud pumps, BOPs

Chapter 2. Production equipment - **2.1** Wellheads and Christmas trees **2.2** Artificial-lift equipment

Chapter 3. Pipelines and compressors - **3.1** Flow assurance

Chapter 4. Maintenance - **4.1** Reliability and inspection

Assessment: Assignment 30%; final 70%

Recommended textbooks

- Lyons and Plisga (eds), Standard Handbook of Petroleum and Natural Gas Engineering (Gulf). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Cooperation Base for Deep Drilling Equipment; TH 40040972 Offshore Petroleum Engineering. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page.

S209 Well Testing

Used in: Petroleum Exploration and Engineering - Sem 6, 2 cr

Syllabus

Chapter 1. Pressure transients - **1.1** Diffusivity equation **1.2** Flow regimes

Chapter 2. Drawdown and build-up - **2.1** Semilog analysis **2.2** Skin

Chapter 3. Type curves - **3.1** Derivative analysis

Chapter 4. Interference tests - **4.1** Reservoir connectivity

Computer practice: Well-test interpretation exercise

Assessment: Lab 40%; final 60%

Recommended textbooks

- Horne, Modern Well Test Analysis (Petroway). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Lee, Well Testing (SPE). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- Mines Petroleum Engineering programme.

S210 Petroleum Exploration and Development Design (capstone)

Used in: Petroleum Exploration and Engineering - Sem 7, 3 cr

Syllabus

Project work: Team capstone: integrate seismic, logs and core to define prospects, estimate resources, propose wells and a development concept with economics; report and presentation

Assessment: Report 50%; presentation 30%; peer review 20%

Recommended textbooks

- Gluyas and Swarbrick, Petroleum Geoscience (Wiley-Blackwell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Colorado School of Mines - GEGN439 Petroleum Exploration Design (3).** Syllabus: Multidisciplinary design course integrating well and 3D seismic data, seismic interpretation (Petrel) and prospect development in exploration plays; team reports and presentations. Link: https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf

S211 Enhanced Oil Recovery

Used in: Petroleum Exploration and Engineering - Sem 7, 3 cr

Syllabus

Chapter 1. Waterflooding - **1.1** Design and performance

Chapter 2. Gas injection - **2.1** Miscible and immiscible flooding

Chapter 3. Chemical EOR - **3.1** Polymer **3.2** Surfactant **3.3** Alkaline

Chapter 4. Thermal EOR - **4.1** Steam injection **4.2** In-situ combustion

Chapter 5. CO2-EOR and storage - **5.1** CO2 utilisation and storage

Assessment: Assignment 30%; final 70%

Recommended textbooks

- Green and Willhite, Enhanced Oil Recovery (SPE). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Lake, Enhanced Oil Recovery (SPE). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- Mines Petroleum Engineering programme.

S212 Petroleum Refining and Processing

Used in: Petroleum Exploration and Engineering - Sem 7, 3 cr

Syllabus

Chapter 1. Crude oil - **1.1** Characterisation and assays

Chapter 2. Separation - **2.1** Atmospheric and vacuum distillation

Chapter 3. Conversion - **3.1** Catalytic cracking **3.2** Hydrocracking **3.3** Reforming

Chapter 4. Treating - **4.1** Hydrotreating and desulfurisation

Chapter 5. Gas processing - **5.1** Sweetening **5.2** NGL recovery

Chapter 6. Petrochemicals - **6.1** Main products

Laboratory practice: ASTM distillation test

Assessment: Lab 30%; final 70%

Recommended textbooks

- Gary, Handwerk and Kaiser, Petroleum Refining: Technology and Economics (CRC). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Speight, The Chemistry and Technology of Petroleum (CRC). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- No direct equivalent in the five benchmark plans reviewed in the geoscience units reviewed (chemical engineering departments).

S213 Natural Gas and Unconventional Resources

Used in: Petroleum Exploration and Engineering - Sem 7, 3 cr

Syllabus

Chapter 1. Natural gas engineering - **1.1** Gas properties **1.2** Gas well deliverability

Chapter 2. Shale gas and oil - **2.1** Characteristics and evaluation

Chapter 3. Coalbed methane - **3.1** Adsorption and production

Chapter 4. Tight reservoirs - **4.1** Development strategies

Chapter 5. Fracturing geomechanics - **5.1** In-situ stress **5.2** Fracture design

Assessment: Assignment 30%; final 70%

Recommended textbooks

- Guo and Ghalambor, Natural Gas Engineering Handbook (Gulf). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Zoback and Kohli, Unconventional Reservoir Geomechanics (Cambridge). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUG-W Resource Exploration Eng. (new-energy class); CUGB School of Energy Resources. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page; CUG (Wuhan) programme list.

S214 Reservoir Simulation

Used in: Petroleum Exploration and Engineering - Sem 7, 2 cr

Syllabus

Chapter 1. Flow equations - **1.1** Discretisation and grids

Chapter 2. Black-oil models - **2.1** Formulation and solution

Chapter 3. History matching - **3.1** Methods and uncertainty

Chapter 4. Simulators - **4.1** Use of a simulator: *e.g. OPM Flow*

Computer practice: Simulation project

Assessment: Project 60%; final 40%

Recommended textbooks

- Ertekin, Abou-Kassem and King, Basic Applied Reservoir Simulation (SPE). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- Mines Petroleum Engineering programme.

S215 Petroleum Economics and HSE

Used in: Petroleum Exploration and Engineering - Sem 7, 2 cr

Syllabus

Chapter 1. Petroleum economics - **1.1** Fiscal systems **1.2** Production-sharing contracts

Chapter 2. Project economics - **2.1** Cash flow **2.2** Risk and decision analysis

Chapter 3. HSE - **3.1** HSE management systems **3.2** Environmental protection

Assessment: Assignment 40%; final 60%

Recommended textbooks

- Johnston, International Petroleum Fiscal Systems and Production Sharing Contracts (PennWell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat
- Mian, Project Economics and Decision Analysis (PennWell). *Commercial textbook - no legal free PDF; borrow or buy.* Library lookup: WorldCat

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH 40031141 Construction HSE Management (HSE part). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF.

S216 Production Internship (5 weeks)

Used in: All ten courses (or most) - Summer 3, 5 cr

Syllabus

Chapter 1. Preparation - **1.1** Placement agreement and safety induction **1.2** Learning objectives

Stages:

- Week 1 induction and site orientation
- Weeks 2-4 supervised technical work
- Week 5 reporting and presentation to the host

Assessment: Daily logbook 30%; supervisor assessment 30%; internship report 40%

Recommended textbooks

No textbook: practice-based subject (faculty guidelines, logbook and report templates).

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Production Field Trip (6 wks) + Mine-Base Training; TH 40030584 Construction Internship; TH 40041003 Production Internship. These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page.

S217 Graduation Design / Bachelor Thesis

Used in: All ten courses (or most) - Sem 8, 8 cr

Syllabus

Chapter 1. Proposal - **1.1** Topic selection with supervisor **1.2** Literature review **1.3** Methodology and work plan

Chapter 2. Execution - **2.1** Data collection, laboratory or design work **2.2** Analysis

Chapter 3. Writing - **3.1** Thesis structure and drafting **3.2** Revision with supervisor

Chapter 4. Defence - **4.1** Oral presentation and defence before committee

Assessment: Supervisor 30%; examiners' evaluation of thesis 50%; defence 20%

Recommended textbooks

No textbook: practice-based subject (faculty guidelines, logbook and report templates).

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** TH Comprehensive Thesis Training (15 cr); CUGB BS Thesis (6 cr, 12 wks). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: Tsinghua 2023 training plan PDF; CUG (Beijing) programme page.

S218 Frontiers Seminar of the Discipline

Used in: All ten courses (or most) - Sem 8, 2 cr

Syllabus

Chapter 1. Seminar series - **1.1** Digitalisation and artificial intelligence in the geosciences **1.2** Critical minerals and energy transition **1.3** Sustainability and ESG in resources and construction **1.4** Advances in space and remote sensing

Assessment: Attendance and two seminar reviews

Recommended textbooks

No textbook: practice-based subject (faculty guidelines, logbook and report templates).

Most similar subjects at benchmark universities

- **Tsinghua University / China University of Geosciences:** CUGB Frontiers of the Discipline (S7). These universities publish course name, credits, hours and semester only; no subject syllabus is published online. Source: CUG (Beijing) programme page.

Appendix A. Open-access textbooks: links and tables of contents

A01 Calculus, Volumes 1-3 - Strang, Herman et al. (OpenStax, Rice University). *CC BY-NC-SA; free PDF and web.*

Link: https://openstax.org/details/books/calculus-volume-1 | Used for subjects: S002, S008

Table of contents: not verified for this report - see the contents page at the link.

A02 Linear Algebra - Jim Hefferon. *Free (CC BY-SA), PDF via OpenIntro.*

Link: https://hefferon.net/linearalgebra/ | Used for subjects: S003

Table of contents: **1** Linear systems and Gauss's method **2** Vector spaces **3** Linear maps and matrices **4** Determinants **5** Eigenvectors and eigenvalues (similarity)

A03 University Physics, Volume 1 - Ling, Sanny, Moebs et al. (OpenStax). *CC BY; free PDF and web.*

Link: https://openstax.org/books/university-physics-volume-1/pages/preface | Used for subjects: S004, S018

Table of contents: **1** Units and Measurement **2** Vectors **3** Motion Along a Straight Line **4** Motion in Two and Three Dimensions **5** Newton's Laws of Motion **6** Applications of Newton's Laws **7** Work and Kinetic Energy **8** Potential Energy and Conservation of Energy **9** Linear Momentum and Collisions **10** Fixed-Axis Rotation **11** Angular Momentum **12** Static Equilibrium and Elasticity **13** Gravitation **14** Fluid Mechanics **15** Oscillations **16** Waves **17** Sound

A04 Chemistry 2e - Flowers, Theopold, Langley, Robinson (OpenStax). *CC BY; free PDF and web (also on LibreTexts).*

Link: https://chem.libretexts.org/Bookshelves/General_Chemistry/Chemistry_2e_%28OpenStax%29 | Used for subjects: S005

Table of contents: **1** Essential Ideas **2** Atoms, Molecules, and Ions **3** Composition of Substances and Solutions **4** Stoichiometry of Chemical Reactions **5** Thermochemistry **6** Electronic Structure and Periodic Properties **7** Chemical Bonding and Molecular Geometry **8** Advanced Theories of Covalent Bonding **9** Gases **10** Liquids and Solids **11** Solutions and Colloids **12** Kinetics **13** Fundamental Equilibrium Concepts **14** Acid-Base Equilibria **15** Equilibria of Other Reaction Classes **16** Thermodynamics **17** Electrochemistry **18** Representative Metals, Metalloids, and Nonmetals **19** Transition Metals and Coordination Chemistry **20** Organic Chemistry **21** Nuclear Chemistry

A05 Think Python: How to Think Like a Computer Scientist (2nd ed.) - Allen B. Downey (Green Tea Press). *CC BY-NC; free online (3rd ed. free online as Jupyter notebooks).*

Link: https://greenteapress.com/thinkpython2/html/ | Used for subjects: S006

Table of contents: **1** The way of the program **2** Variables, expressions and statements **3** Functions **4** Case study: interface design **5** Conditionals and recursion **6** Fruitful functions **7** Iteration **8** Strings **9** Case study: word play **10** Lists **11** Dictionaries **12** Tuples **13** Case study: data structure selection **14** Files **15** Classes and objects **16** Classes and functions **17** Classes and methods **18** Inheritance **19** The Goodies **20** Debugging **21** Analysis of Algorithms

A06 Lessons in Electric Circuits (6 volumes) - Tony R. Kuphaldt. *Design Science / Creative Commons licence; free PDF per volume.*

Link: https://ibiblio.org/kuphaldt/electricCircuits/index.htm | Used for subjects: S010, S019, S021

Table of contents: **1** Volume I - DC **2** Volume II - AC **3** Volume III - Semiconductors **4** Volume IV - Digital **5** Volume V - Reference **6** Volume VI - Experiments

A07 Physical Geology (2nd ed.) - Steven Earle (BCcampus Open Education). *CC BY; free PDF and web.*

Link: https://yc-prod-openlibrary-repo.ecampusontario.ca/jspui/bitstream/123456789/713/3/Physical-Geology-2nd-Edition-1591817334.pdf | Used for subjects: S011, S014, S025

Table of contents: not verified for this report - see the contents page at the link.

A08 Engineering Statics: Open and Interactive - Daniel Baker, William Haynes. *CC BY-NC-SA; free web and PDF.*

Link: https://engineeringstatics.org/pdf/statics.pdf | Used for subjects: S012

Table of contents: **1** Introduction to Statics **2** Forces and Other Vectors **3** Equilibrium of Particles **4** Moments and Static Equivalence **5** Rigid Body Equilibrium **6** Equilibrium of Structures **7** Centroids and Centers of Gravity **8** Internal Loadings **9** Friction **10** Moments of Inertia

A09 Notes on Diffy Qs: Differential Equations for Engineers - Jiri Lebl. *CC BY-SA / CC BY-NC-SA; free PDF.*

Link: https://jirka.org/diffyqs/diffyqs.pdf | Used for subjects: S015

Table of contents: **1** Introduction **2** First-order equations **3** Higher-order linear ODEs **4** Systems of ODEs **5** Fourier series and PDEs **6** More on eigenvalue problems **7** The Laplace transform **8** Power-series methods **9** Nonlinear systems **10** Appendix A: Linear algebra

A10 OpenIntro Statistics (4th ed.) - Diez, Cetinkaya-Rundel, Barr. *CC BY-SA; free PDF (pay-what-you-want, $0 minimum).*

Link: https://leanpub.com/os | Used for subjects: S016

Table of contents: not verified for this report - see the contents page at the link.

A11 Mineralogy - Dexter Perkins et al. (Univ. of North Dakota; LibreTexts). *CC BY-NC-SA; free web and PDF export.*

Link: https://geo.libretexts.org/Bookshelves/Geology/Mineralogy_(Perkins_et_al.) | Used for subjects: S020, S023, S129

Table of contents: **1** Introduction **2** Mineral Chemistry **3** Mineral Properties **4** Crystals and Crystallization **5** Optical Mineralogy **6** Igneous Rocks and Silicate Minerals **7** Sedimentary Minerals and Sedimentary Rocks **8** Metamorphic Minerals and Metamorphic Rocks **9** Ore Deposits and Economic Minerals **10** Crystal Morphology and Symmetry **11** Crystallography **12** X-ray Diffraction and Mineral Analysis **13** Crystal Structures **14** Mineral Descriptions

A12 Geological Structures: a Practical Introduction - John W.F. Waldron, Morgan Snyder (Univ. of Alberta). *CC BY-NC; free web and PDF.*

Link: https://pressbooks.openeducationalberta.ca/introductorystructuralgeology/ | Used for subjects: S024, S026

Table of contents: **1** Part 1: Topics (maps, orientation, stereographic projection, folds, faults, stress and strain, cross-sections) **2** Part 2: Labs (practical exercises for each topic)

A13 Mineral Deposit Models (USGS Bulletin 1693) - D.P. Cox, D.A. Singer (eds), U.S. Geological Survey, 1986. *Public domain; free PDF.*

Link: https://pubs.usgs.gov/bul/b1693/ | Used for subjects: S027, S034, S040

Table of contents: **1** Part 1: 85 descriptive deposit models (geological environment, deposit description, exploration guides) **2** Part 2: 60 grade-tonnage models from over 3,900 deposits

A14 Geophysics for Practicing Geoscientists (GPG) - UBC Geophysical Inversion Facility. *Open web resource (CC BY).*

Link: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm | Used for subjects: S030, S077, S078, S079, S080, S082, S084, S092

Table of contents: **1** Welcome and getting started **2** Foundations: framework and physical properties **3** Geophysical survey methods: magnetics, gravity, DC resistivity and IP, seismic refraction, GPR, terrain conductivity (EM) **4** Inversion concepts **5** Exercises **6** Case histories **7** Software and manuals **8** Resources (glossary, references)

A15 Fundamentals of Remote Sensing (tutorial) - Canada Centre for Remote Sensing / Natural Resources Canada. *Free web tutorial.*

Link: https://natural-resources.canada.ca/science-data/science-research/geomatics/remote-sensing/tutorial-fundamentals-remote-sensing | Used for subjects: S031, S161, S169

Table of contents: not verified for this report - see the contents page at the link.

A16 Essentials of Geographic Information Systems - Jonathan Campbell, Michael Shin. *CC BY-NC-SA; free (Open Textbook Library).*

Link: https://open.umn.edu/opentextbooks/textbooks/67 | Used for subjects: S033, S151

Table of contents: **1** Introduction **2** Map Anatomy **3** Data, Information, and Where to Find Them **4** Data Models for GIS **5** Geospatial Data Management **6** Data Characteristics and Visualization **7** Geospatial Analysis I: Vector Operations **8** Geospatial Analysis II: Raster Data **9** Cartographic Principles **10** GIS Project Management

A17 The JORC Code (2012 edition) - Joint Ore Reserves Committee. *Free PDF (Chinese translation also free).*

Link: https://jorc.org/docs/jorc_code2012.pdf | Used for subjects: S037, S043

Table of contents: not verified for this report - see the contents page at the link.

A18 Groundwater - R. Allen Freeze, John A. Cherry (1979). *Free PDF from The Groundwater Project (preserved book).*

Link: https://gw-project.org/books/groundwater/ | Used for subjects: S039, S101, S120

Table of contents: not verified for this report - see the contents page at the link.

A19 Practical Rock Engineering - Evert Hoek (Rocscience). *Free PDF from publisher (author's course notes).*

Link: https://www.rocscience.com/assets/resources/learning/hoek/Practical-Rock-Engineering-Full-Text.pdf | Used for subjects: S063, S069, S072, S107, S139

Table of contents: **1** What is rock engineering? **2** Intact rock strength **3** Shear strength of discontinuities **4** Rock mass properties **5** In situ and induced stresses **6** When is a rock engineering design acceptable? **7** Structurally controlled instability in tunnels **8** Factor of safety and probability of failure **9** A slope stability problem in Hong Kong **10** Rock slopes in civil and mining engineering **11** Rock mass classification **12** Tunnels in weak rock **13** Design of large underground caverns **14** The Rio Grande project - Argentina **15** Rockbolts and cables **16** Model to demonstrate how rockbolts work **17** Shotcrete support **18** Tensile failure of spalling in tunnels **19** Analysis of rockfall hazards **20** Blasting damage in rock

A20 The Landslide Handbook - A Guide to Understanding Landslides (USGS Circular 1325) - L.M. Highland, P. Bobrowsky (2008). *Public domain; free PDF.*

Link: https://pubs.usgs.gov/publication/cir1325 | Used for subjects: S069, S163, S193

Table of contents: not verified for this report - see the contents page at the link.

A21 Signals and Systems: Theory and Applications - Fawwaz Ulaby, Andrew Yagle (Michigan Publishing). *Free PDF for students (Free Electrical Engineering Textbook Initiative).*

Link: https://ulabyweb.eecs.umich.edu/ftuc | Used for subjects: S074

Table of contents: not verified for this report - see the contents page at the link.

A22 Electromagnetics, Volume 1 - Steven W. Ellingson (Virginia Tech Publishing). *CC BY-SA; free PDF.*

Link: https://vtechworks.lib.vt.edu/handle/10919/84164 | Used for subjects: S075

Table of contents: not verified for this report - see the contents page at the link.

A23 Seismic Data Analysis: Processing, Inversion, and Interpretation of Seismic Data - Oz Yilmaz (SEG). *Open access on the SEG Wiki (web, not PDF).*

Link: https://wiki.seg.org/wiki/Seismic_Data_Analysis | Used for subjects: S080, S085, S090

Table of contents: **1** Introduction **2** Fundamentals of signal processing **3** Deconvolution **4** Velocity analysis and statics corrections **5** Migration **6** Dip-moveout correction and prestack migration **7** Noise and multiple attenuation **8** 3-D seismic exploration **9** Earth imaging in depth **10** Earth modeling in depth **11** Structural inversion

A24 Guidelines for Radioelement Mapping Using Gamma Ray Spectrometry Data (IAEA-TECDOC-1363) - International Atomic Energy Agency (2003). *Free PDF.*

Link: https://www-pub.iaea.org/MTCD/Publications/PDF/te_1363_web/PDF/print%20version.pdf | Used for subjects: S093

Table of contents: not verified for this report - see the contents page at the link.

A25 Analytical Chemistry 2.1 - David Harvey (DePauw University; LibreTexts). *Free; full-book PDF download.*

Link: https://chem.libretexts.org/Bookshelves/Analytical_Chemistry/Analytical_Chemistry_2.1_by_David_Harvey | Used for subjects: S095, S096, S106

Table of contents: **1** Introduction to Analytical Chemistry **2** Basic Tools of Analytical Chemistry **3** The Vocabulary of Analytical Chemistry **4** Evaluating Analytical Data **5** Standardizing Analytical Methods **6** Equilibrium Chemistry **7** Obtaining and Preparing Samples for Analysis **8** Gravimetric Methods **9** Titrimetric Methods **10** Spectroscopic Methods **11** Electrochemical Methods **12** Chromatographic and Electrophoretic Methods **13** Kinetic Methods **14** Developing a Standard Method **15** Quality Assurance

A26 Introduction to Engineering Thermodynamics - Claire Yu Yan (Univ. of British Columbia). *CC BY; free (Open Textbook Library).*

Link: https://open.umn.edu/opentextbooks/textbooks/1252 | Used for subjects: S115

Table of contents: **1** Basic Concepts and Definitions **2** Thermodynamic Properties **3** Ideal and Real Gasses **4** The First Law of Thermodynamics for Closed Systems **5** The First Law of Thermodynamics for a Control Volume **6** Entropy and the Second Law of Thermodynamics

A27 Handbook of Best Practices for Geothermal Drilling - J. Finger, D. Blankenship (Sandia National Laboratories). *Free PDF (U.S. DOE).*

Link: https://www1.eere.energy.gov/geothermal/pdfs/drillinghandbook.pdf | Used for subjects: S127

Table of contents: not verified for this report - see the contents page at the link.

A28 Database Design (2nd ed.) - Adrienne Watt, Nelson Eng (BCcampus). *CC BY; free PDF.*

Link: https://yc-prod-openlibrary-repo.ecampusontario.ca/jspui/bitstream/123456789/247/16/Database-Design-2nd-Edition-1540829576.pdf | Used for subjects: S148, S153

Table of contents: **1** Before the Advent of Database Systems **2** Fundamental Concepts **3** Characteristics and Benefits of a Database **4** Types of Data Models **5** Data Modelling **6** Classification of Database Management Systems **7** The Relational Data Model **8** The Entity Relationship Data Model **9** Integrity Rules and Constraints **10** ER Modelling **11** Functional Dependencies **12** Normalization **13** Database Development Process **14** Database Users **15** SQL Structured Query Language **16** SQL Data Manipulation Language

A29 Dive into Deep Learning - Zhang, Lipton, Li, Smola. *Free online book and PDF (arXiv).*

Link: https://arxiv.org/abs/2106.11342 | Used for subjects: S157

Table of contents: not verified for this report - see the contents page at the link.

A30 Practical Meteorology: An Algebra-based Survey of Atmospheric Science - Roland Stull (Univ. of British Columbia). *CC BY-NC-SA; free whole-book and chapter PDFs.*

Link: https://www.eoas.ubc.ca/books/Practical_Meteorology | Used for subjects: S165, S168, S175, S176, S177

Table of contents: **1** Atmospheric Basics **2** Solar & Infrared Radiation **3** Thermodynamics **4** Water Vapor **5** Atmospheric Stability **6** Clouds **7** Precipitation Processes **8** Satellites & Radar **9** Weather Reports & Map Analysis **10** Atmospheric Forces & Winds **11** General Circulation **12** Fronts & Airmasses **13** Extratropical Cyclones **14** Thunderstorm Fundamentals **15** Thunderstorm Hazards **16** Tropical Cyclones **17** Regional Winds **18** Atmospheric Boundary Layer **19** Pollutant Dispersion **20** Numerical Weather Prediction (NWP) **21** Natural Climate Processes **22** Atmospheric Optics

A31 Basics of Space Flight - NASA Jet Propulsion Laboratory. *Free (U.S. Government work); web tutorial.*

Link: https://science.nasa.gov/learn/basics-of-space-flight/participants-guide | Used for subjects: S167, S173, S174

Table of contents: not verified for this report - see the contents page at the link.

A32 CubeSat 101: Basic Concepts and Processes for First-Time CubeSat Developers - NASA CubeSat Launch Initiative (2017). *Free PDF (U.S. Government work).*

Link: https://www.nasa.gov/wp-content/uploads/2017/03/nasa_csli_cubesat_101_508.pdf | Used for subjects: S173, S178

Table of contents: not verified for this report - see the contents page at the link.

A33 Introduction to Tourism and Hospitality in BC - Morgan Westcott (ed.) (BCcampus). *CC BY; free PDF.*

Link: https://opentextbc.ca/introtourism/open/download?type=pdf | Used for subjects: S183, S195

Table of contents: **1** History and Overview **2** Transportation **3** Accommodation **4** Food and Beverage Services **5** Recreation **6** Entertainment **7** Travel Services **8** Services Marketing **9** Customer Service **10** Environmental Stewardship **11** Risk Management and Legal Liability **12** Aboriginal Tourism **13** Careers and Work Experience **14** Back to the Big Picture: Globalization and Trends

A34 Operational Guidelines for UNESCO Global Geoparks - UNESCO. *Free PDF.*

Link: https://unesco.org.uk/site/assets/files/2129/operational_guidelines_for_unesco_global_geoparks_2015.pdf | Used for subjects: S187

Table of contents: not verified for this report - see the contents page at the link.

Appendix B. Benchmark sources

- MIT OpenCourseWare subject pages (links given in each subject entry), ocw.mit.edu
- Colorado School of Mines catalogue: Geological Engineering degree and course descriptions https://catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/geologicalengineering.pdf
- Colorado School of Mines course catalogue: CEEN https://catalog.mines.edu/coursesaz/civileng/civileng.pdf, MATH https://catalog.mines.edu/coursesaz/math/math.pdf, PHGN https://catalog.mines.edu/coursesaz/physics, Mining Engineering https://facultyhandbook.mines.edu/undergraduate/programs/miningengineering/miningengineering.pdf
- Tsinghua University 2023 training plan, Civil, Hydraulic and Ocean Engineering https://www.tsinghua.edu.cn/jxjywj/bkzy2023/zxzy/10.pdf
- China University of Geosciences (Beijing), Resources Prospecting Engineering programme https://sesr.cugb.edu.cn/c/2020/06/17/619374.shtml
- China University of Geosciences (Wuhan), undergraduate programmes https://eniec.cug.edu.cn/info/1012/1198.htm